ON THE DISTRIBUTION OF PRIME SQUARES AND ITS APPLICATIONS

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ABSTRACT. We prove that, for every $A > 0$, all but $O_A(X(\log X)^{-A})$ integers $n < X$ satisfying $n \equiv 11 \pmod{24}$ admit a representation

$$n = a^2 + b^2 + p^2,$$

where p and $a^2 + b^2 + 1$ are prime and $(a, b) = 1$. For ergodic averages along prime squares, we prove a universal divergence result for integrable functions, while establishing almost-everywhere convergence for near-integrable functions. We also quantify how the bounds used to establish convergence behave as one approaches integrability. Moreover, we prove endpoint improving and sparse bounds for the corresponding discrete averages, with an explicit Orlicz loss.

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1. INTRODUCTION

The object of this paper is the sequence of prime squares

$$\mathbb{P}^2 = \{p^2 : p \in \mathbb{P}\}.$$

We study it from three points of view: improving and sparse inequalities for discrete averages, pointwise ergodic theory at or near the L^1 endpoint, and an exceptional-set problem of Goldbach type.

These questions arise in different areas of analysis and number theory, but three arithmetic features of prime squares recur. First, the distribution of primes in arithmetic progressions enters all three applications. In particular, for each fixed q, it implies that prime squares are asymptotically equidistributed among the nonzero quadratic residues modulo q. Second, one

needs cancellation in quadratic exponential sums. For

$$S_N(\alpha) = \sum_{p \le N} (\log p) e(\alpha p^2),$$

the behaviour near a rational point $\alpha = a/q$ is described by quadratic Gauss sums and primes in arithmetic progressions, while away from such points one needs cancellation. Finally, along suitable square-free moduli q for which the density of the nonzero quadratic residues tends to zero, their normalized gaps are asymptotically Poisson distributed: for every $\theta > 0$, the proportion of consecutive gaps exceeding θ times the mean gap tends to $e^{-\theta}$ [16]. The arguments also use more general arithmetic facts, including the restrictions on the prime factorization of sums of two squares and Dirichlet's approximation theorem.

The sums S_N belong to a classical line of work in the Waring–Goldbach problem going back to Vinogradov and Hua [22, 57]. An influential consequence is Hua's theorem that every sufficiently large integer congruent to 5 modulo 24 is a sum of five prime squares. Subsequent work strengthened the relevant exponential-sum estimates and treated prime squares drawn from thinner sets of primes; see, for example, [1, 18, 44, 48]. The estimates used below belong to this circle-method framework.

We begin with discrete improving inequalities. Define

$$A_{N,\Lambda} f(x) = \frac{1}{N} \sum_{n \le N} \Lambda(n) f(x - n^2),$$

where Λ is the von Mangoldt function. The normalization is chosen so that $A_{N,\Lambda}$ has mass comparable to one and is supported at spatial scale N^2.

The continuous model explains the expected exponents. Convolution with a bounded kernel sends L^1 to L^∞, but averaging over a curved measure also improves integrability even though the measure is singular. For arclength measure $d\sigma$ on $\mathbb{S}^1$, the Littman–Strichartz estimate gives

$$\|f * d\sigma\|_{L^3(\mathbb{R}^2)} \lesssim \|f\|_{L^{3/2}(\mathbb{R}^2)};$$

Such improving estimates arise from the curvature of the sphere and the resulting decay of the Fourier transform of its surface measure. The same ingredients play a central role in the Stein–Tomas restriction theorem [33, 47, 55]. The exponents are sharp, as is seen by taking f to be the indicator of a thin annulus around the circle.

A useful comparison is provided by a Salem measure of full dimension μ supported on $\mathbb{S}^1$. If μ_j denotes its j-th dyadic frequency piece, Fourier decay gives an L^2 estimate for convolution with μ_j, while the Frostman estimate gives an $L^1 \to L^\infty$ bound. Applied to characteristic functions, these estimates give

$$|\langle \mathbf{1}_F * \mu_j, \mathbf{1}_G \rangle| \lesssim \min\big\{ 2^{-j/2+o(j)} (|F||G|)^{1/2}, 2^{j+o(j)} |F||G| \big\}.$$

Balancing the two estimates yields $L^p \to L^{p'}$ improvement for $p > 3/2$. For arclength measure, curvature removes the $o(j)$ losses, so the endpoint $p = 3/2$ is also obtained.

The same critical exponent governs averages along the squares on $\mathbb{Z}$. Indeed, up to spatial scale N^2 there are about N squares, so they form a set of dimension $1/2$. Han, Lacey, and Yang proved the corresponding improving and sparse estimates for $p > 3/2$ [21], and their argument can be refined to reach the endpoint. The endpoint distinction has little effect on the usual ergodic applications. In the prime-square setting, however, establishing the improving and sparse estimates at the endpoint is technically more delicate because the relevant exponential-sum estimates carry additional arithmetic losses. These losses give rise to the Orlicz factor in the following theorem.

For $t > 0$, write

$$\operatorname{Log} t = 2 + |\log t|, \qquad \operatorname{Log}\operatorname{Log} t = 2 + \log(2 + |\log t|),$$

and, for $c > 0$, put

$$\Phi_c(t) = \exp\left(c\frac{(\operatorname{Log} t)^{1/2}}{\operatorname{Log}\operatorname{Log} t}\right).$$

Thus $\Phi_c(t)$ is large when t is small. Our endpoint estimate is as follows.

Theorem 1.1. *Let $I = [-N^2, 2N^2] \cap \mathbb{Z}$. For all $F, G \subseteq I$,*

$$\frac{1}{|I|}\big|\langle A_{N,\Lambda}\mathbf{1}_F, \mathbf{1}_G\rangle\big| \ll \left(\frac{|F|}{|I|}\frac{|G|}{|I|}\right)^{2/3} \Phi_{4\sqrt{3}/9+o(1)}\left(\frac{|F|}{|I|}\frac{|G|}{|I|}\right). \tag{1.2}$$

Consequently, for every $3/2 < p \le 2$ and every $f \in \ell^p(\mathbb{Z})$ supported in I,

$$\left(\frac{1}{|I|}\sum_{x\in\mathbb{Z}} |A_{N,\Lambda}f(x)|^{p'}\right)^{1/p'} \lesssim_p \left(\frac{1}{|I|}\sum_{x\in\mathbb{Z}} |f(x)|^{p}\right)^{1/p}.$$

Here and below the notation $4\sqrt{3}/9 + o(1)$ means that the constant may be replaced by $4\sqrt{3}/9 + \varepsilon$ for every fixed $\varepsilon > 0$.

Thus (1.2) reaches the same critical pair $(p, p') = (3/2, 3)$ as the circle and the unweighted square averages, with the arithmetic loss confined to the Orlicz factor. In other words, up to scale-dependent normalization, convolution with $A_{N,\Lambda}$ softens the "singularities" of a function on the integers more strongly than convolution with a general Salem measure on $\mathbb{S}^1 \subset \mathbb{R}^2$.

The exponent $3/2$ cannot be improved. Indeed, take

$$F = \{-p^2 : p \le N\}, \qquad G = \{0\}.$$

Then $|F| \asymp N/\log N$ and the prime number theorem gives

$$\frac{1}{|I|}\langle A_{N,\Lambda}\mathbf{1}_F, \mathbf{1}_G\rangle \asymp N^{-2} \asymp \left(\frac{|F||G|}{|I|^2}\right)^{2/3}\left(\operatorname{Log}\frac{|F||G|}{|I|^2}\right)^{2/3}.$$

The example also shows that some endpoint loss is necessary. There is still a gap between the logarithmic loss forced here and the quasi-logarithmic function $\Phi_{4\sqrt{3}/9+o(1)}$ in the theorem.

Improving inequalities for discrete squares, primes, and polynomial averages have been developed in [19–21, 30]; Theorem 1.1 is the prime-square endpoint of this circle of questions.

A stronger and more directly applicable result concerns the associated maximal operator, which controls the averages simultaneously over all scales. Put

$$A^*_\Lambda f(x) = \sup_{N\ge 2} |A_{N,\Lambda}f(x)|.$$

Maximal estimates along arithmetic sets originate in Bourgain's work on polynomial averages and Wierdl's work on the primes [3–5, 58]; later developments include [19–21, 40]. A more recent approach to such maximal estimates is sparse domination, developed for discrete operators in [10, 12, 23, 26]. We obtain the endpoint sparse bound corresponding to (1.2).

Theorem 1.3. *Let $F, G \subset \mathbb{Z}$ be finite. There is a sparse collection of intervals $\mathcal{S}$ such that*

$$\big|\langle A^*_\Lambda \mathbf{1}_F, \mathbf{1}_G\rangle\big| \ll \sum_{I\in\mathcal{S}} |I| \left(\frac{|F\cap I||G\cap I|}{|I|^2}\right)^{2/3} \Phi_{4\sqrt{3}/9+o(1)}\left(\frac{|F\cap I||G\cap I|}{|I|^2}\right).$$

Moreover, if $(1/p, 1/q)$ *lies in the interior of the triangle with vertices* $(2/3, 2/3)$, $(1, 0)$, *and* $(0, 1)$, *then there is a sparse collection* $\mathcal{S}_{p,q}$ *such that*

$$\left|\langle A_\Lambda^* \mathbf{1}_F, \mathbf{1}_G\rangle\right| \ll_{p,q} \sum_{I\in\mathcal{S}_{p,q}} |I| \left(\frac{|F\cap I|}{|I|}\right)^{1/p} \left(\frac{|G\cap I|}{|I|}\right)^{1/q}. \tag{1.4}$$

Here sparsity means that each $I \in \mathcal{S}$ contains a set $E_I \subset I$ with $|E_I| \geq |I|/10$, and the sets E_I are pairwise disjoint.

A contrasting endpoint phenomenon arises in ergodic theory. The arithmetic distribution and spacing properties of prime squares again play a role, but through a different mechanism. Given a sequence of integers (n_k), call it L^r-good if, for every measure-preserving probability system (X, μ, T) and every $f \in L^r(X)$, the averages

$$\frac{1}{N}\sum_{k\leq N} f(T^{n_k}x)$$

converge almost everywhere. Birkhoff's theorem shows that $n_k = k$ is L^r-good for every $r \geq 1$ [2]. Bourgain proved L^r-goodness for the squares and, more generally, for integer polynomial sequences when $r > 1$, while Wierdl proved the corresponding result for the primes [5, 58]. Nair extended this to $n_k = P(p_k)$ for every integer-valued polynomial P, and Mehlhop obtained a multidimensional extension [38, 43].

The endpoint L^1 behaves differently. Buczolich and Mauldin proved that the squares are universally L^1-bad [7]. LaVictoire proved persistent universal L^1-badness for the monomials $n_k = k^d$, $d \geq 2$, and for the primes [31]; see also the weighted divergence result of Buczolich and Eisner [6]. Recent work of Fragkos, the author, and Stokolosa shows a related endpoint failure for polynomial averages over short intervals along every infinite subsequence [11]. For prime squares, we use the distribution and spacing of the squares of units modulo suitable square-free integers. These properties yield a sequence of finite periodic measure-preserving systems for which the prime-square maximal operators have arbitrarily large weak $(1, 1)$ constants.

There is also a quantitative question on the positive side. The usual proof of pointwise convergence for L^p functions, $p > 1$, combines convergence on a dense class with a maximal inequality. As $p \downarrow 1$, the growth of the maximal constant measures the loss in this approach to the endpoint. For averages along the primes, Lacey, the author, and Rahimi proved sparse bounds for indicator functions with factors $(p-1)^{-2}$ unconditionally and $(p-1)^{-1}$ under GRH [30, Corollary 1.4]. Such growth alone does not imply L^1-divergence, but its rate gives information beyond qualitative badness: it quantifies how close to L^1 the argument based on maximal estimates remains effective and helps describe the classes of unbounded functions for which convergence survives. We establish both persistent divergence and an explicit maximal bound for prime squares.

Theorem 1.5. *Let* p_k *denote the* k*-th prime, and put* $n_k = p_k^2$.

(i) *The sequence* (n_k) *is persistently universally* L^1*-bad. More precisely, for every nonatomic ergodic measure-preserving probability system* (X, μ, T) *and every infinite set* $\mathcal{S} \subset \mathbb{N}$, *there is an* $f \in L^1(X)$ *for which*

$$\frac{1}{N}\sum_{k\leq N} f(T^{n_k}x), \qquad N \in \mathcal{S},$$

diverges on a set of positive measure.

(ii) *There is an absolute constant $C > 0$ such that, for every invertible measure-preserving probability system (X, μ, T), every $1 < p \le 2$, and every $f \in L^p(X)$,*

$$\left\| \sup_{N \ge 1} \left| \frac{1}{N} \sum_{k \le N} f(T^{n_k} x) \right| \right\|_{L^p(X)} \le \exp\left(\frac{C}{p-1} \log \frac{e}{p-1} \right) \|f\|_{L^p(X)}.$$

We do not expect the upper bound in (ii) to be sharp. Determining the optimal dependence on $p-1$ remains open, and the available arguments do not suggest a precise conjecture for its order of growth. Section 7 proves a stronger modular maximal estimate and obtains convergence for an explicit Orlicz class near L^1.

We next turn to an additive application. Let $E(X)$ denote the number of even integers $n \le X$ that are not sums of two primes. The classical circle method gives $E(X) = o(X)$, while Montgomery and Vaughan proved the quantitative bound $E(X) \ll X^{1-\delta}$ for some $\delta > 0$ [42]. Goldbach problems in which one or more prime variables are restricted to thin sets have a parallel history. Linnik proved that the set

$$\mathcal{P} = \{a^2 + b^2 + 1 \in \mathbb{P} : (a, b) = 1\}$$

is infinite [32]. Its distribution in short intervals and its additive properties were subsequently studied by Wu, Matomäki, Tolev, and others [34,35,52–54,59]. In particular, Matomäki proved a binary Goldbach theorem with one prime in $\mathcal{P}$, with the number of exceptions up to X bounded by $\ll X(\log X)^{-c}$ for some $c > 0$ [35]. Teräväinen later proved an almost-all binary theorem and a ternary Goldbach theorem with all prime variables in $\mathcal{P}$ [51] with exceptional set of size $o(X)$. Matomäki and Shao's work on Vinogradov's theorem with almost twin primes provides another example of sieve and transference methods in Goldbach problems with constrained prime variables [37].

Our prime-square variant has a forced local condition. If $r = a^2 + b^2 + 1 \in \mathcal{P}$ and $r > 3$, then a and b are odd, so $r \equiv 3 \pmod 8$. Modulo 3, exactly one of a, b is divisible by 3, and hence $r \equiv 2 \pmod 3$. Thus $r \equiv 11 \pmod{24}$. Since $p^2 \equiv 1 \pmod{24}$ for every prime $p > 3$, one must have $r + p^2 \equiv 12 \pmod{24}$. This is the only local progression relevant to the theorem, apart from the harmless identity $12 = 3 + 3^2$.

Theorem 1.6 (Quantitative exceptional set)**.** *Almost all integers congruent to* 12 *modulo* 24 *can be written as*

$$a^2 + b^2 + p^2 + 1, \qquad a^2 + b^2 + 1 \in \mathcal{P}, \quad p \in \mathbb{P}.$$

More precisely, if

$$E_2(X) = \{1 \le n < X : n \equiv 12 \pmod{24},\ n \notin \mathcal{P} + \mathbb{P}^2\},$$

then, for every $A > 0$,

$$|E_2(X)| \ll_A X(\log X)^{-A}.$$

The logarithmic saving of arbitrary order is an essential part of the result. While $|E_2(X)| = o(X)$ would follow from substantially softer input, the bound $|E_2(X)| \ll_A X(\log X)^{-A}$ requires circle-method estimates that are uniform in the moduli and auxiliary parameters arising from the sieve.

Several questions remain open. For a polynomial $P \in \mathbb{Z}[x]$ of degree at least two, one may ask for endpoint improving inequalities and sparse bounds for averages along $P(p)$, with the critical exponent dictated by the degree and the geometry of P. Even for $P(p) = p^2$, it would be interesting to replace the loss $\Phi_{4\sqrt{3}/9+o(1)}$ by a logarithmic loss closer to the lower-bound

scale $(\mathrm{Log}(1/t))^{2/3}$. At the L^1 endpoint, a natural next problem is to prove persistent universal L^1-badness for nonmonomial polynomial sequences $P(n)$. The corresponding question for $P(p)$ is the prime analogue suggested by the present paper. On the additive side, a power-saving estimate $|E_2(X)| \ll X^{1-\delta}$, or an asymptotic formula for the number of representations, would require stronger uniform information than is presently used here.

1.1. **Ideas of the proofs.** The harmonic-analytic argument is organized around a general high–low principle proved in Sections 4 and 5. In the prime-square application, the model estimates have the form

$$\| \mathrm{Low}_{Q,N} \|_{\ell^1 \to \ell^\infty} \lesssim N^{-2} Q \log Q,$$
$$\| \mathrm{High}_{Q,N} \|_{\ell^2 \to \ell^2} \lesssim Q^{-1/2} \exp\left((2+o(1)) \frac{\sqrt{\log Q}}{\log\log Q} \right).$$

The same high estimate holds maximally in N. Balancing these two bounds on characteristic functions produces the power $2/3$ and the Orlicz constant in Theorem 1.1. This abstract form of the argument also isolates the local low estimate needed by the stopping-time proof of Theorem 1.3.

The decomposition is proved in Section 6. A circle-method approximation expresses the multiplier of the annular prime-square average as a sum of terms localized near reduced fractions a/q. The ordinary main terms contain the normalized incomplete Gauss sums $\mathfrak{s}(a,q)$. If a real exceptional character exists, Page's theorem produces an additional term involving $\mathfrak{s}_{\chi_q}(a,q)$ and its exceptional zero[1] β. We then apply the high–low decomposition at denominator Q, using pointwise kernel bounds for the low part and multifrequency and square-function estimates for the high part.

For the main terms, our arithmetic input is

$$|\mathfrak{s}(a,q)| \le q^{-1/2} \exp\left((2+o(1)) \frac{\sqrt{\log q}}{\log\log q} \right).$$

For incomplete quadratic Gauss sums over reduced residues, this subexponential factor is part of the natural maximal estimate, up to the value of the constant. Since the high estimate is uniform in the denominator, the same loss appears in the endpoint Orlicz factor. Removing it would require input beyond pointwise bounds for the individual Gauss sums.

The exceptional-character term is too large to be absorbed into the remainder and must itself be divided between the low and high parts. Write $q = q_0 d$, where q_0 is the exceptional conductor. Two complementary arithmetic estimates are used: a bound for the complete sum over $a \in \mathbb{A}_q$, suited to the pointwise kernel estimate, and a bound for each coefficient $\mathfrak{s}_{\chi_q}(a,q)$, which decays like $d^{-1/2+o(1)}$. We then assign the exceptional terms to the low or high part according to the size of d.

The multifrequency maximal estimate requires the smoothing cutoff to be fixed on each denominator scale, rather than depending on q or d. We therefore replace the varying cutoff by a common one. The term with the common cutoff is controlled by the multifrequency maximal estimate, and the difference between the two cutoffs by a square-function estimate. Together these estimates give

$$\|E_{Q,N} f\|_2 \lesssim_A Q^{-1/2} (\log Q)^{O_A(1)} (1+\log N)^{-A} \|f\|_2,$$

[1]For the recent exclusion of Landau–Siegel zeros through AI-assisted work, see [17, 50] and the discussion in Section 2.

which is strong enough for the maximal stopping-time argument. The multifrequency and high–low framework follows the work on square and prime averages in [20, 21, 30]; the exceptional-character decomposition and the scale-dependent cutoffs are specific to the prime-square setting.

For the near-L^1 estimate in Theorem 1.5(ii), we first prove a restricted weak-type bound with an Orlicz loss. We seek a decomposition into a model with a maximal $\ell^p(\mathbb{Z})$ bound for p close to one, and an error with small maximal $\ell^2(\mathbb{Z})$ norm. The circle-method approximation supplies the initial model, while the Gauss-sum bounds and maximal $\ell^2(\mathbb{Z})$ estimates from Section 6 control the error. To estimate the model, we represent its arithmetic coefficients by measures on finite cyclic groups. Tao's quantitative Ionescu–Wainger multiplier and sampling theorems [49] then allow us to transfer continuous estimates to $\mathbb{Z}$ without estimating each rational frequency separately. The formulation for canonical major arcs [27] is unnecessary here, and its stated bounds leave the dependence on p unspecified, making Tao's quantitative formulation better suited to our Orlicz estimates.

The additional work is to control the supremum over averaging scales while tracking the constants from Ionescu–Wainger theory and the arithmetic estimates. We use dyadic increments and direct estimates for a finite range of scales. For larger scales, we narrow the frequency cutoff and apply the sampling theorem to a continuous maximal operator. Fourier decay controls the errors introduced by changing the cutoffs. We also divide the exceptional-character contributions according to their conductors: the possible small-conductor character is included in the model, while large-conductor terms are controlled in the ℓ^2 error. All estimates remain uniform in the possible Siegel zero. Balancing the model and error estimates on indicators and optimizing the parameters gives the restricted Orlicz bound. Standard harmonic analysis arguments and Calderón's transference principle then give the modular inequality, the stated dependence on $p-1$, and the corresponding ergodic estimates.

The proof of Theorem 1.5(i) uses LaVictoire's criterion for persistent universal L^1-badness [31]. For an odd modulus q, let

$$\Lambda_q = \{b^2 \pmod q : (b,q)=1\}.$$

For each fixed q, the prime number theorem in arithmetic progressions shows that every element of Λ_q occurs with the same limiting frequency among the prime squares. Consequently, their distribution modulo q converges to the normalized counting measure on Λ_q as the averaging scale tends to infinity. By a theorem of Granville and Kurlberg, along suitable square-free moduli the gaps between consecutive elements of Λ_q, after division by their mean, have the exponential distribution associated with Poisson spacings [16]. This gives the density, thickening, and difference estimates required in LaVictoire's criterion [31, Section 3]. Conze's transference principle and Sawyer's Banach principle then convert the resulting finite cyclic maximal estimates into Theorem 1.5(i).

For Theorem 1.6, consider representations $n = r + p^2$, where r and p are prime and r is required to belong to $\mathcal{P}$. By the elementary criterion for primitive representations as a sum of two squares, this amounts to excluding prime divisors of $r-1$ that are congruent to $3 \pmod 4$.

Fix a sifting threshold z, and let A count the representations for which $r-1$ has no such prime divisor below z. Every desired representation is counted by A, but A may also count values of r for which all the offending prime factors of $r-1$ lie above z. Since $r \equiv 11 \pmod{24}$, the number of prime factors congruent to $3 \pmod 4$, counted with multiplicity, is even. Thus any such unwanted value contains at least a pair of these larger prime factors. A second count B, which may overcount, is constructed to include every representation arising from such a pair. Consequently, the desired number of representations is at least $A-B$. We apply a lower-bound

sieve to A and an upper-bound sieve to B. It therefore remains to show that the main term supplied by the lower sieve is larger than the correction supplied by the upper sieve.

The analytic input is a Bombieri–Vinogradov approximation, in mean square over the represented integer n, for the equation

$$m^2 + \ell t = n,$$

where m, t are prime and t lies in an arithmetic progression. On the minor arcs, a prime-square exponential-sum estimate and Bessel's inequality reduce the required mean-square bound to a combinatorial count. The diagonal and off-diagonal contributions are then estimated separately.

On the major arcs, Siegel–Walfisz and Bombieri–Vinogradov estimates approximate both the quadratic prime sum, whose phase is $m^2\theta$, and the prime sum in an arithmetic progression, whose phase is $\ell t\theta$. Here $R = (\log N)^C$ is the major-arc cutoff, so the initial approximation uses reduced fractions with denominator $q \le R$. We use the same parameter in a common smooth cutoff with nonnegative inverse Fourier transform, producing a positive archimedean factor $H_{\ell,R}$. The corresponding arithmetic factor is the truncated local product G_R. All of these quantities are defined precisely in Sections 10 and 11.

The main new point is to pass from denominators $q \le R$ to all R-smooth moduli. In the corresponding argument of Matomäki [35], stronger decay of the arithmetic coefficients permits absolute summation. For prime squares, the Gauss sums have only $q^{-1/2+o(1)}$ decay, which is not sufficient for this approach. Instead, we retain the frequency localization. At each dyadic scale $q \sim 2^s \le N/(100R^{1/2})$, the localized multipliers centered at distinct reduced fractions have disjoint supports. Plancherel therefore expresses the squared global $\ell^2(\mathbb{Z})$ norm as the sum of the squared norms of the individual frequency terms, using frequency separation in the same spirit as the large sieve. Consequently, the Gauss-sum coefficients are summed through their squares rather than their absolute values, which provides sufficient decay for the range $R < q \le \frac{N}{100R^{1/2}}$.

For the remaining R-smooth moduli $q > \frac{N}{100R^{1/2}}$, a Rankin trick argument and multiplicativity show that their total contribution is negligible. Using the same parameter R in $H_{\ell,R}$ and G_R preserves the positive main term and the local factors needed in the sieve.

The sieve arguments of Wu and Matomäki proceed with a half-dimensional lower sieve for A and a linear upper sieve for B. The same direct approach does not work in our setting. For the small primes $p \le R$, the sieve-density products associated with A and B do not satisfy the product estimates required for sieve dimensions $1/2$ and 1, respectively. Consequently, a single sieve over all primes $p < z$ is justified only with the uniform dimensions $2/3$ for A and $3/2$ for B. These estimates make the lower bound for A too small relative to the upper bound for B, and the resulting numerical lower bound for $A - B$ is not positive.

We therefore separate the small primes $p \le R$ from the intermediate primes $R < p < z$. The first range is removed by preliminary Rosser–Iwaniec weights of level $D_0 = \exp\big((\log\log N)^3\big)$. Since $R = (\log N)^C$, one has

$$\frac{\log D_0}{\log R} \longrightarrow \infty, \qquad \log D_0 = o(\log D(\ell))$$

uniformly throughout the retained ℓ-range, where $D(\ell)$ denotes the available sieve level. Indeed, throughout this range $D(\ell) \ge N^{\vartheta}$ for some fixed $\vartheta > 0$. The fundamental lemma therefore gives the exact local product over $p \le R$ with relative error $o(1)$.

We therefore first sieve the primes $p \le R$, taking $R = (\log N)^C$. The fundamental lemma of the Rosser–Iwaniec sieve shows that the preliminary upper and lower sums both equal the product over $p \le R$, up to a relative error $O_J((\log N)^{-J})$ for every fixed $J > 0$. At the same

time, division by D_0 reduces the remaining sieve level by only an $N^{o(1)}$ factor and therefore changes the outer sieve parameter by $o(1)$. Since the outer weights have both signs, the upper or lower preliminary weights are selected according to the sign of the corresponding outer weight.

The outer sieve is then applied only to the intermediate primes $R < p < z$. In this range the lower sieve for A has dimension $1/2$, while the upper sieve for B has dimension 1, matching the dimensions used in Matomäki's argument [34].

To compare the resulting main terms, we sum the upper-sieve contributions over $\ell = 2up_2$. Here $p_2 \equiv 3 \pmod 4$ is one of the two large prime factors isolated above, while every prime factor of u is congruent to $1 \pmod 4$. The local products over $p \le R$, which have already been included in the preliminary sieve, cancel explicitly against the corresponding factors in the lower-sieve main term. After this cancellation, the dependence on u is given by a multiplicative weight. Its mean value is estimated using the theorem of Granville–Koukoulopoulos, while Shiu's theorem gives the required uniformity in the ranges arising from the sum over p_2 [15,45]. Summation over p_2 then produces the Wu–Matomäki integral factor $\mathcal{W}(\alpha)$. The error from replacing $H_{\ell,R}$ by a uniform Archimedean multiplier H_ℓ is summable over ℓ and is absorbed into the $o(1)$ term.

It follows that the difference between the lower bound for A and the upper bound for B has main coefficient

$$f_{1/2}(\alpha) - c_\alpha F_1(2)\mathcal{W}(\alpha), \qquad c_\alpha = \frac{16e^{-\gamma/2}}{\sqrt{\pi\alpha}}.$$

Here f_κ and F_κ are the lower and upper Rosser–Iwaniec sieve functions, respectively, α is the sieve parameter, and γ is Euler's constant. For $\alpha = 9/8$,

$$f_{1/2}(9/8) - c_{9/8}F_1(2)\mathcal{W}(9/8) > 0.17,$$

so the lower-sieve contribution A exceeds the correction term B. In summary, the pre-sieving of $p \le R$ and the resulting weighted summation over ℓ are the principal differences from the earlier sieving arguments for $\mathcal{P}$ in [34,35,51,59].

1.2. **Organization of the paper.** Section 2 fixes notation and records the standard analytic and arithmetic inputs. Section 3 proves the quadratic Gauss-sum and correlation estimates used throughout. Sections 4 and 5 establish the abstract improving and sparse principles, and Section 6 proves the prime-square high–low decomposition. Section 7 proves the Orlicz maximal inequalities and quantifies the growth of the maximal L^p bounds as $p \downarrow 1$. Section 8 establishes persistent universal L^1-badness for prime squares. The Goldbach argument begins with the reductions in Section 9; the circle-method approximation is proved in Section 10, and the lower and upper sieves are completed in Section 11.

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2. Notation and preliminaries

This section collects the notation and standard analytic and arithmetic facts used throughout the paper.

2.1. **General conventions.** We write $e(t) = e^{2\pi it}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, equipped with normalized Lebesgue measure. For $X \geq 1$, denote $[X] = \{1, \dots, \lfloor X \rfloor\}$. Intervals in $\mathbb{Z}$ are finite sets of consecutive integers. For an odd integer $c \geq 1$, cI denotes the interval with the same centre as I and c times its length. The indicator of a set E, or of a statement E, is denoted by $\mathbf{1}_E$. If E is finite and nonempty, put

$$\langle f \rangle_{E,p} = \left(\frac{1}{|E|} \sum_{x \in E} |f(x)|^p \right)^{1/p}, \qquad \langle f \rangle_{E,\infty} = \sup_{x \in E} |f(x)|.$$

All unnormalized ℓ^p-norms and inner products are taken with respect to counting measure:

$$\|f\|_p = \left(\sum_{x \in \mathbb{Z}} |f(x)|^p \right)^{1/p}, \qquad \langle f, g \rangle = \sum_{x \in \mathbb{Z}} f(x)\overline{g(x)}.$$

As usual, p' is determined by $1/p + 1/p' = 1$.

The relations $X \ll Y$, $Y \gg X$, and $X = O(Y)$ mean that $|X| \leq CY$. Dependence on parameters is indicated by subscripts, as in $\ll_A$. We write $X \sim Y$ if $X \ll Y \ll X$, and $X \asymp Y$ if $X/Y \to 1$. The symbol $o(1)$ always refers to the main asymptotic parameter tending to infinity. The value of a positive absolute constant may change from line to line.

For $x > 0$, define

$$\operatorname{Log} x = 2 + |\log x|, \qquad \operatorname{Log}\operatorname{Log} x = 2 + \log(2 + |\log x|). \tag{2.1}$$

Thus $\operatorname{Log} x \asymp \log x$ for large x, while $\operatorname{Log} x \asymp \log(x^{-1})$ for $0 < x \leq 1/2$. For $\alpha > 0$ and $0 < \beta < 1$, put

$$\Phi_{\alpha,\beta}(x) = \exp\left(\alpha \frac{(\operatorname{Log} x)^\beta}{\operatorname{Log}\operatorname{Log} x} \right), \qquad \Phi_\alpha = \Phi_{\alpha,1/2}. \tag{2.2}$$

2.2. **Arithmetic notation.** When used as arithmetic variables, the letters $p, p_1, p_2, \dots$ denote primes; p may also denote an exponent in L^p or ℓ^p. We write $\mathbb{P}$ for the primes and $\mathbb{P}^2 = \{p^2 : p \in \mathbb{P}\}$. The set

$$\mathcal{P} = \{x^2 + y^2 + 1 \in \mathbb{P} : (x, y) = 1\} \tag{2.3}$$

is fixed throughout. We use (m, n) and $[m, n]$ for the greatest common divisor and least common multiple. The notation $p^a \parallel n$ means that $p^a \mid n$ but $p^{a+1} \nmid n$. Further, $P^+(n)$ is the largest prime factor of n, with $P^+(1) = 1$, $\omega(n)$ is the number of distinct prime factors, and $\tau(n) = \sum_{m|n} 1$. The Möbius, Euler totient, and von Mangoldt functions are denoted by μ, ϕ, and Λ, respectively. Thus

$$\phi(q) = q \prod_{p \mid q} \left(1 - \frac{1}{p}\right), \qquad \Lambda(n) = \begin{cases} \log p, & n = p^j,\ j \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

We use without further comment the standard estimates

$$\phi(q) \gg \frac{q}{\log\log(3q)}, \qquad \omega(q) \ll \frac{\log(2q)}{\log\log(3q)}, \qquad \tau(q) \ll_\varepsilon q^\varepsilon.$$

Given a prime p and a nonzero integer n, let

$$v_p(n) = \max\{\nu \geq 0 : p^\nu \mid n\}$$

denote the p-adic valuation of n.

For an odd prime p, $(\frac{n}{p})$ is the Legendre symbol; the Jacobi symbol is used for odd composite moduli. In particular the symbol is zero when the numerator is not coprime to the modulus.

Put $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ and

$$\mathbb{A}_q = \mathbb{Z}_q^\times = \{a \pmod q : (a,q) = 1\}.$$

Unless another range is displayed, $\sum_{a\in\mathbb{A}_q}$ means a sum over $0 \le a < q$ with $(a,q) = 1$. The Ramanujan sum is

$$c_q(n) = \sum_{a\in\mathbb{A}_q} e(an/q). \tag{2.4}$$

We shall repeatedly use

$$c_q(n) = \sum_{d|(q,n)} d\mu(q/d) = \mu\left(\frac{q}{(q,n)}\right)\frac{\phi(q)}{\phi(q/(q,n))}.$$

In particular, $c_q(n) = \mu(q)$ when $(q,n) = 1$, and for prime powers

$$c_{p^a}(n) = \begin{cases} 0, & p^{a-1} \nmid n, \\ -p^{a-1}, & p^{a-1} \parallel n, \\ \phi(p^a), & p^a \mid n. \end{cases}$$

Cohen's identity, in the sign convention (2.4), is

$$\sum_{r\in\mathbb{A}_q} c_q(n-r) = \mu(q)c_q(n);$$

see [9, Corollary 7.2]. The quadratic analogue required in the paper is proved in Lemma 3.1.

If χ is a Dirichlet character modulo q, extended by zero away from the units, its conductor is denoted by q^* and its inducing primitive character by χ^*. We use the normalized linear Gauss sum

$$G(\chi,a) = \frac{1}{\phi(q)}\sum_{r \bmod q} \chi(r)e(ar/q).$$

If χ is primitive and $(a,q) = 1$, then

$$G(\chi,a) = \overline{\chi}(a)G(\chi,1), \qquad |G(\chi,1)| = \frac{\sqrt{q}}{\phi(q)}.$$

If $\chi \bmod q$ is induced by a primitive character χ^* of conductor q_χ, we use the explicit formulas in [56, (3.3)–(3.5)], which express $G(\chi,a)$ in terms of χ^*, q_χ, and q/q_χ.

2.3. **Fourier analysis and arithmetic averages.** For $f \in \ell^1(\mathbb{Z})$, we use the Fourier transform

$$\widehat{f}(\xi) = \sum_{x\in\mathbb{Z}} f(x)e(-x\xi), \qquad f(x) = \int_{\mathbb{T}} \widehat{f}(\xi)e(x\xi)\,d\xi.$$

For $F \in L^1(\mathbb{R})$, we write

$$\widehat{F}(\xi) = \int_{\mathbb{R}} F(x)e(-x\xi)\,dx, \qquad F^\vee(x) = \int_{\mathbb{R}} F(\xi)e(x\xi)\,d\xi.$$

Thus $F^\vee$ denotes the inverse Euclidean Fourier transform. For $f : \mathbb{Z}_q \to \mathbb{C}$, we use the normalization

$$\widehat{f}(a) = \frac{1}{q}\sum_{x\in\mathbb{Z}_q} f(x)e(-ax/q), \qquad f(x) = \sum_{a\in\mathbb{Z}_q} \widehat{f}(a)e(ax/q).$$

These normalizations give

$$\frac{1}{q}\sum_{x\in\mathbb{Z}_q} c_q(x)e(-ax/q) = \mathbf{1}_{(a,q)=1}.$$

For functions on $\mathbb{Z}$, convolution is $(f*g)(x)=\sum_y f(y)g(x-y)$. We use Plancherel's identity and the multiplier estimate

$$\|f\|_2 = \|\widehat{f}\|_{L^2(\mathbb{T})}, \qquad \|T_m f\|_2 \le \|m\|_\infty \|f\|_2, \qquad \widehat{T_m f} = m\widehat{f}.$$

The two principal averaging operators are

$$A_{N,\Lambda}f(x) = \frac{1}{N}\sum_{n\le N}\Lambda(n)f(x-n^2),$$
$$A_{N,\mathbb{N}^2}f(x) = \frac{1}{N}\sum_{n\le N} f(x-n^2).$$

Their kernels are supported in an interval of length N^2. Thus in abstract improving or sparse statements the spatial scale corresponding to these operators is N^2, not N. For a family of operators T_N, write

$$T^*f(x) = \sup_N |T_N f(x)|.$$

2.4. **Primes in arithmetic progressions.** We recall the Landau–Page theorem. There is an absolute $c_0>0$ such that, for every $Q\ge 3$, among primitive real characters of conductor at most Q there is at most one character χ_0 for which $L(s,\chi_0)$ has a zero

$$\beta > 1 - \frac{c_0}{\log Q}. \tag{2.5}$$

If it exists, this zero is real and simple. We call χ_0 and β the exceptional character and exceptional zero at level Q. This is the Landau–Page theorem; see [25, Corollary 11.10].

Recent work of OpenAI rules out Landau–Siegel zeros; see Green's reformulation, obtained using ChatGPT, [17], and Tao's exposition [50]. For completeness, we retain the classical exceptional-character and exceptional-zero conventions throughout, although the exceptional case does not occur once c_0 is chosen sufficiently small.

We use the following consequence of Page's theorem and the prime number theorem in arithmetic progressions. Put

$$Q_x = \exp(c\sqrt{\log x}),$$

where $c>0$ is sufficiently small. If an exceptional character exists at level Q_x, denote it by χ_0, its conductor by q_0, and its exceptional zero by β. Then, uniformly for $q\le Q_x$ and $(a,q)=1$,

$$\sum_{\substack{n\le x\\ n\equiv a \pmod q}} \Lambda(n) = \frac{x}{\phi(q)} - \mathbf{1}_{q_0\mid q}\frac{\chi_0(a)x^\beta}{\beta\phi(q)} + O\left(xe^{-C\sqrt{\log x}}\right), \tag{2.6}$$

for some absolute $C>0$. If no exceptional character exists at level Q_x, the second term is omitted. This follows from the standard zero-free region and the Landau–Page theorem; see [25, Theorem 5.27]. Some constants arising from (2.6) are ineffective, and we do not track effectivity.

Write $\pi(x;q,a)=|\{p\le x : p\equiv a \pmod q\}|$ and $\mathrm{Li}(x)=\int_2^x dt/\log t$. We shall also use the Brun–Titchmarsh inequality. If $x>q$ and $(a,q)=1$, then

$$\pi(x;q,a) \le \frac{2x}{\phi(q)\log(x/q)}.$$

In particular, when $q \le x^{1/2}(\log x)^{-L}$, this gives

$$\pi(x;q,a) \ll \frac{x}{\phi(q)\log x}.$$

We shall use the corresponding uniform consequence

$$\max_{\substack{(a,q)=1\\2\le y\le x}} \left|\pi(y;q,a) - \frac{\mathrm{Li}(y)}{\phi(q)}\right| \ll \frac{x}{\phi(q)\log x}, \qquad q \le x^{1/2}(\log x)^{-L}.$$

Here the ranges $y \le x(\log x)^{-2}$ and $y > x(\log x)^{-2}$ are handled respectively by the trivial estimate and the Brun–Titchmarsh inequality.

The Bombieri–Vinogradov theorem states that, for every $A > 0$ and $K \ge 0$, there is $B = B(A,K) > 0$ such that

$$\sum_{q\le x^{1/2}(\log x)^{-B}} \tau(q)^K \max_{\substack{(a,q)=1\\2\le y\le x}} \left|\sum_{\substack{n\le y\\ n\equiv a \pmod q}} \Lambda(n) - \frac{y}{\phi(q)}\right| \ll_{A,K} x(\log x)^{-A};$$

see [25, Theorem 17.1].

2.5. **Circle-method conventions.** Dirichlet's approximation theorem ensures that, for every $\xi \in \mathbb{T}$ and $H \in \mathbb{N}$, there are $a \in \mathbb{A}_q$ and $1 \le q \le H$ such that

$$\left|\xi - \frac{a}{q}\right| \le \frac{1}{qH}.$$

For quadratic averages with prime parameter N, the natural major-arc width is N^{-2}. Given $1 \le R \le N$, we therefore set

$$\mathfrak{M}(R;N) = \bigcup_{q\le R}\bigcup_{a\in\mathbb{A}_q} \left\{\xi\in\mathbb{T} : \left|\xi - \frac{a}{q}\right| \le \frac{R}{qN^2}\right\},$$
$$\mathfrak{m}(R;N) = \mathbb{T}\setminus\mathfrak{M}(R;N).$$

When a different width is required, it is displayed explicitly. In the unconditional high–low decomposition, the auxiliary parameter Q is restricted by

$$\log Q \le c(\log N)^{1/2} \tag{2.7}$$

with $c > 0$ sufficiently small. Under GRH a polynomial range is available, but all results stated as unconditional use (2.7).

We shall also use the standard quadratic Weyl estimate. If $(a,q) = 1$, $q \le T$, and $|\lambda_2 - a/q| \le q^{-2}$, then, uniformly in λ_1,

$$\left|\frac{1}{T}\sum_{n\le T} e(\lambda_2 n^2 + \lambda_1 n)\right| \ll \left(\frac{1}{q} + \frac{\log(2q)}{T} + \frac{q\log(2q)}{T^2}\right)^{1/2},$$

see [25, Section 8.2] for more detail.

2.6. **Real-variable estimates.** We shall use two standard real-variable estimates. The one-sided Hardy–Littlewood maximal function is

$$M_{\mathrm{HL}}^{+} f(x) = \sup_{h>0} \frac{1}{h}\int_0^h |f(x-y)|\,dy.$$

It satisfies

$$\|M_{\mathrm{HL}}^{+} f\|_{L^{1,\infty}(\mathbb{R})} \ll \|f\|_{L^1(\mathbb{R})}, \qquad \|M_{\mathrm{HL}}^{+} f\|_{L^p(\mathbb{R})} \ll \frac{1}{p-1}\|f\|_{L^p(\mathbb{R})}, \quad 1 < p \le 2.$$

The same bounds hold for a convolution operator $T_K f = K * f$ whose $L^2(\mathbb{R})$-operator norm is $O(1)$ and whose kernel satisfies

$$|K(x)| \ll |x|^{-1}, \qquad \sup_{h\neq 0} \int_{|x|>2|h|} |K(x-h) - K(x)|\, dx \ll 1.$$

These statements follow from the Hardy–Littlewood maximal theorem, the Calderón–Zygmund decomposition, and interpolation; see [46, Chapter I, Sections 3–4].

We shall use Khintchine's inequality in the following form. If (ϵ_j) are independent Rademacher variables, then, uniformly for $1 \le p \le 2$,

$$\left(\sum_j |z_j|^2\right)^{1/2} \ll \left(\mathbb{E}\left|\sum_j \epsilon_j z_j\right|^p\right)^{1/p}.$$

Consequently, for functions g_j on a measure space,

$$\left\|\left(\sum_j |g_j|^2\right)^{1/2}\right\|_{L^p} \ll \left(\mathbb{E}\left\|\sum_j \epsilon_j g_j\right\|_{L^p}^p\right)^{1/p}.$$

2.7. **Sparse forms and stopping times.** We introduce the sparse forms and stopping-time linearization used in the proof of Theorem 1.3.

Definition 2.8. A collection $\mathcal{S}$ of intervals is sparse if, for every $I \in \mathcal{S}$, there is $E_I \subset I$ such that $|E_I| \ge |I|/4$ and the sets E_I, $I \in \mathcal{S}$, are pairwise disjoint.

To linearize a dyadic maximal operator on an interval I_0, we use the following stopping-time convention.

Definition 2.9. Let $f = \mathbf{1}_F$ be supported in $3I_0$. A map $\tau : I_0 \longrightarrow \{2^j : 1 \le 2^j \le |I_0|\}$ is admissible if, for every $x \in I_0$,

$$\sup_{M \ge \tau(x)} \frac{1}{M} \sum_{|y-x|\le CM} f(y) \le C_{\mathrm{adm}} \frac{|F|}{|I_0|}. \tag{2.10}$$

where $C > 1$ is a fixed radius constant and $C_{\mathrm{adm}} > 1$ is a sufficiently large fixed constant depending on C. Changing the fixed radius changes only the required admissibility constant. One may also replace the symmetric local average by the maximum of the left and right one-sided local averages; these quantities are comparable with a factor of at most two.

Condition (2.10) bounds the local density $|F \cap [x - CM, x + CM]|/M$ by the parent density $|F|/|I_0|$ at every scale $M \ge \tau(x)$.

2.8. **Ionescu–Wainger notation.** We record the notation needed for the Ionescu–Wainger and sampling theorems. We use the one-dimensional specialization of [49, Definition 1.3]. For $P \ge 1$, let

$$\mathbb{T}[P] = \{\alpha \in \mathbb{T} : P\alpha = 0\} = P^{-1}\mathbb{Z}/\mathbb{Z}.$$

For $\alpha \in \mathbb{Q}/\mathbb{Z}$, write

$$\mathrm{den}(\alpha) = \min\{q \ge 1 : q\alpha = 0\}.$$

A major-arc parameter set is a triple $(k, \mathcal{Q}, \varepsilon)$, where $k \ge 1$, $\varepsilon > 0$, and $\mathcal{Q}$ is a finite set of pairwise coprime positive integers. For $A \subseteq \mathcal{Q}$, put

$$Q_A = \prod_{q\in A} q, \qquad Q_\varnothing = 1,$$

and define

$$\Sigma_{\le k} = \bigcup_{\substack{A\subseteq\mathcal{Q}\\ |A|\le k}} \mathbb{T}[Q_A].$$

Thus, on putting $P = \prod_{q\in\mathcal{Q}} q$, we have $\Sigma_{\le k} \subseteq \mathbb{T}[P]$. The associated major-arc set is

$$\mathfrak{M}_{\mathrm{IW}}(k,\mathcal{Q},\varepsilon) = \Sigma_{\le k} + [-\varepsilon,\varepsilon] \subseteq \mathbb{T}.$$

If $r \ge 1$, $0 < c < 1$, and $q_{\max} \ge \max_{q\in\mathcal{Q}} q$, the parameter set is called (r,c)-good if

$$\varepsilon < \frac{c}{2rq_{\max}^{2rk}}.$$

We next introduce the adelic notation used in the sampling theorems. Let

$$\widehat{\mathbb{Z}} = \varprojlim_{q} \mathbb{Z}_q, \qquad \mathbb{A}_{\mathbb{Z}} = \mathbb{R}\times\widehat{\mathbb{Z}}.$$

The Haar measure on $\widehat{\mathbb{Z}}$ is normalized to have total mass one. Its dual group is $\mathbb{Q}/\mathbb{Z}$, and hence the dual of $\mathbb{A}_{\mathbb{Z}}$ is $\mathbb{R}\times\mathbb{Q}/\mathbb{Z}$.

In all applications below, the arithmetic Fourier support is contained in $\mathbb{T}[P]$. The corresponding functions on $\mathbb{A}_{\mathbb{Z}}$ depend on the second variable only through its residue class modulo P, and may therefore be identified with functions on $Y_P = \mathbb{R}\times\mathbb{Z}_P$, equipped with Lebesgue measure in the first variable and normalized counting measure in the second. For $F \in L^1(Y_P)$, define

$$\widehat{F}(\theta,\alpha) = \frac{1}{P}\sum_{y\in\mathbb{Z}_P}\int_{\mathbb{R}} F(x,y)e(-x\theta - y\alpha)\,dx, \qquad (\theta,\alpha)\in\mathbb{R}\times\mathbb{T}[P].$$

The inversion formula is

$$F(x,y) = \sum_{\alpha\in\mathbb{T}[P]}\int_{\mathbb{R}} \widehat{F}(\theta,\alpha)e(x\theta + y\alpha)\,d\theta.$$

For a bounded function $M : \mathbb{R}\times\mathbb{T}[P]\to\mathbb{C}$, let $T_M^{Y_P}$ denote the multiplier operator on Y_P defined by

$$\widehat{T_M^{Y_P}F}(\theta,\alpha) = M(\theta,\alpha)\widehat{F}(\theta,\alpha).$$

Suppose that $M(\theta,\alpha) = 0$ unless $\alpha\in\Sigma\subseteq\mathbb{T}[P]$ and $|\theta|\le\varepsilon$. The corresponding multiplier on $\mathbb{Z}$ is

$$M_\Sigma^{\mathbb{Z}}(\xi) = \sum_{\alpha\in\Sigma}\sum_{j\in\mathbb{Z}} M(\xi-\alpha+j,\alpha), \qquad \xi\in\mathbb{T},$$

and $T_{M;\Sigma}^{\mathbb{Z}}$ denotes the operator satisfying $\widehat{T_{M;\Sigma}^{\mathbb{Z}}f} = M_\Sigma^{\mathbb{Z}}\widehat{f}$. If $M(\theta,\alpha) = m(\theta)$ is independent of α, we write $T_{m;\Sigma}^{\mathbb{Z}}$. For a complex measure ν on $\mathbb{Z}_P$, put

$$\widehat{\nu}(\alpha) = \sum_{y\in\mathbb{Z}_P} e(-y\alpha)\nu(\{y\}), \qquad \|\nu\|_{\mathrm{TV}} = \sum_{y\in\mathbb{Z}_P}|\nu(\{y\})|.$$

The symbols used below have the product form $M(\theta,\alpha) = m(\theta)\widehat{\nu}(\alpha)$.

It remains to describe the sampling maps. Define

$$\pi : \mathbb{R}\times\mathbb{T}[P]\longrightarrow\mathbb{T}, \qquad \pi(\theta,\alpha) = \theta+\alpha \pmod 1.$$

Suppose that $\Sigma\subseteq\mathbb{T}[P]$ satisfies

$$\|\alpha-\alpha'\|_{\mathbb{T}} > 2\varepsilon \qquad (\alpha,\alpha'\in\Sigma,\ \alpha\ne\alpha').$$

Then the intervals $\alpha+[-\varepsilon,\varepsilon]$ with $\alpha\in\Sigma$ are pairwise disjoint in $\mathbb{T}$. Consequently, every $\xi\in\pi([-\varepsilon,\varepsilon]\times\Sigma)$ has a unique representation

$$\xi=\theta+\alpha\pmod 1,\qquad |\theta|\le\varepsilon,\quad \alpha\in\Sigma.$$

We write

$$\ell^2(\mathbb{Z})_{\pi(\Omega)}=\Big\{f\in\ell^2(\mathbb{Z}):\operatorname{supp}\widehat{f}\subseteq\pi(\Omega)\Big\},\qquad L^2(Y_P)_\Omega=\Big\{F\in L^2(Y_P):\operatorname{supp}\widehat{F}\subseteq\Omega\Big\},$$

where $\Omega=[-\varepsilon,\varepsilon]\times\Sigma$ and $\pi(\Omega)=\bigcup_{\alpha\in\Sigma}\big(\alpha+[-\varepsilon,\varepsilon]\big)$.

The inverse sampling map is

$$\mathcal{S}_\Omega^{-1}f(x,y)=\sum_{\alpha\in\Sigma}\int_{-\varepsilon}^{\varepsilon}\widehat{f}(\alpha+\theta)e(x\theta+y\alpha)\,d\theta.$$

The diagonal sampling map is

$$\mathcal{S}_PF(n)=F(n,n\bmod P),\qquad n\in\mathbb{Z}.$$

With these definitions, $\mathcal{S}_P\mathcal{S}_\Omega^{-1}f=f$ for every f whose Fourier transform is supported in $\pi(\Omega)$. The major-arc sampling theorem compares the $\ell^p(\mathbb{Z})$-norm of f with the $L^p(Y_P)$-norm of $\mathcal{S}_\Omega^{-1}f$, while the quantitative Shannon sampling theorem controls $\mathcal{S}_PF$ when the real Fourier support of F is sufficiently small relative to P^{-1}.

2.9. **Dynamical systems and transference.** A measure-preserving system (X,μ,T) consists of a probability space (X,μ) and a measurable map $T:X\to X$ satisfying $\mu(T^{-1}E)=\mu(E)$ for every measurable E. We write $T^nf=f\circ T^n$. For an increasing convex function Ψ with $\Psi(0)=0$ and $\Psi(t)\to\infty$, let $L^\Psi(X)$ consist of the measurable functions satisfying $\int_X\Psi(|f|/a)\,d\mu<\infty$ for some $a>0$. We call Ψ doubling if $\Psi(2t)\ll\Psi(t)$.

For invertible T, a convolution family $A_Nf=K_N*f$ with finitely supported kernels has ergodic counterparts $A_{N,T}f=\sum_m K_N(m)f\circ T^m$; write $A_T^*f=\sup_{N\ge2}|A_{N,T}f|$. Calderón's transference principle [8] transfers strong $\ell^p(\mathbb{Z})$ maximal estimates to $L^p(X)$, with the same constants. The same argument applies to modular estimates: if $|\{A^*g>\lambda\}|\le C\sum_m\Psi(|g(m)|/\lambda)$ for every finitely supported g and every $\lambda>0$, then

$$\mu\{A_T^*f>\lambda\}\le C\int_X\Psi(|f|/\lambda)\,d\mu.\tag{2.11}$$

Indeed, restrict to finitely many kernels supported in $[-R,R]$ and test on $k\mapsto\mathbf{1}_{[-R,L-1+R]}(k)f(T^{-k}x)$. Summing over $0\le k<L$ and integrating in x gives (2.11) with the additional factor $1+2R/L$. Let $L\to\infty$, then enlarge the finite family.

2.10. **Sieve notation.** For a finite set $\mathcal{F}\subset\mathbb{Z}$, a set of primes $\mathfrak{P}$, and $z\ge2$, put

$$P_{\mathfrak{P}}(z)=\prod_{\substack{p<z\\p\in\mathfrak{P}}}p,\tag{2.12}$$
$$\mathcal{S}(\mathcal{F},\mathfrak{P},z)=\{a\in\mathcal{F}:(a,P_{\mathfrak{P}}(z))=1\},$$
$$S(\mathcal{F},\mathfrak{P},z)=|\mathcal{S}(\mathcal{F},\mathfrak{P},z)|.$$

If $\mathcal{F}_d=\{a\in\mathcal{F}:d\mid a\}$ admits an approximation

$$|\mathcal{F}_d|=X\frac{\omega_{\mathcal{F}}(d)}{d}+r_d$$

for every square-free d all of whose prime divisors belong to $\mathfrak{P}$, then

$$V(z)=\prod_{\substack{p<z\\p\in\mathfrak{P}}}\left(1-\frac{\omega_{\mathcal{F}}(p)}{p}\right)$$

is the corresponding sieve product. The dimension is the exponent κ for which, uniformly in the required range,

$$\prod_{\substack{w\le p<z\\ p\in\mathfrak{P}}}\left(1-\frac{\omega_{\mathcal{F}}(p)}{p}\right)^{-1}\ll\left(\frac{\log z}{\log w}\right)^{\kappa}.$$

We denote by f_κ and F_κ the lower and upper Rosser–Iwaniec sieve functions of dimension κ, respectively, with the normalization of [24, (1.8)–(1.10)]. The specific functions used below are $f_{1/2}$ and F_1; their required values and the relevant fundamental-lemma estimates are recorded in Section 11.

3. Elementary number-theoretic estimates

This section evaluates the local Ramanujan and quadratic exponential sums that occur in the major-arc and sieve arguments. These calculations determine both the low-frequency kernel in the high–low decomposition and the local densities in the sieve.

We first prove a quadratic analogue of Cohen's identity by reducing the Ramanujan-sum correlation to prime powers and evaluating the odd and 2-adic factors separately. We then combine this identity with congruence restrictions to obtain the Euler products used later. Finally, we prove pointwise bounds for the untwisted and character-twisted quadratic Gauss sums. All residue-class sums in this section are over complete systems of residues.

3.1. Cohen's identity for square shifts.

The usual Cohen identity is

$$\sum_{a\in\mathbb{A}_q}c_q(n-a)=\mu(q)c_q(n).$$

We need the following quadratic version.

Lemma 3.1 (Cohen's identity with square shifts)**.** *Let $2^a\parallel q$. Then*

$$\sum_{r\in\mathbb{A}_q}c_q(n-r^2)=|\mu(q/2^a)|T_a(n)\prod_{\substack{p\mid q\\ p>2}}\begin{cases}p+1, & \left(\frac{n}{p}\right)=1,\\ 1-p, & \left(\frac{n}{p}\right)\neq 1,\end{cases}\tag{3.2}$$

where

$$T_a(n)=\begin{cases}1, & a=0,\\ (-1)^{n+1}, & a=1,\\ 0, & a=2,3 \text{ and } n\not\equiv 1 \pmod{2^{a-1}},\\ \phi(2^a)c_{2^a}(n-1)=\pm\phi(2^a)^2, & a=2,3 \text{ and } n\equiv 1 \pmod{2^{a-1}},\\ 0, & a\ge 4.\end{cases}\tag{3.3}$$

Consequently,

$$\left|\sum_{r\in\mathbb{A}_q}c_q(n-r^2)\right|\ll|\mu(q/2^a)|\prod_{p\mid q}(p+1).$$

Here $(\frac{n}{p})$ is the Legendre symbol, with value zero when $p\mid n$.

Proof. Put

$$C_q(n)=\sum_{r\in\mathbb{A}_q}c_q(n-r^2).$$

If $(q_0, q_1) = 1$, the Chinese remainder theorem implies that $\mathbb{A}_{q_0q_1} \simeq \mathbb{A}_{q_0} \times \mathbb{A}_{q_1}$. Since $c_{q_0q_1}(m) = c_{q_0}(m)c_{q_1}(m)$, and each factor depends only on the corresponding component of r, we obtain

$$C_{q_0q_1}(n) = C_{q_0}(n)C_{q_1}(n). \tag{3.4}$$

It therefore suffices to evaluate $C_{p^k}(n)$. For every prime p and $k \geq 1$,

$$c_{p^k}(m) = \begin{cases} 0, & p^{k-1} \nmid m, \\ -p^{k-1}, & p^{k-1} \mid m, p^k \nmid m, \\ \phi(p^k), & p^k \mid m. \end{cases} \tag{3.5}$$

Let first p be odd. For $k = 1$, the congruence $r^2 \equiv n \pmod p$ has two solutions in $\mathbb{A}_p$ if $(\frac{n}{p}) = 1$ and no solution otherwise. Hence

$$C_p(n) = \begin{cases} 2(p-1) - (p-3) = p+1, & (\frac{n}{p}) = 1, \\ -(p-1) = 1-p, & (\frac{n}{p}) \neq 1. \end{cases} \tag{3.6}$$

Suppose that $k \geq 2$. If $(\frac{n}{p}) \neq 1$, then $p^{k-1} \nmid n - r^2$ for every $r \in \mathbb{A}_{p^k}$, so (3.5) gives $C_{p^k}(n) = 0$. If $(\frac{n}{p}) = 1$, the two roots modulo p lift, by Hensel's lemma, to exactly two roots modulo every p^j. There are consequently two classes modulo p^k for which $p^k \mid n - r^2$, and $2(p-1)$ further classes modulo p^k for which $p^{k-1} \parallel n - r^2$. Thus

$$C_{p^k}(n) = 2\phi(p^k) - 2(p-1)p^{k-1} = 0.$$

Together with (3.6), this gives the odd part of (3.2).

For $a \leq 3$, every odd square is 1 modulo 2^a, whence

$$C_{2^a}(n) = \phi(2^a)c_{2^a}(n-1).$$

Equation (3.5) gives the first four cases in (3.3). Finally let $a \geq 4$. The map

$$r \longmapsto r + 2^{a-2}\bar{r} \pmod{2^a},$$

where $r\bar{r} \equiv 1 \pmod{2^a}$, is a bijection of $\mathbb{A}_{2^a}$ and changes r^2 by 2^{a-1} modulo 2^a. Indeed, on each class $r_0 \pmod{2^{a-2}}$ it sends the four lifts $r_0 + 2^{a-2}j$ to the same four lifts in a permuted order, since $\bar{r}$ is odd. By (3.5),

$$c_{2^a}(m) + c_{2^a}(m - 2^{a-1}) = 0$$

for every m. Pairing the terms proves $C_{2^a}(n) = 0$. The lemma now follows from (3.4). □

3.2. **Two exponential sums and their correlations.** We shall frequently use the following two exponential sums, whose basic properties we establish in this subsection.

Definition 3.7 (Exponential sums along squares)**.** For a Dirichlet character χ modulo q, put

$$\mathfrak{s}_\chi(a, q) = \frac{1}{\phi(q)} \sum_{r \in \mathbb{A}_q} \chi(r) e\left(-\frac{ar^2}{q}\right).$$

For the principal character we write

$$\mathfrak{s}(a, q) = \frac{1}{\phi(q)} \sum_{r \in \mathbb{A}_q} e\left(-\frac{ar^2}{q}\right). \tag{3.8}$$

Definition 3.9. Let $g = (q, y)$ and $(b, g) = 1$. Define

$$\Upsilon(a, q, b, y) = \frac{\phi(y)}{\phi([y, q])} \sum_{\substack{r \in \mathbb{A}_q \\ r \equiv b \pmod g}} e\left(-\frac{ar}{q}\right). \tag{3.10}$$

We shall use the following progression form of Cohen's identity.

Lemma 3.11. *Let $g \mid q$ and $(b,g)=1$. Then*

$$\sum_{\substack{r\in\mathbb{A}_q\\ r\equiv b \pmod g}} c_q(n-r) = \begin{cases} 0, & (g,q/g)>1,\\ \mu(q/g)c_{q/g}(n)c_g(n-b), & (g,q/g)=1. \end{cases}$$

Proof. This is [14, Lemma 11]. □

Next, we evaluate the correlation between Υ and $\mathfrak{s}$ over the reduced residue classes $\mathbb{A}_q$.

Lemma 3.12. *Let $g=(q,y)$ and $(b,g)=1$. Then*

$$\frac{\phi(q)\phi([y,q])}{\phi(y)} \sum_{a\in\mathbb{A}_q} \Upsilon(a,q,b,y)\mathfrak{s}(a,q)e\left(\frac{ax}{q}\right)$$
$$= \mu(q/g)\mathbf{1}_{(g,q/g)=1} \sum_{u\in\mathbb{A}_q} c_{q/g}(x-u^2)c_g(x-b-u^2). \tag{3.13}$$

In particular, the right-hand side vanishes unless q/g is square-free; if $(g,q/g)=1$, it factors over the prime powers dividing q.

Proof. Opening (3.10) and (3.8), and then summing over $a\in\mathbb{A}_q$, gives

$$\frac{\phi(q)\phi([y,q])}{\phi(y)} \sum_{a\in\mathbb{A}_q} \Upsilon(a,q,b,y)\mathfrak{s}(a,q)e(ax/q)$$
$$= \sum_{u\in\mathbb{A}_q} \sum_{\substack{r\in\mathbb{A}_q\\ r\equiv b \pmod g}} c_q(x-u^2-r).$$

Apply Lemma 3.11 with $n=x-u^2$ to obtain (3.13). If $q=q_0q_1$, $(q_0,q_1)=1$, and $g=g_0g_1$ with $g_i \mid q_i$, the Chinese remainder theorem and the multiplicativity of Ramanujan sums split the sum over u into the product of the corresponding sums modulo q_0 and q_1. This proves the final assertion. □

The preceding identity gives the Euler product needed later.

Lemma 3.14. *Let $Q>2$, let $k\geq 1$ be an integer, and let $x,b\in\mathbb{Z}$ satisfy $(b,k)=1$. Then*

$$\sum_{P^+(q)<Q} \sum_{a\in\mathbb{A}_q} \Upsilon(a,q,b,k)\mathfrak{s}(a,q)e(ax/q)$$
$$= \mathcal{T}_2 \prod_{\substack{2<p<Q\\ p\mid k\\ (\frac{x-b}{p})=1}} \left(1+\frac{p+1}{p-1}\right) \prod_{\substack{2<p<Q\\ p\mid k\\ (\frac{x-b}{p})\neq 1}} 0 \prod_{\substack{2<p<Q\\ p\nmid k\\ (\frac{x}{p})=1}} \left(1-\frac{p+1}{(p-1)^2}\right) \prod_{\substack{2<p<Q\\ p\nmid k\\ (\frac{x}{p})\neq 1}} \left(1+\frac{1}{p-1}\right), \tag{3.15}$$

where, writing $h=v_2(k)$,

$$\mathcal{T}_2 = \begin{cases} 1+(-1)^x, & h=0,\\ 2^{\min\{h,3\}}\mathbf{1}_{\{x-b\equiv 1 \pmod{2^{\min\{h,3\}}}\}}, & h\geq 1. \end{cases}$$

Thus the product is zero if some odd prime $p<Q$ dividing k satisfies $\left(\frac{x-b}{p}\right)\neq 1$.

Proof. Apply Lemma 3.12 with $y=k$. At every odd prime, only the exponents 0 and 1 contribute. At 2, the exponent 1 may always contribute, while the exponents 2 and 3 may contribute only when $4\mid k$ and $8\mid k$, respectively; all higher exponents give zero.

For fixed k, x, b, the summand is multiplicative as a function of q. Indeed, if $q = q_0q_1$ with $(q_0, q_1) = 1$ and $g_i = (q_i, k)$, then $g = (q, k) = g_0g_1$. By Lemma 3.12, the Ramanujan-sum correlation factors into its components modulo q_0 and q_1. Moreover,

$$\frac{\phi(k)}{\phi(q)\phi([k,q])} = \prod_{p^\nu \| q} \frac{\phi(p^{v_p(k)})}{\phi(p^\nu)\phi(p^{\max\{\nu, v_p(k)\}})},$$

where a factor with $v_p(k) = 0$ is interpreted in the evident way. Hence the complete summand factors over the prime powers dividing q, and the sum over $P^+(q) < Q$ is an Euler product.

For an odd prime $p \nmid k$, the local denominator is $\phi(p)^2$, while Lemma 3.1 and the factor $\mu(p) = -1$ give the local coefficient

$$\begin{cases} -(p+1)/(p-1)^2, & (\frac{x}{p}) = 1, \\ 1/(p-1), & (\frac{x}{p}) \neq 1. \end{cases}$$

For $p \mid k$, the local denominator is $\phi(p)$ and the local coefficient is

$$\begin{cases} (p+1)/(p-1), & (\frac{x-b}{p}) = 1, \\ -1, & (\frac{x-b}{p}) \neq 1. \end{cases}$$

Adding 1 to each coefficient gives the four odd-prime Euler products in (3.15). At $p = 2$, all terms with $\nu \geq 4$ vanish. If $2 \nmid k$, only the term with $\nu = 1$ may remain, so the contribution at 2 is $1 + (-1)^x$. If $2 \mid k$, the contribution at 2 is

$$1 + \sum_{\nu=1}^{\min\{v_2(k),3\}} c_{2^\nu}(x - b - 1),$$

which equals $\mathcal{T}_2$ by (3.3). Multiplying the contributions over all primes proves the lemma. □

Corollary 3.16. *Let $Q > 2$, let $k \geq 1$ be an integer, and let $\ell, n, b \in \mathbb{Z}$ satisfy $(b, k) = 1$. Then*

$$\sum_{P^+(q)<Q} \sum_{a \in \mathbb{A}_q} \Upsilon(a\ell, q, b, k)\mathfrak{s}(a, q)e(an/q)$$
$$= \mathcal{T}_2(\ell, k, n; b) \prod_{2<p<Q} \frac{p\phi((p,k))}{(p-1)^2} \left| \left\{ (u, v) \in \mathbb{A}_p^2 : \begin{array}{ll} u^2 + \ell v \equiv n & (\mathrm{mod}\ p), \\ v \equiv b & (\mathrm{mod}\ (p,k)) \end{array} \right\} \right|, \quad (3.17)$$

where

$$\mathcal{T}_2(\ell, k, n; b) := 1 + \sum_{\nu=1}^{3} \frac{\phi(k)}{\phi(2^\nu)\phi([k, 2^\nu])} \sum_{\substack{u,v \in \mathbb{A}_{2^\nu} \\ v \equiv b \ (\mathrm{mod}\ (2^\nu, k))}} c_{2^\nu}(n - u^2 - \ell v). \quad (3.18)$$

Proof. Opening the definitions of Υ and $\mathfrak{s}$, and summing first over a, gives

$$\sum_{a \in \mathbb{A}_q} \Upsilon(a\ell, q, b, k)\mathfrak{s}(a, q)e(an/q) = \frac{\phi(k)}{\phi(q)\phi([k,q])} \sum_{\substack{u,v \in \mathbb{A}_q \\ v \equiv b \ (\mathrm{mod}\ (q,k))}} c_q(n - u^2 - \ell v). \quad (3.19)$$

This expression is multiplicative in q. Let $p > 2$, and denote by N_p the cardinality appearing on the right-hand side of (3.17). The quadratic sum modulo p^ν vanishes for $\nu \geq 2$, so only the modulus p contributes. Since

$$c_p(m) = p\mathbf{1}_{p|m} - 1, \qquad \frac{\phi(k)}{\phi(p)\phi([k,p])} = \frac{\phi((p,k))}{(p-1)^2},$$

and the number of pairs satisfying only $v \equiv b \pmod{(p,k)}$ is $(p-1)^2/\phi((p,k))$, we obtain

$$1+\frac{\phi(k)}{\phi(p)\phi([k,p])}\sum_{\substack{u,v\in\mathbb{A}_p\\ v\equiv b \pmod{(p,k)}}} c_p(n-u^2-\ell v) = 1+\frac{\phi((p,k))}{(p-1)^2}\left(pN_p-\frac{(p-1)^2}{\phi((p,k))}\right)$$
$$= \frac{p\phi((p,k))}{(p-1)^2}N_p.$$

At $p=2$, the terms with modulus 2^ν, $\nu\ge 4$, vanish, while the sum of the remaining terms is $\mathcal{T}_2(\ell,k,n;b)$. Multiplying the contributions over all primes proves (3.17). □

Lemma 3.20. *For $Q\ge 3$ and every integer n,*

$$\sum_{P^+(q)<Q}\frac{1}{q\phi(q)}\left|\sum_{r\in\mathbb{A}_q} c_q(n-r^2)\right| \ll \log Q.$$

Proof. Write $q=2^a q_1$ with q_1 odd. By Lemma 3.1, the summand vanishes unless $a\le 3$ and q_1 is square-free. The contribution from 2^a is an absolute constant, and hence

$$\sum_{P^+(q)<Q}\frac{1}{q\phi(q)}\left|\sum_{r\in\mathbb{A}_q} c_q(n-r^2)\right| \ll \prod_{2<p<Q}\left(1+\frac{p+1}{p(p-1)}\right)$$
$$\ll \exp\left(\sum_{p<Q}\frac{1}{p}+O(1)\right)\ll \log Q,$$

by Mertens' theorem. □

3.3. **Pointwise Gauss-sum bounds.** We now record the pointwise bounds for the quadratic Gauss sums appearing in the major-arc multipliers.

Lemma 3.21. *If $(a,q)=1$, then*

$$|\mathfrak{s}(a,q)|\le q^{-1/2}\exp\left((2+o(1))\frac{(\log q)^{1/2}}{\log\log q}\right).$$

In particular, the right-hand side may be replaced by $q^{-1/2}\exp(3(\log q)^{1/2}/\log\log q)$ for all sufficiently large q.

Proof. The normalized sum is multiplicative in the modulus. Let p be odd. Writing

$$G(a;p^j)=\sum_{r \bmod p^j} e(-ar^2/p^j),$$

the classical quadratic Gauss-sum evaluation gives $|G(a;p^j)|=p^{j/2}$ and $G(a;p^j)=pG(a;p^{j-2})$ for $j\ge 2$; see, for example, [25, Chapter 3]. Therefore

$$\sum_{r\in\mathbb{A}_{p^j}} e(-ar^2/p^j)=G(a;p^j)-pG(a;p^{j-2})=0, \qquad j\ge 2. \tag{3.22}$$

For $j=1$ its absolute value is at most $\sqrt{p}+1$. The same calculation at 2 shows that the unit sum vanishes for 2^j with $j\ge 4$; the three remaining powers contribute only an absolute constant. Consequently $\mathfrak{s}(a,q)=0$ unless $q=2^j q_0$, $0\le j\le 3$, and q_0 is odd and square-free, and in the latter case

$$|\mathfrak{s}(a,q)|\ll q^{-1/2}\prod_{p\mid q_0}(1-p^{-1/2})^{-1}. \tag{3.23}$$

Taking logarithms, the additional factor is

$$\exp\left(\sum_{p\mid q_0} p^{-1/2} + O\left(\sum_{p\mid q_0} p^{-1}\right)\right).$$

The sum is maximal, subject to $\prod_{p\mid q_0} p \le q$, by taking the smallest primes. The prime number theorem and partial summation give

$$\sum_{p\mid q_0} p^{-1/2} \le (2+o(1))\frac{(\log q)^{1/2}}{\log\log q}, \qquad \sum_{p\mid q_0} p^{-1} = O(\log\log\log q).$$

Substitution in (3.23) proves the assertion. □

Lemma 3.24. *Let χ be a real Dirichlet character modulo q, and let $(a,q)=1$. Then, for every $\varepsilon>0$,*

$$|\mathfrak{s}_\chi(a,q)| \ll_\varepsilon q^{-1/2+\varepsilon}. \tag{3.25}$$

Equivalently, $|\mathfrak{s}_\chi(a,q)| \ll q^{-1/2+o(1)}$.

Proof. By the Chinese remainder theorem the numerator of $\mathfrak{s}_\chi(a,q)$ factors over the prime powers $p^\nu \parallel q$. Thus it is enough to prove, uniformly for $(b,p)=1$,

$$\left|\sum_{r\in\mathbb{A}_{p^\nu}} \chi_p(r)e(-br^2/p^\nu)\right| \ll Cp^{\nu/2},$$

where χ_p is the local component of χ and C is absolute.

Let p be odd. If χ_p is principal, then

$$\sum_{r\in\mathbb{A}_{p^\nu}} e(-br^2/p^\nu) = \sum_{r \bmod p^\nu} e(-br^2/p^\nu) - \sum_{\substack{r \bmod p^\nu \\ p\mid r}} e(-br^2/p^\nu).$$

The first sum is a complete quadratic Gauss sum modulo p^ν, and the second is p times a complete quadratic Gauss sum modulo $p^{\nu-2}$ when $\nu\ge 2$. Hence both are $O(p^{\nu/2})$. If χ_p is non-principal, then $\chi_p(r)=(r/p)$. For $\nu=1$, Weil's bound gives

$$\left|\sum_{r \bmod p}\left(\frac{r}{p}\right)e(-br^2/p)\right| \ll p^{1/2}.$$

For $\nu\ge 2$, write $r = u + p^{\nu-1}t$, with $u \bmod p^{\nu-1}$, $p\nmid u$, and $t \bmod p$. Since χ_p is unchanged under this replacement,

$$(u+p^{\nu-1}t)^2 \equiv u^2 + 2up^{\nu-1}t \pmod{p^\nu},$$

and the inner sum over t is

$$\sum_{t \bmod p} e(-2but/p) = 0.$$

Thus the local sum vanishes for $\nu\ge 2$.

For $p=2$, every real character modulo 2^ν is induced from a character of conductor dividing 8. For $\nu\le 5$ the desired bound is trivial. For $\nu\ge 6$, write $r = u + 2^{\nu-3}t$, with $u \bmod 2^{\nu-3}$, u odd, and $t \bmod 8$. The character is unchanged, and

$$(u+2^{\nu-3}t)^2 \equiv u^2 + u2^{\nu-2}t \pmod{2^\nu}.$$

Therefore the inner sum over t is

$$\sum_{t \bmod 8} e(-but/4) = 0,$$

since bu is odd. Hence the local sum is zero for $\nu \geq 6$, and the bound follows also at $p = 2$.

Multiplying the local estimates gives

$$\left| \sum_{r \in \mathbb{A}_q} \chi(r) e(-ar^2/q) \right| \ll C^{\omega(q)} q^{1/2}.$$

Dividing by $\phi(q)$, we obtain

$$|\mathfrak{s}_\chi(a,q)| \ll C^{\omega(q)} q^{1/2} \phi(q)^{-1} \ll C^{\omega(q)} q^{-1/2} \frac{q}{\phi(q)} \ll_\varepsilon q^{-1/2+\varepsilon}, \tag{3.26}$$

since $C^{\omega(q)} q/\phi(q) \ll_\varepsilon q^\varepsilon$. This proves (3.25). □

We shall use the following refinement of Lemma 3.24 in Section 6.

Lemma 3.27. *Let $x \in \mathbb{Z}$. Let χ_0 be a primitive real character of conductor q_0, and for each multiple q of q_0 let χ_q denote the character modulo q induced by χ_0. For $0 \leq j \leq 5$, let $\chi_{2^j q_0}$ have the same meaning, and set*

$$\sigma_0 := \max_{0 \leq j \leq 5} \sup_{a \in \mathbb{A}_{2^j q_0}} |\mathfrak{s}_{\chi_{2^j q_0}}(a, 2^j q_0)|, \qquad \rho(d) := \prod_{p \mid d} (1 - p^{-1/2})^{-1}.$$

There exists $\mathcal{B}_0 \geq 1$, depending only on q_0, such that, for every $q = q_0 d$,

$$\begin{aligned} \left| \sum_{a \in \mathbb{A}_q} \mathfrak{s}_{\chi_q}(a,q) e(ax/q) \right| &\ll \mathcal{B}_0 \prod_{p \mid d} \frac{p+1}{p-1}, \\ |\mathfrak{s}_{\chi_q}(a,q)| &\ll \sigma_0 d^{-1/2} \rho(d), \qquad \text{for } a \in \mathbb{A}_q \\ \sigma_0 \mathcal{B}_0^{1/2} &\ll q_0^{-1/25}. \end{aligned} \tag{3.28}$$

In particular,

$$(1 + \log q_0) \sigma_0 \mathcal{B}_0^{1/2} \ll 1.$$

Moreover, if $a \mapsto \mathfrak{s}_{\chi_q}(a,q)$ is not identically zero, then

$$d = 2^\nu d_1, \qquad 0 \leq \nu \leq 5, \qquad \mu^2(d_1) = 1, \qquad (d_1, 2q_0) = 1. \tag{3.29}$$

Proof. The first estimate in (3.28) follows by factoring the complete sum over prime powers. The second estimate uses the vanishing of induced quadratic sums at higher prime powers. The final bound on $\sigma_0 \mathcal{B}_0^{1/2}$ comes from the restriction $p \equiv 1 \pmod 4$ on the odd primes in the conductor.

The Chinese remainder theorem factors both expressions in (3.28) over prime powers. If $p \nmid q_0$, the local factors in the first and second estimates are bounded, respectively, by

$$\frac{p+1}{p-1} \qquad \text{and} \qquad p^{-1/2}(1 - p^{-1/2})^{-1}.$$

If $p \mid q_0$ is odd and $\chi_p(-1) = -1$, then the terms indexed by r and $-r$ cancel. Hence

$$\sum_{r \in \mathbb{A}_p} \chi_p(r) e(-br^2/p) = 0.$$

Thus a nonzero local quadratic sum requires $\chi_p(-1) = 1$, which for the quadratic character modulo p is equivalent to $p \equiv 1 \pmod 4$. Weil's bound gives

$$\left|\sum_{r\in\mathbb{A}_p} \chi_p(r)e(-br^2/p)\right| \le 2p^{1/2},$$

so the normalized local factor in the second estimate of (3.28) is at most $2p^{1/2}/(p-1)$. For the first estimate, Fourier inversion gives the local factor

$$\frac{1}{p-1}\sum_{r\in\mathbb{A}_p} \chi_p(r)c_p(x-r^2).$$

Since $\sum_{r\in\mathbb{A}_p} \chi_p(r) = 0$ and

$$c_p(n) = \begin{cases} p-1, & p \mid n, \\ -1, & p \nmid n, \end{cases}$$

this equals

$$\frac{p}{p-1} \sum_{\substack{r\in\mathbb{A}_p \\ r^2\equiv x \pmod p}} \chi_p(r).$$

The congruence $r^2 \equiv x \pmod p$ has at most two solutions, so this factor has absolute value at most $2p/(p-1)$.

At $p = 2$, every primitive real character has conductor dividing 8. Moreover, the local quadratic sum vanishes once the exponent of 2 in the induced modulus is sufficiently large, as shown in the proof of Lemma 3.24. Only finitely many 2-adic factors remain, and all of them are bounded by an absolute constant. We may therefore take

$$\mathcal{B}_0 \asymp \prod_{\substack{p\mid q_0 \\ p>2}} \frac{2p}{p-1},$$

which gives the first estimate in (3.28).

Let p be odd. If $p \nmid q_0$, then $v_p(q) = v_p(d)$, and the local character modulo $p^{v_p(q)}$ is principal. By (3.22), the corresponding local quadratic sum vanishes when $v_p(d) \ge 2$. Hence

$$p \nmid q_0 \quad\Longrightarrow\quad v_p(d) \in \{0, 1\}.$$

Now suppose that $p \mid q_0$. Since q_0 is the conductor of a primitive real character, $v_p(q_0) = 1$. If $p \mid d$, then $\nu := v_p(q) \ge 2$, and the local character modulo p^ν is induced from the quadratic character modulo p. Writing $r = u + p^{\nu-1}t$, we obtain

$$\sum_{\substack{r \pmod{p^\nu} \\ p\nmid r}} \chi_p(r)e(-br^2/p^\nu) = \sum_{\substack{u \pmod{p^{\nu-1}} \\ p\nmid u}} \chi_p(u)e(-bu^2/p^\nu) \sum_{t \pmod p} e(-2but/p) = 0.$$

Thus

$$p \mid q_0 \quad\Longrightarrow\quad v_p(d) = 0.$$

At $p = 2$, the calculation in the proof of Lemma 3.24 shows that the local sum vanishes when $v_2(q) \ge 6$. Consequently, whenever the induced quadratic sum is not identically zero, one may write

$$d = 2^\nu d_1, \qquad 0 \le \nu \le 5, \qquad \mu^2(d_1) = 1, \qquad (d_1, 2q_0) = 1,$$

which is (3.29). Since $q = (2^\nu q_0)d_1$ and $(2^\nu q_0, d_1) = 1$, the Chinese remainder theorem gives, for suitable $b_0 \in \mathbb{A}_{2^\nu q_0}$ and $b_1 \in \mathbb{A}_{d_1}$,

$$\mathfrak{s}_{\chi_q}(a,q) = \mathfrak{s}_{\chi_{2^\nu q_0}}(b_0, 2^\nu q_0)\mathfrak{s}(b_1, d_1).$$

For every $p \mid d_1$, the local character is principal and $v_p(d_1) = 1$. Hence the corresponding normalized local sum satisfies

$$\frac{1}{\phi(p)}\left|\sum_{r\in\mathbb{A}_p} e(-b_p r^2/p)\right| \le \frac{\sqrt{p}+1}{p-1} = p^{-1/2}(1-p^{-1/2})^{-1}.$$

Multiplying these estimates over $p \mid d_1$ and using the definition of σ_0, we obtain

$$|\mathfrak{s}_{\chi_q}(a,q)| \le \sigma_0 d_1^{-1/2} \prod_{p\mid d_1}(1-p^{-1/2})^{-1}.$$

Since $d = 2^\nu d_1$ with $0 \le \nu \le 5$, the bounded 2-adic factor may be absorbed into the implied constant. Therefore

$$|\mathfrak{s}_{\chi_q}(a,q)| \ll \sigma_0 d^{-1/2}\rho(d),$$

which proves the second estimate in (3.28).

It remains to prove the last estimate in (3.28). The maximum over $0 \le j \le 5$ in the definition of σ_0 affects only the 2-adic local factor and therefore changes the estimate by an absolute constant. If $\sigma_0 = 0$, there is nothing to prove. Otherwise, factorization over prime powers and pairing r with $-r$ show that, if $\chi_p(-1) = -1$ for some odd prime $p \mid q_0$, then the corresponding local quadratic sum vanishes. Thus a nonzero term requires $\chi_p(-1) = 1$ for every odd prime $p \mid q_0$. Since χ_p is the quadratic character modulo p, this is equivalent to $p \equiv 1 \pmod 4$. At such a prime, the local contribution to σ_0 is at most $2p^{1/2}/(p-1)$, while the corresponding factor in $\mathcal{B}_0^{1/2}$ is $(2p/(p-1))^{1/2}$. Multiplying these bounds over $p \mid q_0$ gives

$$\sigma_0\mathcal{B}_0^{1/2} \ll \prod_{\substack{p\mid q_0\\ p>2}} \frac{2\sqrt{2}\,p}{(p-1)^{3/2}}.$$

The factor corresponding to $p = 5$ is bounded absolutely, and for every $p \ge 13$,

$$\frac{2\sqrt{2}\,p}{(p-1)^{3/2}} \le p^{-1/25}.$$

The 2-adic factor is also bounded absolutely. Since every remaining odd prime divisor of q_0 is at least 13, we conclude that

$$\sigma_0\mathcal{B}_0^{1/2} \ll q_0^{-1/25},$$

as required. □

4. Proof of the improving inequalities

We prove the fixed-scale improving estimates for the prime-square averages. The argument is separated into an abstract high–low principle and the number-theoretic decomposition proved later in Section 6. The low part is estimated from ℓ^1 to ℓ^∞; the high part is estimated on ℓ^2. Balancing these two bounds gives the endpoint restricted estimate, and interpolation with the ℓ^2 bound gives the strong ℓ^p-improving estimates for $p > 3/2$.

We use the logarithmic and Orlicz conventions in (2.1) and (2.2); in particular, $\Phi_c = \Phi_{c,1/2}$.

Theorem 4.1. *Let A_N and B_N be positive convolution operators whose kernels are supported in an interval of length $O(N)$. Suppose that*

$$|A_N f(x)| \le (\log N)^a B_N|f|(x), \tag{4.2}$$

$$|B_N f(x)| \ll N^{-v}\|f\mathbf{1}_{[x-CN,x+CN]}\|_1, \qquad 0 \le v < 1, \tag{4.3}$$

and that $s \ge 1-v$, $r > 0$, and, whenever $2 \le J \le N^{(1-v)/s}/10$,

$$B_N = Lo_{J,N} + Hi_{J,N}, \qquad \|Lo_{J,N} f\|_\infty \ll N^{-1}J^s(\log J)^\gamma\|f\|_1, \tag{4.4}$$

$$\|Hi_{J,N} f\|_2 \ll J^{-r}(\log J)^\beta\|f\|_2, \tag{4.5}$$

where $\gamma \ge \beta \ge 0$. Suppose also that, for $2 \le Q$ and $\log Q \le c_0(\log N)^c$,

$$A_N = \mathrm{Low}_{Q,N} + \mathrm{High}_{Q,N} + E_{Q,N}, \tag{4.6}$$

$$\|\mathrm{Low}_{Q,N} f\|_\infty \ll N^{-1}Q^s(\log Q)^\alpha\|f\|_1,$$

$$\|\mathrm{High}_{Q,N} f\|_2 \ll Q^{-r}\exp\left(c_1\frac{(\log Q)^\kappa}{\log\log Q}\right)\|f\|_2, \tag{4.7}$$

$$\|E_{Q,N} f\|_2 \ll_A Q^{-r}(\log Q)^{C_A}(1+\log N)^{-A}\|f\|_2 \qquad (A > 0), \tag{4.8}$$

where $0 < \kappa < 1$ and $a, \alpha, c_0, c_1, c > 0$. In (4.8), the constant C_A may depend on A. Then, for F, G contained in an interval of length N,

$$\frac{|\langle B_N\mathbf{1}_F, \mathbf{1}_G\rangle|}{N} \ll (|F||G|/N^2)^{1-\frac{s}{2(s+r)}}(\mathrm{Log}(|F||G|/N^2))^{\gamma+\frac{s(\beta-\gamma)}{s+r}}, \tag{4.9}$$

$$\frac{|\langle A_N\mathbf{1}_F, \mathbf{1}_G\rangle|}{N} \ll (|F||G|/N^2)^{1-\frac{s}{2(s+r)}}\Phi_{c^*,\kappa}(|F||G|/N^2), \tag{4.10}$$

where

$$c^* = \frac{c_1 s}{2^\kappa(s+r)^{\kappa+1}} + o(1). \tag{4.11}$$

Here and below the $o(1)$ in a subscript means that the estimate holds with c^* replaced by any fixed larger constant.

Proof. Let $\delta := |F||G|/N^2$. After enlarging the implied constant, we may assume that $\delta \le \delta_0$, where $\delta_0 > 0$ depends only on the fixed parameters; otherwise we take $J = Q = 2$. From (4.4), (4.5), the triangle inequality, and Cauchy–Schwarz,

$$\begin{aligned}\frac{|\langle B_N\mathbf{1}_F, \mathbf{1}_G\rangle|}{N} &\ll N^{-1}\|Lo_{J,N}\mathbf{1}_F\|_\infty|G| + N^{-1}\|Hi_{J,N}\mathbf{1}_F\|_2|G|^{1/2} \\ &\ll \delta J^s(\log J)^\gamma + \delta^{1/2}J^{-r}(\log J)^\beta.\end{aligned} \tag{4.12}$$

Choose a dyadic J within a factor 2 of

$$\delta^{-1/(2(s+r))}(\mathrm{Log}\,\delta)^{(\beta-\gamma)/(s+r)}.$$

If this is at most $N^{(1-v)/s}/10$, substitution in (4.12) gives (4.9). Otherwise use (4.3). Since $s \ge 1-v$, the estimate

$$\delta^{-1/(2(s+r))}(\mathrm{Log}\,\delta)^{(\beta-\gamma)/(s+r)} > N^{(1-v)/s}/10$$

implies that

$$N^{-1-v}|F||G| \ll \delta^{1-\frac{s}{2(s+r)}}(\mathrm{Log}\,\delta)^{\frac{s(\beta-\gamma)}{s+r}}.$$

Because $\gamma \ge 0$ and $\mathrm{Log}\,\delta \ge 1$, this is bounded by the right-hand side of (4.9).

We turn to A_N. If $\mathrm{Log}\,\delta \ge c_3(\log N)^c$, where $c_3 > 0$ is sufficiently small, then (4.2) and (4.9) give

$$N^{-1}|\langle A_N\mathbf{1}_F, \mathbf{1}_G\rangle| \ll (\log N)^a N^{-1}|\langle B_N\mathbf{1}_F, \mathbf{1}_G\rangle| \ll \delta^{1-\frac{s}{2(s+r)}}(\mathrm{Log}\,\delta)^{\gamma+a/c+\frac{s(\beta-\gamma)}{s+r}}$$

$$\ll \delta^{1-\frac{s}{2(s+r)}}\Phi_{c^*,\kappa}(\delta).$$

The last inequality follows because $\Phi_{c^*,\kappa}(t)$ dominates every fixed power of $\operatorname{Log} t$ as $t \downarrow 0$.

Suppose now that $\operatorname{Log}\delta < c_3(\log N)^c$. Choose Q dyadic. From (4.6)–(4.8),

$$\frac{|\langle A_N \mathbf{1}_F, \mathbf{1}_G\rangle|}{N} \ll N^{-1}\|\operatorname{Low}_{Q,N}\mathbf{1}_F\|_\infty|G| + N^{-1}\|\operatorname{High}_{Q,N}\mathbf{1}_F\|_2|G|^{1/2} + N^{-1}\|E_{Q,N}\mathbf{1}_F\|_2|G|^{1/2} \tag{4.13}$$

$$\ll \delta Q^s(\log Q)^\alpha + \delta^{1/2}Q^{-r}\exp\left(c_1\frac{(\log Q)^\kappa}{\log\log Q}\right) + \delta^{1/2}Q^{-r}(\log Q)^{C_A}(1+\log N)^{-A}.$$

Take

$$\log Q = \frac{\operatorname{Log}\delta}{2(s+r)} + \frac{c_1}{s+r}\frac{((\operatorname{Log}\delta)/(2(s+r)))^\kappa}{\log\operatorname{Log}\delta} + O(\log\operatorname{Log}\delta). \tag{4.14}$$

The constant in the $O(\log\operatorname{Log}\delta)$ is chosen so that the two first terms in (4.13) are comparable. Since $\operatorname{Log}\delta < c_3(\log N)^c$, c_3 may be chosen so that $\log Q \le c_0(\log N)^c$; thus the decomposition is available. Moreover,

$$Q^s = \delta^{-\frac{s}{2(s+r)}}\exp\left(\left(\frac{c_1 s}{2^\kappa(s+r)^{\kappa+1}} + o(1)\right)\frac{(\operatorname{Log}\delta)^\kappa}{\log\operatorname{Log}\delta}\right).$$

The powers of $\log Q$ in the first two terms of (4.13) are absorbed by the $o(1)$ in the exponential. For fixed A, the factor $(\log Q)^{C_A}$ in the error term is dominated, for sufficiently large Q, by the subexponential factor in the high term. The bounded range of Q is absorbed into the implied constant. Since $(1+\log N)^{-A} \le 1$, the error term is also absorbed by the high term. This proves (4.10) and (4.11). □

The above proof does not use that F and G lie in a common interval, except to ensure that $|F||G| \le N^2$. Hence (4.9) and (4.10) remain valid for arbitrary finite sets $F, G \subset \mathbb{Z}$ satisfying $|F||G| \le N^2$.

Corollary 4.15. *Assume in addition that A_N and B_N are uniformly bounded on ℓ^2. If*

$$\frac{2(r+s)}{2r+s} < p \le 2,$$

then, for $f \in \ell^p(\mathbb{Z})$,

$$\|B_N f\|_{p'} + \|A_N f\|_{p'} \ll_p N^{1/p'-1/p}\|f\|_p. \tag{4.16}$$

Proof. Set $p_0 = \frac{2(r+s)}{2r+s}$ and assume that f is supported on an interval of length $O(N)$ first. We remove the logarithmic losses in (4.9) and (4.10). Indeed, for every $\varepsilon > 0$ and every $0 < \delta \le 1$, $\big(\log(2/\delta)\big)^{O(1)} \ll_\varepsilon \delta^{-\varepsilon}$. Since every fixed logarithmic or subexponential factor occurring in (4.9) and (4.10) is bounded by $C_\varepsilon\delta^{-\varepsilon}$, these estimates imply that, for every $p_0 < \widetilde{p} < 2$,

$$|\langle T_N\mathbf{1}_F, \mathbf{1}_G\rangle| \ll_{\widetilde{p}} N^{1/\widetilde{p}'-1/\widetilde{p}}|F|^{1/\widetilde{p}}|G|^{1/\widetilde{p}}, \qquad T_N \in \{A_N, B_N\}. \tag{4.17}$$

Thus A_N and B_N are uniformly of restricted weak type $(\widetilde{p}, \widetilde{p}')$, with the required scaling in N.

Now fix $p_0 < p < 2$ and choose $\widetilde{p}$ with $p_0 < \widetilde{p} < p$. Interpolating (4.17) with the uniform $\ell^2 \to \ell^2$ estimate gives (4.16) when f is supported in an interval of length $O(N)$, see [28, Chapter 1]. The case $p = 2$ is precisely the assumed uniform ℓ^2 estimate.

Finally, partition $\mathbb{Z}$ into intervals I of length comparable to N. Since the kernels are supported in $[-CN, CN]$, $T_N(f\mathbf{1}_I)$ is supported in $I + [-CN, CN]$. The intervals may therefore be divided

into finitely many families such that these enlarged intervals are pairwise disjoint within each family. For each such family $\mathcal{I}$, the local estimate and $p' \geq p$ give

$$\Big\| \sum_{I\in\mathcal{I}} T_N(f\mathbf{1}_I) \Big\|_{p'} \ll_p N^{1/p'-1/p} \left(\sum_{I\in\mathcal{I}} \|f\mathbf{1}_I\|_p^{p'} \right)^{1/p'} \leq N^{1/p'-1/p} \left(\sum_{I\in\mathcal{I}} \|f\mathbf{1}_I\|_p^{p} \right)^{1/p}.$$

Summing over the finitely many families proves (4.16) for every $f \in \ell^p(\mathbb{Z})$. □

We shall prove in Section 6 that the annular prime-square averages admit the decomposition required below. For dyadic N, put

$$\widetilde{A}_{N,\Lambda} = A_{N,\Lambda} - \frac{1}{2}A_{N/2,\Lambda}, \qquad \widetilde{A}_{N,\mathbb{N}^2} = A_{N,\mathbb{N}^2} - \frac{1}{2}A_{N/2,\mathbb{N}^2}.$$

Thus

$$A_{N,\Lambda} = \sum_{\substack{j\geq 0 \\ 2^{-j}N\geq 2}} 2^{-j}\widetilde{A}_{2^{-j}N,\Lambda}, \tag{4.18}$$

For the square averages, telescoping gives

$$A_{N,\mathbb{N}^2}f = \sum_{\substack{j\geq 0 \\ 2^{-j}N\geq 2}} 2^{-j}\widetilde{A}_{2^{-j}N,\mathbb{N}^2}f + \frac{1}{N}f(\cdot - 1).$$

Proposition 4.19. *Let N be dyadic, let $Q \geq 2$, and suppose that*

$$\log Q \leq c_0(\log N)^{1/2},$$

where $c_0 > 0$ is a sufficiently small absolute constant. Then

$$\widetilde{A}_{N,\Lambda} = \mathrm{Low}_{Q,N} + \mathrm{High}_{Q,N} + E_{Q,N}, \tag{4.20}$$

where

$$\begin{aligned} \|\mathrm{Low}_{Q,N} f\|_\infty &\ll N^{-2}Q\log(2Q)\,\|f\|_1, \\ \|\mathrm{High}_{Q,N} f\|_2 &\ll Q^{-1/2}\exp\left((2+o(1))\frac{(\log Q)^{1/2}}{\log\log Q}\right)\|f\|_2, \\ \|E_{Q,N}f\|_2 &\ll_A Q^{-1/2}(\log(2Q))^{O_A(1)}(1+\log N)^{-A}\|f\|_2 \end{aligned} \tag{4.21}$$

for every $A > 0$. More precisely, for some absolute $C > 0$, the low term satisfies

$$|\mathrm{Low}_{Q,N} f(x)| \ll Q\log(2Q)\sup_{M\geq N} M^{-2}\sum_{|y-x|\leq CM^2}|f(y)|. \tag{4.22}$$

The maximal estimate needed in Section 5 is

$$\Big\| \sup_{\substack{M\geq N_0 \\ M\in 2^{\mathbb{N}}}} |\mathrm{High}_{Q,M} f| \Big\|_2 \ll Q^{-1/2}\exp\left((2+o(1))\frac{(\log Q)^{1/2}}{\log\log Q}\right)\|f\|_2, \tag{4.23}$$

provided

$$\log Q \leq c_0(\log N_0)^{1/2}.$$

Here $o(1)$ is taken as $Q\to\infty$, uniformly in the displayed ranges of N and N_0; bounded values of Q are absorbed into the implicit constants. The operators in (4.20) are defined in Section 6.

Proof of Theorem 1.1. By positivity, it is enough to consider dyadic N. Indeed, if $N \leq N' < 2N$ and N' is dyadic, then

$$|A_{N,\Lambda}f| \leq 2A_{N',\Lambda}|f|,$$

and the spatial scales N^2 and $(N')^2$ are comparable. We first apply the high–low argument of Theorem 4.1 to the annular operators $\widetilde{A}_{N,\Lambda}$ and $\widetilde{A}_{N,\mathbb{N}^2}$, with spatial scale N^2 and

$$r=\tfrac{1}{2},\qquad s=1,\qquad v=\tfrac{1}{2},\qquad \gamma=2,\qquad \beta=1,\qquad \alpha=1,\qquad \kappa=\tfrac{1}{2},\qquad c_1=2.$$

The estimates required for $\widetilde{A}_{N,\mathbb{N}^2}$ follow from [21, Lemma 2.2 and (2.5)], applied at the scales N and $N/2$. The estimates for $\widetilde{A}_{N,\Lambda}$ are given by Proposition 4.19, while

$$|\widetilde{A}_{N,\Lambda}f|\le(\log N)\widetilde{A}_{N,\mathbb{N}^2}|f|.$$

Both annular operators are uniformly bounded on ℓ^2. For every fixed $\varepsilon>0$, the high estimate in Proposition 4.19 satisfies the hypothesis (4.7) with $c_1=2+\varepsilon$, for all sufficiently large Q. Applying Theorem 4.1 and then absorbing ε into the final $o(1)$ gives the asserted endpoint estimate for $\widetilde{A}_{N,\Lambda}$.

It remains to pass from the annular averages to $A_{N,\Lambda}$. Since $|I|\asymp N^2$, put $\delta=\frac{|F||G|}{N^4}$. The case $\delta\gtrsim 1$ follows directly from the uniform ℓ^2 bound, so we may assume that $\delta\le 1$. Put $M_j=2^{-j}N$, and let j_0 be the first index for which $2^{4j_0}\delta>1$, with $j_0=\infty$ if there is no such index.

For $j<j_0$, we have $|F||G|\le M_j^4$. Applying the annular endpoint estimate with $c_1=2+\varepsilon$, and using

$$1-\frac{s}{2(s+r)}=\frac{2}{3},\qquad c_\varepsilon:=\frac{2+\varepsilon}{\sqrt{2}(3/2)^{3/2}}=\frac{4\sqrt{3}}{9}+O(\varepsilon),$$

gives

$$\begin{aligned}\frac{2^{-j}}{N^2}\big|\langle\widetilde{A}_{M_j,\Lambda}\mathbf{1}_F,\mathbf{1}_G\rangle\big|&\ll_\varepsilon 2^{-j/3}\delta^{2/3}\Phi_{c_\varepsilon+o(1)}(2^{4j}\delta)\\&\ll_\varepsilon 2^{-j/3}\delta^{2/3}\Phi_{c_\varepsilon+o(1)}(\delta).\end{aligned}$$

Here we used $2^{4j}\delta\le 1$ for $j<j_0$, so $\Phi_{c_\varepsilon+o(1)}(2^{4j}\delta)\le\Phi_{c_\varepsilon+o(1)}(\delta)$, after increasing the implicit constant if necessary.

For $j\ge j_0$, the uniform ℓ^2 estimate gives

$$\frac{2^{-j}}{N^2}\big|\langle\widetilde{A}_{M_j,\Lambda}\mathbf{1}_F,\mathbf{1}_G\rangle\big|\ll 2^{-j}\delta^{1/2}.$$

Therefore

$$\begin{aligned}\frac{1}{N^2}\big|\langle A_{N,\Lambda}\mathbf{1}_F,\mathbf{1}_G\rangle\big|&\ll_\varepsilon \delta^{2/3}\Phi_{c_\varepsilon+o(1)}(\delta)\sum_{j<j_0}2^{-j/3}+\delta^{1/2}\sum_{j\ge j_0}2^{-j}\\&\ll_\varepsilon \delta^{2/3}\Phi_{c_\varepsilon+o(1)}(\delta)+\delta^{3/4}.\end{aligned}$$

Since $\delta^{3/4}\le\delta^{2/3}$ and $\Phi_{c_\varepsilon+o(1)}(\delta)\ge 1$, this proves the endpoint estimate with Orlicz constant $c_\varepsilon+o(1)$. Letting $\varepsilon\downarrow 0$ gives the stated constant $4\sqrt{3}/9+o(1)$ for $A_{N,\Lambda}$.

This completes the proof of the endpoint case. The estimates for $p>3/2$ follow in the same way from Corollary 4.15. □

5. Proof of the sparse bounds

We pass from the fixed-scale improving estimates to sparse bounds for the prime-square maximal operator. The new point is that, after linearizing the maximal function, the scale depends on the spatial point. Thus the fixed-scale ℓ^2 estimate for the high part must be replaced by a maximal ℓ^2 estimate. We then produce a sparse family through iteration.

Notice that there is no maximal-function loss for the low term, since

$$\left\| \sup_{N>M} |Lo_{J,N} f| \right\|_\infty \le \sup_{N>M} \|Lo_{J,N} f\|_\infty, \qquad M > 0.$$

We next estimate the low and high parts at admissible stopping times.

Lemma 5.1. *Fix constants $C \ge 1$ and $c > 2C+1$. Let (B_M) be a family of positive convolution operators satisfying*

$$|B_M f(x)| \ll M^{-v} \|f \mathbf{1}_{[x-CM, x+CM]}\|_1, \qquad 0 \le v < 1. \tag{5.2}$$

Let $s \ge 1-v$, $r > 0$, and $\gamma \ge \beta \ge 0$, and suppose that, whenever M is dyadic and $2 \le J \le M^{(1-v)/s}/10$,

$$B_M = Lo_{J,M} + Hi_{J,M} + Er_{J,M}. \tag{5.3}$$

Let I_0 be an interval, and let $f = \mathbf{1}_F$ and $g = \mathbf{1}_G$, where $F \subset cI_0$ and $G \subset I_0$. Suppose that

$$\tau : I_0 \longrightarrow \{2^k : 2^k \le |I_0|\}$$

satisfies

$$\sup_{M \ge \tau(x)} M^{-1} \sum_{|y-x| \le C_0 M} f(y) \ll \langle f \rangle_{cI_0,1}, \qquad x \in I_0. \tag{5.4}$$

Write $B_\tau f(x) = B_{\tau(x)} f(x)$.

Assume that, for every $J \ge 2$ and every $x \in I_0$ such that $\tau(x) \ge (10J)^{s/(1-v)}$,

$$|Lo_{J,\tau(x)} f(x)| \ll J^s (\log J)^\gamma \sup_{M \ge \tau(x)} M^{-1} \sum_{|y-x| \le CM} f(y), \tag{5.5}$$

and that, for every $h \in \ell^2(\mathbb{Z})$,

$$\left\| \sup_{\substack{M \ge (10J)^{s/(1-v)} \\ M \text{ dyadic}}} |Hi_{J,M} h| \right\|_2 \ll J^{-r} (\log J)^\beta \|h\|_2. \tag{5.6}$$

Finally, suppose that, for some fixed $A > 1/2$, uniformly in dyadic M,

$$\|\widehat{Er}_{J,M}\|_{L^\infty(\mathbb{T})} \ll_A J^{-r} (\log J)^\beta (1 + \log M)^{-A}. \tag{5.7}$$

Then

$$|I_0|^{-1} |\langle B_\tau f, g \rangle| \ll_c \left(\frac{|F||G|}{|I_0|^2} \right)^{1 - \frac{s}{2(s+r)}} \left(\operatorname{Log} \frac{|F||G|}{|I_0|^2} \right)^{\gamma + \frac{s(\beta-\gamma)}{s+r}}. \tag{5.8}$$

Proof. Put $\delta_0 = \frac{|F||G|}{|I_0|^2}$. We may assume that $\delta_0 \le 1/4$, since otherwise the result follows by choosing the implicit constant in (5.8) sufficiently large. Choose a dyadic J within a fixed factor of

$$\delta_0^{-1/(2(s+r))} (\operatorname{Log} \delta_0)^{(\beta-\gamma)/(s+r)} \tag{5.9}$$

and put $T = (10J)^{s/(1-v)}$.

We first consider the set $\{\tau < T\}$. By (5.2) and admissibility,

$$B_{\tau(x)} f(x) \ll \tau(x)^{-v} \sum_{|y-x| \le C\tau(x)} f(y) \ll \tau(x)^{1-v} \langle f \rangle_{cI_0,1} \ll T^{1-v} \langle f \rangle_{cI_0,1}.$$

Since $T^{1-v} = (10J)^s$, it follows that

$$|I_0|^{-1} \left| \langle \mathbf{1}_{\{\tau < T\}} B_\tau f, g \rangle \right| \ll J^s \delta_0.$$

On $\{\tau \geq T\}$ we have $J \leq \tau(x)^{(1-v)/s}/10$, so the fixed-scale decomposition (5.3) is available at the scale $M = \tau(x)$. For each dyadic M put

$$E_M := \{x \in I_0 : \tau(x) = M\}.$$

Then, pointwise on I_0,

$$\mathbf{1}_{\{\tau \geq T\}} B_\tau f = \mathbf{1}_{\{\tau \geq T\}} Lo_{J,\tau} f + \mathbf{1}_{\{\tau \geq T\}} Hi_{J,\tau} f + \sum_{\substack{M \geq T \\ M \text{ dyadic}}} \mathbf{1}_{E_M} Er_{J,M} f.$$

The pointwise estimate (5.5) and admissibility give

$$|I_0|^{-1} \big|\langle \mathbf{1}_{\{\tau \geq T\}} Lo_{J,\tau} f, g\rangle\big| \ll \frac{J^s (\log J)^\gamma}{|I_0|} \sum_{\substack{x \in G \\ \tau(x) \geq T}} \sup_{M \geq \tau(x)} M^{-1} \sum_{|y-x| \leq CM} f(y) \ll \delta_0 J^s (\log J)^\gamma.$$

For the high-frequency part we use the multifrequency estimate (5.6). Thus Cauchy–Schwarz gives

$$\begin{aligned} |I_0|^{-1} \big|\langle \mathbf{1}_{\{\tau \geq T\}} Hi_{J,\tau} f, g\rangle\big| &\leq |I_0|^{-1} \|g\|_2 \Big\| \sup_{M \geq T} |Hi_{J,M} f| \Big\|_2 \\ &\ll |I_0|^{-1} |G|^{1/2} J^{-r} (\log J)^\beta |F|^{1/2} = \delta_0^{1/2} J^{-r} (\log J)^\beta. \end{aligned}$$

The error term is treated scale by scale. By Cauchy–Schwarz and Plancherel,

$$\begin{aligned} |I_0|^{-1} \sum_{\substack{M \geq T \\ M \text{ dyadic}}} \big|\langle \mathbf{1}_{E_M} Er_{J,M} f, g\rangle\big| &\leq |I_0|^{-1} \sum_{\substack{M \geq T \\ M \text{ dyadic}}} |G \cap E_M|^{1/2} \|Er_{J,M} f\|_2 \\ &\leq |I_0|^{-1} |F|^{1/2} \sum_{\substack{M \geq T \\ M \text{ dyadic}}} |G \cap E_M|^{1/2} \|\widehat{Er}_{J,M}\|_\infty. \end{aligned}$$

Using (5.7) and Cauchy–Schwarz over the dyadic scales,

$$\begin{aligned} &|I_0|^{-1} \sum_{\substack{M \geq T \\ M \text{ dyadic}}} \big|\langle \mathbf{1}_{E_M} Er_{J,M} f, g\rangle\big| \\ &\ll_A J^{-r} (\log J)^\beta |I_0|^{-1} |F|^{1/2} \left(\sum_M |G \cap E_M| \right)^{1/2} \left(\sum_{M \text{ dyadic}, M \geq T} (1 + \log M)^{-2A} \right)^{1/2} \\ &\ll_A J^{-r} (\log J)^\beta \delta_0^{1/2} (\log T)^{1/2 - A}, \end{aligned}$$

since the sets $G \cap E_M$ are disjoint subsets of G.

Combining the small-τ, low, high, and error estimates gives

$$|I_0|^{-1} |\langle B_\tau f, g\rangle| \ll \delta_0 J^s (\log J)^\gamma + \delta_0^{1/2} J^{-r} (\log J)^\beta.$$

Substituting (5.9) and using $\log J \sim \operatorname{Log} \delta_0$ gives

$$|I_0|^{-1} |\langle B_\tau f, g\rangle| \ll \delta_0^{1 - \frac{s}{2(s+r)}} (\operatorname{Log} \delta_0)^{\gamma + \frac{s(\beta-\gamma)}{s+r}},$$

which proves (5.8). □

Choose a fixed odd integer $c > 2C + 1$, where C is the constant in (5.4). The proof of the following lemma is a standard argument in harmonic analysis. We include the details for completeness.

Lemma 5.10. *Suppose that every admissible τ satisfies*

$$|\langle T_\tau \mathbf{1}_F, \mathbf{1}_G\rangle| \ll |I_0|\,\Psi\left(\frac{|F|}{|I_0|}, \frac{|G|}{|I_0|}\right) \tag{5.11}$$

for $F \subset cI_0$ and $G \subset I_0$, where $(T_N)_{N\in 2^{\mathbb{N}}}$ is a family of positive convolution operators whose kernel is supported in $[-CN, CN]$ for some fixed $C > 0$. Then there is a sparse family $\mathcal{S}$ such that

$$\langle \sup_{N\in 2^{\mathbb{N}}} |T_N \mathbf{1}_F|, \mathbf{1}_G\rangle \ll \sum_{I\in\mathcal{S}} |I|\,\Psi\left(\frac{|F\cap cI|}{|I|}, \frac{|G\cap I|}{|I|}\right). \tag{5.12}$$

Since the operators are positive and the input is nonnegative, the absolute values in $\sup_N |T_N\mathbf{1}_F|$ may be omitted. We keep them only to match the maximal-operator notation.

Proof. All scale suprema in this proof are over dyadic scales. Fix a dyadic interval I_0, with $F \subset cI_0$ and $G \subset I_0$, and let $\mathcal{I}(I_0)$ be the maximal dyadic subintervals $I \subset I_0$ for which

$$\langle \mathbf{1}_F\rangle_{cI,1} > K\langle \mathbf{1}_F\rangle_{cI_0,1} \quad \text{or} \quad \langle \mathbf{1}_G\rangle_{I,1} > K\langle \mathbf{1}_G\rangle_{I_0,1}, \tag{5.13}$$

where K is sufficiently large. The weak $(1,1)$ estimates for the Hardy–Littlewood and dyadic maximal operators give

$$\sum_{I\in\mathcal{I}(I_0)} |I| \ll_c K^{-1}|I_0|.$$

Thus K may be chosen so that

$$\sum_{I\in\mathcal{I}(I_0)} |I| \le \tfrac{1}{4}|I_0|. \tag{5.14}$$

For $x \in I_0$, let $I(x)$ be the interval in $\mathcal{I}(I_0)$ containing x, if one exists, and put $I(x) = \varnothing$ otherwise. Define

$$\mathcal{S}_{I_0}(x) = \begin{cases} \{N \in 2^{\mathbb{N}} : |I(x)| < N \le |I_0|\}, & I(x) \neq \varnothing, \\ \{N \in 2^{\mathbb{N}} : N \le |I_0|\}, & I(x) = \varnothing. \end{cases}$$

Choose $\tau(x) \in \mathcal{S}_{I_0}(x)$ such that

$$T_{\tau(x)}\mathbf{1}_F(x) = \sup_{N\in\mathcal{S}_{I_0}(x)} T_N\mathbf{1}_F(x).$$

We verify that τ is admissible. Let $N \ge \tau(x)$ be dyadic. If $N < |I_0|$, let $I \subset I_0$ be the dyadic interval containing x with $N \le |I| < 2N$. By maximality of the stopping intervals, (5.13) fails for I, while $c > 2C + 1$ gives $[x - CN, x + CN] \subset cI$. Consequently,

$$N^{-1} \sum_{|y-x|\le CN} \mathbf{1}_F(y) \lesssim_c \langle \mathbf{1}_F\rangle_{cI,1} \lesssim \langle \mathbf{1}_F\rangle_{cI_0,1}.$$

If $N \ge |I_0|$, then

$$N^{-1} \sum_{|y-x|\le CN} \mathbf{1}_F(y) \le N^{-1}|F| \lesssim_c \langle \mathbf{1}_F\rangle_{cI_0,1}.$$

Hence τ satisfies (5.4).

By (5.11),

$$\Big\langle \sup_{N\in\mathcal{S}_{I_0}(x)} |T_N\mathbf{1}_F|, \mathbf{1}_G\Big\rangle \ll |I_0|\Psi\big(\langle \mathbf{1}_F\rangle_{cI_0,1}, \langle \mathbf{1}_G\rangle_{I_0,1}\big). \tag{5.15}$$

For $x \in I \in \mathcal{I}(I_0)$, the remaining scales are $N \le |I|$. If K_N denotes the kernel of T_N, then $K_N(x, y) = 0$ unless $|x - y| \le CN$. Hence, for $x \in I$ and $N \le |I|$,

$$T_N\mathbf{1}_F(x) = T_N\mathbf{1}_{F\cap cI}(x).$$

Therefore

$$\sup_{N\le|I_0|}|T_N\mathbf{1}_F(x)|\le\sup_{N\in\mathcal{S}_{I_0}(x)}|T_N\mathbf{1}_F(x)|+\sum_{I\in\mathcal{I}(I_0)}\mathbf{1}_I(x)\sup_{N\le|I|}|T_N\mathbf{1}_{F\cap cI}(x)|.$$

Pairing with $\mathbf{1}_G$ and using (5.15), we obtain

$$\Big\langle\sup_{N\le|I_0|}|T_N\mathbf{1}_F|,\mathbf{1}_G\Big\rangle\ll|I_0|\Psi\big(\langle\mathbf{1}_F\rangle_{cI_0,1},\langle\mathbf{1}_G\rangle_{I_0,1}\big)+\sum_{I\in\mathcal{I}(I_0)}\Big\langle\sup_{N\le|I|}|T_N\mathbf{1}_{F\cap cI}|,\mathbf{1}_{G\cap I}\Big\rangle.\tag{5.16}$$

Iterating (5.16) and using (5.14) at every generation gives a sparse collection $\mathcal{S}$ of dyadic intervals such that

$$\Big\langle\sup_{N\le|I_0|}|T_N\mathbf{1}_F|,\mathbf{1}_G\Big\rangle\ll\sum_{I\in\mathcal{S}}|I|\Psi\big(\langle\mathbf{1}_F\rangle_{cI,1},\langle\mathbf{1}_G\rangle_{I,1}\big).$$

The constants are uniform in the truncation $N\le|I_0|$. Letting the truncation increase and applying monotone convergence gives (5.12). □

We now apply the stopping-time argument of Lemmas 5.1 and 5.10 to the annular prime-square averages. At small stopping scales we use their domination by the square averages, while at large scales we use the decomposition in Proposition 4.19.

Lemma 5.17. *Let $f=\mathbf{1}_F$ and $g=\mathbf{1}_G$, where $F\subset cI_0$ and $G\subset I_0$, and put $\delta_0=\frac{|F||G|}{|I_0|^2}$. Let $\tau:I_0\to\{2^k:(2^k)^2\le|I_0|\}$ be admissible at the spatial scale $\tau(x)^2$, in the sense that*

$$\sup_{M\ge\tau(x)}M^{-2}\sum_{|y-x|\le CM^2}f(y)\ll\langle f\rangle_{cI_0,1}.$$

Under the hypotheses of Proposition 4.19,

$$|I_0|^{-1}\big|\langle\widetilde{A}_{\tau,\Lambda}\mathbf{1}_F,\mathbf{1}_G\rangle\big|\ll\delta_0^{2/3}\Phi_{4\sqrt{3}/9+o(1)}(\delta_0).\tag{5.18}$$

Proof. Put $L=\operatorname{Log}\delta_0^{-1}$ and

$$E_0=\{x\in I_0:\log\tau(x)\le C_0L^2\},\qquad g_0=\mathbf{1}_{E_0}g.$$

By positivity,

$$\mathbf{1}_{E_0}\widetilde{A}_{\tau,\Lambda}f\le C_0L^2A_{\tau,\mathbb{N}^2}f.$$

The square averages satisfy the hypotheses of Lemma 5.1, with

$$r=\tfrac12,\qquad s=1,\qquad v=\tfrac12,\qquad\gamma=2,\qquad\beta=1,$$

by [21, Lemmas 5.2–5.6]. Applying that lemma to f, g_0, and the same admissible stopping time τ, and using $|\operatorname{supp}g_0|\le|G|$, gives

$$|I_0|^{-1}\left|\left\langle\mathbf{1}_{E_0}\widetilde{A}_{\tau,\Lambda}f,g\right\rangle\right|\ll L^2\delta_0^{2/3}L^{4/3}\ll\delta_0^{2/3}L^{10/3}.\tag{5.19}$$

For every fixed $\varepsilon>0$, every fixed power of L is bounded by $\Phi_\varepsilon(\delta_0)$, after enlarging the implicit constant. This gives the bound on the right-hand side of (5.18).

It remains to consider $\{\log\tau>C_0L^2\}$. Choose Q as in (4.14), with

$$s=1,\qquad r=\frac12,\qquad\kappa=\frac12,\qquad c_1=2+\varepsilon,$$

and round it to a dyadic integer. Then $\log Q\ll L$. By taking C_0 sufficiently large, we have $\log Q\le c(\log\tau(x))^{1/2}$ whenever $\log\tau(x)>C_0L^2$. Proposition 4.19 therefore gives

$$\widetilde{A}_{\tau(x),\Lambda}f(x)=\operatorname{Low}_{Q,\tau(x)}f(x)+\operatorname{High}_{Q,\tau(x)}f(x)+E_{Q,\tau(x)}f(x).\tag{5.20}$$

By (4.22) and admissibility,

$$|\operatorname{Low}_{Q,\tau(x)} f(x)| \ll Q\log Q \sup_{M\ge\tau(x)} M^{-2} \sum_{|y-x|\le CM^2} f(y) \ll Q\log Q\,\langle f\rangle_{cI_0,1}.$$

Consequently,

$$\left|\left\langle \mathbf{1}_{\{\log\tau>C_0L^2\}} \operatorname{Low}_{Q,\tau} f, g\right\rangle\right| \ll |I_0|\delta_0 Q\log Q. \tag{5.21}$$

For the high term, Cauchy–Schwarz and (4.23) give

$$\begin{aligned}\left|\left\langle \mathbf{1}_{\{\log\tau>C_0L^2\}} \operatorname{High}_{Q,\tau} f, g\right\rangle\right| &\le \|g\|_2 \left\| \sup_{M>\exp(C_0L^2)} |\operatorname{High}_{Q,M} f|\right\|_2 \\ &\ll |I_0|\delta_0^{1/2} Q^{-1/2} \exp\left((2+\varepsilon)\frac{(\log Q)^{1/2}}{\log\log Q}\right).\end{aligned} \tag{5.22}$$

For the error term, (4.21) gives

$$\begin{aligned}\left\| \sup_{\log M>C_0L^2} |E_{Q,M} f|\right\|_2^2 &\le \sum_{\log M>C_0L^2} \|E_{Q,M} f\|_2^2 \\ &\ll_A Q^{-1}(\log Q)^{2C_A} \sum_{\log M>C_0L^2} (1+\log M)^{-2A}\|f\|_2^2 \\ &\ll_A Q^{-1}(\log Q)^{2C_A} L^{2-4A}\|f\|_2^2,\end{aligned}$$

where the sums are over dyadic M. Hence

$$\left|\left\langle \mathbf{1}_{\{\log\tau>C_0L^2\}} E_{Q,\tau} f, g\right\rangle\right| \ll_A |I_0|\delta_0^{1/2} Q^{-1/2}(\log Q)^{C_A} L^{1-2A}. \tag{5.23}$$

For any fixed $A > 1/2$, the remaining factor $(\log Q)^{C_A} L^{1-2A}$ is a fixed power of L. Since $\log Q \asymp L$, it is absorbed by

$$\exp\left(\varepsilon\frac{(\log Q)^{1/2}}{\log\log Q}\right)$$

for every fixed $\varepsilon > 0$. Thus the error is absorbed into the high contribution after replacing $2+\varepsilon$ by $2+2\varepsilon$.

Combining the estimates (5.19), (5.20), (5.21), (5.22), and (5.23), we obtain

$$|I_0|^{-1}\left|\langle \widetilde{A}_{\tau,\Lambda} f, g\rangle\right| \ll \delta_0^{2/3}\Phi_\varepsilon(\delta_0) + \delta_0 Q\log Q + \delta_0^{1/2} Q^{-1/2}\exp\left((2+2\varepsilon)\frac{(\log Q)^{1/2}}{\log\log Q}\right).$$

By the choice of Q in (4.14), with $c_1 = 2+2\varepsilon$, the last two terms are bounded by $\delta_0^{2/3}\Phi_{4\sqrt{3}/9+O(\varepsilon)+o(1)}(\delta_0)$. The first term in (5.19) is also absorbed into this bound after fixing $\varepsilon > 0$. Letting $\varepsilon \downarrow 0$ proves (5.18). □

Proof of Theorem 1.3. For the square averages $A_{N,\mathbb{N}^2}$, the local estimates required in Lemma 5.1 are proved in [21, Lemmas 5.2–5.6]: Lemma 5.2 controls the tails of the smoothed kernel, Lemmas 5.3 and 5.4 give the two high contributions, and Lemma 5.6 uses precisely the admissibility condition (5.4). Their parameters are

$$r = \tfrac{1}{2}, \qquad s = 1, \qquad v = \tfrac{1}{2}, \qquad \gamma = 2, \qquad \beta = 1.$$

Lemmas 5.1 and 5.10 therefore give the asserted endpoint sparse form for $A_{N,\mathbb{N}^2}$.

For the annular prime-square averages, Lemmas 5.17 and 5.10, applied at the spatial scale N^2, give the corresponding sparse bound over dyadic arithmetic cutoffs. For $f \ge 0$, positivity

and annular reconstruction give

$$A_\Lambda^* f \le 4 \sup_{\substack{N\in 2^{\mathbb{N}}\\ N\ge 2}} |\widetilde{A}_{N,\Lambda} f|.$$

We therefore obtain

$$\left\langle \sup_N |A_{N,\Lambda}\mathbf{1}_F|, \mathbf{1}_G \right\rangle \ll \sum_{I\in\mathcal{S}} |I| \left(\frac{|F\cap cI||G\cap I|}{|I|^2}\right)^{2/3} \Phi_{4\sqrt{3}/9+o(1)}\left(\frac{|F\cap cI||G\cap I|}{|I|^2}\right). \tag{5.24}$$

It remains to obtain the open sparse region. First, $\sup_N |A_{N,\Lambda}\mathbf{1}_F(x)| \le C$ uniformly in x, and hence

$$\left|\left\langle \sup_N |A_{N,\Lambda}\mathbf{1}_F|, \mathbf{1}_G \right\rangle\right| \le C|G|.$$

This is the local restricted estimate corresponding to $(1/p, 1/q) = (0,1)$. Next, by [38, Theorem 1],

$$\left\| \sup_N |A_{N,\Lambda} f| \right\|_u \ll_u \|f\|_u, \qquad u > 1.$$

Therefore Hölder's inequality gives

$$\left|\left\langle \sup_N |A_{N,\Lambda}\mathbf{1}_F|, \mathbf{1}_G \right\rangle\right| \ll_u |F|^{1/u}|G|^{1/u'}.$$

After localization to an interval I, this is the restricted local estimate at

$$\left(\frac{1}{p},\frac{1}{q}\right) = \left(\frac{1}{u},\frac{1}{u'}\right), \qquad u > 1.$$

As u ranges over $(1,\infty)$, these points fill the open segment between $(1,0)$ and $(0,1)$. Interpolating them with (5.24) gives every $(1/p, 1/q)$ in the interior of the triangle with vertices

$$(2/3, 2/3), \qquad (1,0), \qquad (0,1).$$

Applying Lemma 5.10 to those interpolated local estimates gives (1.4). □

6. The high–low estimates

In this section we prove Proposition 4.19. We work with the annular averages

$$\widetilde{A}_{N,\Lambda} f = A_{N,\Lambda} f - \tfrac{1}{2} A_{N/2,\Lambda} f = \frac{1}{N} \sum_{N/2<n\le N} \Lambda(n) f(\cdot - n^2), \tag{6.1}$$

where N is dyadic. For $0 < \beta \le 1$, put

$$\widehat{\widetilde{M}}_{N,\beta}(\theta) = \frac{1}{N}\int_{N/2}^{N} t^{\beta-1} e(-t^2\theta)\, dt, \qquad \widetilde{M}_N = \widetilde{M}_{N,1}. \tag{6.2}$$

Splitting the integral at $\|\theta\|^{-1/2}$ and integrating by parts gives, uniformly in $0 < \beta \le 1$,

$$|\widehat{\widetilde{M}}_{N,\beta}(\theta)| \ll N^{\beta-1} \min\{1, (N^2\|\theta\|)^{-1/2}\} + N^{-1}. \tag{6.3}$$

Fix an even smooth function $0 \le \eta \le 1$, equal to one on $[-1/2, 1/2]$ and supported on $[-1,1]$. For $t \in \mathbb{R}$, write $\eta_t(\theta) = \eta(16^t\theta)$. In denominator sums, $q \sim 2^s$ means $2^s \le q < 2^{s+1}$, with $s \in \mathbb{Z}_{\ge 0}$. Set

$$R = \max\{\exp((\log N)^{1/10}), Q^8\}. \tag{6.4}$$

The assumption $\log Q \le c_0 (\log N)^{1/2}$, with $c_0 > 0$ sufficiently small, ensures that Page's theorem applies up to $R^{11/10}$. It also gives $QR^4 = o(N^2)$, so the narrower cutoffs introduced below are contained in the region where $\eta_s = 1$.

6.1. **The major-arc model and its error.** We use the following annular form of the standard prime-square circle-method input.

Lemma 6.5. *If $(a,q)=1$, $q\le R^{11/10}$, and $|\theta|\le R/(qN^2)$, then*

$$\widehat{\widetilde{A}}_{N,\Lambda}(a/q+\theta)=\mathfrak{s}(a,q)\widehat{\widetilde{M}}_N(\theta)-\mathbf{1}_{q_0\mid q}\mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta}(\theta)+O(R^{-5}). \tag{6.6}$$

Here $q_0=q_0(N)$, $\beta=\beta(N)$, and $\chi_0=\chi_0(N)$ are the exceptional conductor, zero, and primitive character supplied by the Landau–Page theorem at scale N, and χ_q is induced by χ_0. These quantities may change when N changes. The second term is omitted if no exceptional character exists. If ξ lies outside all major arcs

$$\|\xi-a/q\|\le\frac{R}{qN^2},\qquad q\le R,\quad (a,q)=1,$$

where distance is taken modulo one, then

$$|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)|\ll R^{-1/4}(\log N)^{C_0}. \tag{6.7}$$

Here $R\ge 1$, N is sufficiently large, and $R^4\le N/2$, as is the case for the choice in (6.4).

Proof. Since Λ is supported on prime powers, the condition $(n,q)>1$ in (6.1) forces $n=p^\alpha$ with $p\mid q$. The terms with $\alpha=1$ contribute $O(qN^{-1}(\log N)^2)$, while all higher prime powers contribute $O(N^{-1/2}(\log N)^2)$. Thus the total error from these terms is

$$O\big(N^{-1/2}(\log N)^2+qN^{-1}(\log N)^2\big).$$

Grouping the remaining terms into reduced residue classes modulo q therefore gives

$$\widehat{\widetilde{A}}_{N,\Lambda}(a/q+\theta)=\frac{1}{N}\sum_{r\in\mathbb{A}_q}e(-ar^2/q)\sum_{\substack{N/2<n\le N\\ n\equiv r\ (\mathrm{mod}\ q)}}\Lambda(n)e(-n^2\theta)+O\big(N^{-1/2}(\log N)^2+qN^{-1}(\log N)^2\big).$$

By (2.6), uniformly for $N/2\le x\le N$,

$$\sum_{\substack{n\le x\\ n\equiv r\ (\mathrm{mod}\ q)}}\Lambda(n)=\frac{x}{\phi(q)}-\mathbf{1}_{q_0\mid q}\frac{\chi_q(r)x^\beta}{\beta\phi(q)}+O\big(xe^{-c\sqrt{\log N}}\big).$$

Partial summation, using $|\frac{d}{dt}e(-t^2\theta)|\ll|\theta|t$, now yields

$$\frac{1}{N}\sum_{\substack{N/2<n\le N\\ n\equiv r\ (\mathrm{mod}\ q)}}\Lambda(n)e(-n^2\theta)=\frac{1}{N\phi(q)}\int_{N/2}^{N}e(-t^2\theta)\,dt-\mathbf{1}_{q_0\mid q}\frac{\chi_q(r)}{N\phi(q)}\int_{N/2}^{N}t^{\beta-1}e(-t^2\theta)\,dt+O\left(e^{-c\sqrt{\log N}}\big(1+|\theta|N^2\big)\right).$$

Here $t^{\beta-1}$ arises from differentiating t^β/β. Summing over $r\in\mathbb{A}_q$ produces the two Gauss sums and gives

$$\widehat{\widetilde{A}}_{N,\Lambda}(a/q+\theta)=\mathfrak{s}(a,q)\widehat{\widetilde{M}}_N(\theta)-\mathbf{1}_{q_0\mid q}\mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta}(\theta)+O\left(N^{-1/2}(\log N)^2+qN^{-1}(\log N)^2+qe^{-c\sqrt{\log N}}\big(1+|\theta|N^2\big)\right).$$

Since $q\le R^{11/10}$ and $|\theta|N^2\le R/q$, the error is

$$\ll N^{-1/2}(\log N)^2+R^{11/10}N^{-1}(\log N)^2+R^2e^{-c\sqrt{\log N}}\ll R^{-5},$$

by (6.4) and the sufficiently small choice of c_0. This proves (6.6).

For the minor arcs put $x = N/2$. Choose a reduced approximation a/q with $q \le x^{3/2}$,

$$|\xi - a/q| \le (qx^{3/2})^{-1}, \qquad |\xi - a/q| < q^{-2}. \tag{6.8}$$

If $q \le x^{1/2}$, apply [44, Theorem 1.1] directly to the annular sum. With $T = q(1 + |\xi - a/q|x^2)$, being outside the major arcs gives $T \ge R/4$, while (6.8) gives $T \ll x^{1/2}$. Absorbing $\tau(q)^{3/2} \ll q^{1/12}$ in this range yields the bound

$$|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)| \ll (\log N)^{C_0} R^{-5/12}.$$

For $x^{1/2} \le q \le x^{3/2}$, use Ghosh's quadratic prime exponential-sum estimate [13], with its fixed exponent $1/16$. Its normalized bound is $O(x^{-1/16})$, and $R^4 \le x$ implies $x^{-1/16} \le R^{-1/4}$. Combining the two cases proves (6.7). □

Define the major-arc model by

$$\widehat{V}_N(\xi) = \sum_{q \le R} \sum_{a \in \mathbb{A}_q} \big(\mathfrak{s}(a,q)\widehat{\widetilde{M}}_N(\xi - a/q) - \mathbf{1}_{q_0|q}\mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta}(\xi - a/q)\big)\eta_s(\xi - a/q), \tag{6.9}$$

and put

$$E_{Q,N} = \widetilde{A}_{N,\Lambda} - V_N.$$

Lemma 6.10 (Approximation)**.** *For every $A > 0$,*

$$\|E_{Q,N} f\|_2 \ll_A Q^{-1/2}(1 + \log N)^{-A}\|f\|_2. \tag{6.11}$$

Proof. Fix $\xi \in \mathbb{T}$. The localized products in (6.9) are understood to be periodized. Distinct reduced fractions with $q, q' \sim 2^s$ are separated by at least $(qq')^{-1} \ge 2^{-2s-2}$, whereas $\eta_s(\xi - a/q)$ can be nonzero only when $\|\xi - a/q\| \le 16^{-s}$. Thus only $O(1)$ fractions contribute at each dyadic denominator scale, each with $O(1)$ active periodic lifts.

Consider the terms in $\widehat{V}_N(\xi)$ for which

$$\|\xi - a/q\| > \frac{R}{qN^2}.$$

Write $\theta = \xi - a/q$. By (6.3), uniformly in $0 < \beta \le 1$ and $j \in \mathbb{Z}$,

$$|\widehat{\widetilde{M}}_N(\theta + j)| + |\widehat{\widetilde{M}}_{N,\beta}(\theta + j)| \ll (q/R)^{1/2} + N^{-1}.$$

Lemmas 3.21 and 3.24, with $\varepsilon = 1/12$, bound both Gauss coefficients by $O(q^{-5/12})$. Each such fraction therefore contributes $O(R^{-1/2}q^{1/12} + N^{-1})$. Summing over the $O(\log(eR))$ denominator scales gives a total contribution of size

$$O\big((R^{-5/12} + N^{-1})\log(eR)\big) = O(R^{-1/3}), \tag{6.12}$$

where we use $R = N^{o(1)}$ in the permitted range of Q.

If $\xi \in \mathfrak{M}(R; N)$, there is a reduced fraction a/q, with $q \le R$, such that

$$\left\|\xi - \frac{a}{q}\right\| \le \frac{R}{qN^2}.$$

Such a fraction is unique by rational separation and $R^3 = o(N^2)$. Write $\xi = a/q + \theta \pmod 1$, with $|\theta| \le R/(qN^2)$. If $q \sim 2^s$, then

$$16^s|\theta| \le q^4 \frac{R}{qN^2} \le \frac{R^4}{N^2} < \frac{1}{2},$$

so $\eta_s(\theta) = 1$. Lemma 6.5 gives

$$\widehat{\widetilde{A}}_{N,\Lambda}(\xi) = \mathfrak{s}(a,q)\widehat{\widetilde{M}}_N(\theta) - \mathbf{1}_{q_0|q}\mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta}(\theta) + O(R^{-5}). \tag{6.13}$$

The two main terms are the $j=0$ terms in the contribution of a/q to $\widehat{V}_N(\xi)$. The other active lifts satisfy $|\theta+j|\geq 1/2$ and contribute $O(N^{-1})$ in total by integration by parts in (6.2). Every other contributing fraction lies outside its corresponding major arc. Thus (6.12) and (6.13) give

$$|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)-\widehat{V}_N(\xi)|\ll R^{-5}+N^{-1}+R^{-1/3}\ll R^{-1/3}.$$

On the complementary minor arcs $\xi\in\mathfrak{m}(R;N)$, (6.7) gives

$$|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)|\ll R^{-1/4}(\log N)^{C_0}\ll R^{-1/6}.$$

The last inequality follows from $R\geq\exp((\log N)^{1/10})$, for sufficiently large N. Every fraction contributing to $\widehat{V}_N(\xi)$ now lies outside its corresponding major arc, so (6.12) gives $|\widehat{V}_N(\xi)|\ll R^{-1/3}$. Consequently,

$$|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)-\widehat{V}_N(\xi)|\leq|\widehat{\widetilde{A}}_{N,\Lambda}(\xi)|+|\widehat{V}_N(\xi)|\ll R^{-1/6}.$$

Combining the two cases, we obtain $\|\widehat{E}_{Q,N}\|_{L^\infty(\mathbb{T})}\ll R^{-1/6}$. Finally, (6.4) gives

$$R^{-1/6}=R^{-1/16}R^{-5/48}\leq Q^{-1/2}\exp\left(-\frac{5}{48}(\log N)^{1/10}\right)\ll_A Q^{-1/2}(1+\log N)^{-A}.$$

Plancherel proves (6.11). The remaining bounded scales are absorbed into the implied constant, since Q is then bounded. □

6.2. **The decomposition.** For $q\leq Q$, set

$$m_q=\frac{\log(N^2q/Q)}{\log 16},\qquad 16^{-m_q}=\frac{Q}{qN^2}.$$

If an exceptional primitive real character $\chi_0=\chi_0(N)$ of conductor q_0 exists at the current scale N, let $\mathcal{B}_0$ be the constant from Lemma 3.27. Put

$$D=\frac{Q}{\mathcal{B}_0},\qquad \ell_d=\frac{\log(\mathcal{B}_0dN^2/Q)}{\log 16},\qquad 16^{-\ell_d}=\frac{Q}{\mathcal{B}_0dN^2}\quad(1\leq d\leq D).$$

In the exceptional sums below, $q=q_0d$ and χ_q denotes the character modulo q induced by χ_0. When $D<1$, the corresponding low sum is empty. If no exceptional character exists, all sums involving q_0 are omitted.

Define

$$\begin{aligned}\widehat{\mathrm{Low}}_{Q,N}(\xi)=&\sum_{q\leq Q}\sum_{a\in\mathbb{A}_q}\mathfrak{s}(a,q)\widehat{\widetilde{M}}_N(\xi-a/q)\eta_{m_q}(\xi-a/q)\\&-\sum_{\substack{q=q_0d\leq R\\ d\leq D}}\sum_{a\in\mathbb{A}_q}\mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta}(\xi-a/q)\eta_{\ell_d}(\xi-a/q).\end{aligned}$$

The high multiplier is the complementary part of V_N in (6.9):

$$\begin{aligned}\widehat{\mathrm{High}}_{Q,N}(\xi)=&\sum_{Q<q\leq R}\sum_{a\in\mathbb{A}_q}\mathfrak{s}(a,q)\widehat{\widetilde{M}}_N(\xi-a/q)\eta_s(\xi-a/q)\\&+\sum_{q\leq Q}\sum_{a\in\mathbb{A}_q}\mathfrak{s}(a,q)\widehat{\widetilde{M}}_N(\xi-a/q)(\eta_s-\eta_{m_q})(\xi-a/q)\\&-\sum_{\substack{q=q_0d\leq R\\ d>D}}\sum_{a\in\mathbb{A}_q}\mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta}(\xi-a/q)\eta_s(\xi-a/q)\end{aligned}$$

$$-\sum_{\substack{q=q_0d\le R\\ d\le D}}\sum_{a\in\mathbb{A}_q}\mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta}(\xi-a/q)(\eta_s-\eta_{\ell_d})(\xi-a/q). \tag{6.14}$$

For each ordinary term with $q\le Q$, the low part contains the cutoff η_{m_q}, while the high part contains the complementary cutoff $\eta_s-\eta_{m_q}$. Similarly, for each exceptional term $q=q_0d$ with $d\le D$, the two cutoffs are η_{ℓ_d} and $\eta_s-\eta_{\ell_d}$. Thus

$$\eta_{m_q}+(\eta_s-\eta_{m_q})=\eta_s,\qquad \eta_{\ell_d}+(\eta_s-\eta_{\ell_d})=\eta_s.$$

The ordinary terms with $q>Q$ and the exceptional terms with $d>D$ occur entirely in the high part, with cutoff η_s. Consequently, $\mathrm{Low}_{Q,N}+\mathrm{High}_{Q,N}=V_N$. Since $E_{Q,N}=\widetilde{A}_{N,\Lambda}-V_N$, we obtain

$$\widetilde{A}_{N,\Lambda}=\mathrm{Low}_{Q,N}+\mathrm{High}_{Q,N}+E_{Q,N}. \tag{6.15}$$

6.3. **The low estimate.** We prove the $\ell^1\to\ell^\infty$ bound for the low part.

Proposition 6.16. *For nonnegative f,*

$$|\mathrm{Low}_{Q,N}f(x)|\ll Q\log(2Q)\sup_{M\ge N}M^{-2}\sum_{|y-x|\le CM^2}f(y). \tag{6.17}$$

Consequently,

$$\|\mathrm{Low}_{Q,N}f\|_\infty\ll N^{-2}Q\log(2Q)\|f\|_1.$$

Proof. Fourier inversion and Lemma 3.20 give, uniformly in x,

$$\sum_{q\le Q}\frac{1}{q}\left|\sum_{a\in\mathbb{A}_q}\mathfrak{s}(a,q)e(ax/q)\right|\ll\log(2Q).$$

The cutoff η_{m_q} has spatial scale $qN^2/Q\le N^2$. By scaling and the rapid decay of $\eta^\vee$, for every $K\ge1$,

$$|\eta^\vee_{m_q}(x)|\ll_K 16^{-m_q}(1+16^{-m_q}|x|)^{-K}=\frac{Q}{qN^2}\left(1+\frac{Q|x|}{qN^2}\right)^{-K}.$$

The measure corresponding to $\widehat{\widetilde{M}}_N$ has total mass $O(1)$ and is supported in an interval J_N of length $O(N^2)$. Since $qN^2/Q\le N^2$, convolution with this measure gives

$$|(\widetilde{M}_N*\eta^\vee_{m_q})(x)|\ll\frac{Q}{qN^2}\left(1+\frac{\mathrm{dist}(x,J_N)}{N^2}\right)^{-10}\ll\frac{Q}{qN^2}\left(1+\frac{|x|}{N^2}\right)^{-10},$$

where the last inequality uses $J_N\subset[-CN^2,CN^2]$. Consequently, Fourier inversion and Lemma 3.20 give

$$\begin{aligned}\Big|\sum_{q\le Q}\sum_{a\in\mathbb{A}_q}\mathfrak{s}(a,q)e(ax/q)(\widetilde{M}_N*\eta^\vee_{m_q})(x)\Big|&\ll\frac{Q}{N^2}\left(1+\frac{|x|}{N^2}\right)^{-10}\sum_{q\le Q}\frac{1}{q}\left|\sum_{a\in\mathbb{A}_q}\mathfrak{s}(a,q)e(ax/q)\right|\\&\ll\frac{Q\log(2Q)}{N^2}\left(1+\frac{|x|}{N^2}\right)^{-10}.\end{aligned}$$

We now estimate the exceptional part of the low kernel. By Fourier inversion, this kernel is

$$-\sum_{\substack{q=q_0d\le R\\ d\le D}}\left(\sum_{a\in\mathbb{A}_q}\mathfrak{s}_{\chi_q}(a,q)e(ax/q)\right)(\widetilde{M}_{N,\beta}*\eta^\vee_{\ell_d})(x).$$

The measure corresponding to $\widehat{\widetilde{M}}_{N,\beta}$ is supported in $[N^2/4, N^2]$, and its total mass satisfies

$$\frac{1}{N}\int_{N/2}^{N} t^{\beta-1}\,dt \ll N^{\beta-1} \le 1.$$

Moreover, since $16^{-\ell_d} = Q/(\mathcal{B}_0 dN^2)$, rapid decay of $\eta^\vee$ gives

$$|\eta_{\ell_d}^\vee(x)| \ll \frac{Q}{\mathcal{B}_0 dN^2}\left(1+\frac{Q|x|}{\mathcal{B}_0 dN^2}\right)^{-10}.$$

For $d \le D = Q/\mathcal{B}_0$, the corresponding spatial scale $\mathcal{B}_0 dN^2/Q$ is at most N^2. Convolving with $\widetilde{M}_{N,\beta}$ therefore yields

$$|(\widetilde{M}_{N,\beta} * \eta_{\ell_d}^\vee)(x)| \ll \frac{Q}{\mathcal{B}_0 dN^2}\left(1+\frac{|x|}{N^2}\right)^{-10},$$

after enlarging the absolute constant to account for the support $[N^2/4, N^2]$.

If the arithmetic sum corresponding to $q = q_0 d$ is nonzero, Lemma 3.27 gives

$$\left|\sum_{a\in\mathbb{A}_{q_0 d}} \mathfrak{s}_{\chi_{q_0 d}}(a, q_0 d) e(ax/(q_0 d))\right| \ll \mathcal{B}_0 \prod_{p\mid d}\frac{p+1}{p-1}.$$

Consequently,

$$\Big|\sum_{\substack{q=q_0 d\le R\\ d\le D}}\left(\sum_{a\in\mathbb{A}_q}\mathfrak{s}_{\chi_q}(a,q)e(ax/q)\right)(\widetilde{M}_{N,\beta}*\eta_{\ell_d}^\vee)(x)\Big| \ll \frac{Q}{N^2}\left(1+\frac{|x|}{N^2}\right)^{-10}\sum_{\substack{d\le D\\ \mathfrak{s}_{\chi_{q_0 d}}\not\equiv 0}}\frac{1}{d}\prod_{p\mid d}\frac{p+1}{p-1}.$$

Here we have discarded the additional restriction $q_0 d \le R$.

It remains to estimate the sum over d. By Lemma 3.27, every contributing d can be written as

$$d = 2^\nu d_1, \qquad 0\le\nu\le 5, \qquad \mu^2(d_1)=1, \qquad (d_1, 2q_0)=1.$$

The finitely many possible powers of 2 contribute only an absolute constant. Hence

$$\sum_{\substack{d\le D\\ \mathfrak{s}_{\chi_{q_0 d}}\not\equiv 0}}\frac{1}{d}\prod_{p\mid d}\frac{p+1}{p-1} \ll \sum_{\substack{d_1\le D\\ \mu^2(d_1)=1\\ (d_1,2q_0)=1}}\frac{1}{d_1}\prod_{p\mid d_1}\frac{p+1}{p-1} \le \prod_{\substack{p\le D\\ p\nmid 2q_0}}\left(1+\frac{p+1}{p(p-1)}\right).$$

Since

$$\frac{p+1}{p(p-1)} = \frac{1}{p} + O(p^{-2}),$$

Mertens' theorem gives

$$\prod_{\substack{p\le D\\ p\nmid 2q_0}}\left(1+\frac{p+1}{p(p-1)}\right) \ll \exp\left(\sum_{p\le D}\frac{1}{p}+O(1)\right) \ll \log(2D) \le \log(2Q).$$

We have therefore proved

$$\Big|\sum_{\substack{q=q_0 d\le R\\ d\le D}}\left(\sum_{a\in\mathbb{A}_q}\mathfrak{s}_{\chi_q}(a,q)e(ax/q)\right)(\widetilde{M}_{N,\beta}*\eta_{\ell_d}^\vee)(x)\Big| \ll \frac{Q\log(2Q)}{N^2}\left(1+\frac{|x|}{N^2}\right)^{-10}.$$

Combining this with the ordinary low-kernel estimate gives the same bound for the full low kernel.

Finally, for $f \ge 0$, split the convolution into the regions

$$|x-y| \le CN^2 \quad \text{and} \quad 2^j N^2 < |x-y| \le 2^{j+1}N^2, \qquad 2^j \ge C.$$

On the j-th region the kernel is $O(Q\log(2Q)N^{-2}2^{-10j})$. Taking $M = 2^{j/2}N$, up to a fixed factor, gives

$$N^{-2}2^{-10j} \sum_{|x-y|\le C2^jN^2} f(y) \ll 2^{-9j} \sup_{M\ge N} M^{-2} \sum_{|x-y|\le CM^2} f(y).$$

Summing the geometric series proves (6.17). Since

$$\sup_{M\ge N} M^{-2} \sum_{|x-y|\le CM^2} f(y) \le N^{-2}\|f\|_1,$$

the global $\ell^1 \to \ell^\infty$ estimate follows. □

6.4. **The high estimate.** We use the following standard multifrequency estimate for the terms in (6.14) with cutoff η_s. We include its proof for completeness, including the uniformity in $0 < \beta \le 1$ and in the coefficients. The two terms containing $\eta_s - \eta_{m_q}$ and $\eta_s - \eta_{\ell_d}$ are treated separately by square-function estimates.

Lemma 6.18 (Multifrequency estimate)**.** *Let $0 < \beta \le 1$, and let $\mathcal{R}_s$ be any subset of the reduced fractions with denominators $q \sim 2^s$. Then*

$$\left\| \sup_{\substack{N\in 2^{\mathbb{N}}\\ N\gtrsim 2^s}} \left| \left(\sum_{a/q\in\mathcal{R}_s} c_{a,q} \widehat{\widetilde{M}}_{N,\beta}(\xi - a/q)\eta_s(\xi-a/q)\widehat{f}(\xi) \right)^\vee \right| \right\|_2 \ll (1+s) \sup_{a/q\in\mathcal{R}_s} |c_{a,q}|\, \|f\|_2. \tag{6.19}$$

Proof. Write $\mathcal{R}_s = \{\theta_1, \dots, \theta_J\}$, where $\theta_j = a_j/q_j$. Distinct reduced fractions with denominators $q_j \sim 2^s$ satisfy

$$\|\theta_j - \theta_k\|_{\mathbb{T}} \gg 2^{-2s}, \qquad J \ll 2^{2s}, \qquad \log(2+J) \ll 1+s.$$

Put $\widehat{f_j}(\theta) = \eta_s(\theta)\widehat{f}(\theta+\theta_j)$. The expression inside the supremum in (6.19) equals

$$\sum_{j=1}^J c_{a_j,q_j} e(\theta_j x)\big(\widetilde{M}_{N,\beta} * f_j\big)(x).$$

For $\beta = 1$, apply the separated-frequency maximal inequality [29, Theorem 1.1], followed by transference to $\ell^2(\mathbb{Z})$, to the family $g_j = c_{a_j,q_j} f_j$. Since the coefficients are independent of x and N, linearity gives

$$\begin{aligned} \left\| \sup_{N\gtrsim 2^s} \left| \sum_{j=1}^J c_{a_j,q_j} e(\theta_j x)\big(\widetilde{M}_N * f_j\big)(x) \right| \right\|_2 &\ll \log(2+J) \left(\sum_{j=1}^J |c_{a_j,q_j}|^2 \|f_j\|_2^2 \right)^{1/2} \\ &\le \log(2+J) \sup_j |c_{a_j,q_j}| \left(\sum_{j=1}^J \|f_j\|_2^2 \right)^{1/2}. \end{aligned} \tag{6.20}$$

This is bounded by the right-hand side of (6.19), using $\sum_{j=1}^J \|f_j\|_2^2 \ll \|f\|_2^2$.

The estimate (6.20) also holds uniformly for $0 < \beta \le 1$. Indeed, on $N/2 \le t \le N$, we can write $t^{\beta-1} = N^{\beta-1}(t/N)^{\beta-1}$, and the functions $u \mapsto u^{\beta-1}$ have uniformly bounded size and variation on $[1/2, 1]$. Abel summation in the defining integral for $\widehat{\widetilde{M}}_{N,\beta}$ therefore reduces the required multifrequency estimate to the case $\widetilde{M}_N = \widetilde{M}_{N,1}$, with an additional factor $N^{\beta-1}$, which is bounded by 1. □

Corollary 6.21. *Let $0 < \beta \le 1$, $s \ge 0$, and $|c_{a,q}| \le b$. Then*

$$\left\| \sup_{\substack{N\in 2^{\mathbb{N}}\\ N\ge 2^s}} \sup_{K} \left| \left(\sum_{\substack{q\sim 2^s\\ q\le K}} \sum_{a\in\mathbb{A}_q} c_{a,q} \widehat{\widetilde{M}}_{N,\beta}(\xi - a/q)\eta_s(\xi - a/q)\widehat{f}(\xi) \right)^{\vee} \right| \right\|_2 \ll (1+s)^2 b \|f\|_2. \quad (6.22)$$

The same estimate holds if each summand is multiplied by a fixed multiplier of modulus at most one supported where $\eta_s(\xi - a/q) \neq 0$.

Proof. Order the denominators $q \sim 2^s$ increasingly. Every initial segment is a disjoint union of at most one dyadic block of each possible length. If $\mathcal{D}_j$ denotes the family of blocks of one fixed length, the Rademacher–Menshov decomposition gives

$$\sup_K \sup_{N\ge 2^s} \left| \sum_{\substack{q\sim 2^s\\ q\le K}} T_{N,q} f \right| \le \sum_j \left(\sum_{I\in\mathcal{D}_j} \sup_{N\ge 2^s} \left| \sum_{q\in I} T_{N,q} f \right|^2 \right)^{1/2},$$

where $T_{N,q}$ denotes the sum over $a \in \mathbb{A}_q$ in (6.22).

For each block I, Lemma 6.18 gives

$$\left\| \sup_{N\ge 2^s} \left| \sum_{q\in I} T_{N,q} f \right| \right\|_2 \ll (1+s) b \|f_I\|_2,$$

where $\widehat{f_I}$ is the restriction of $\widehat{f}$ to the frequency supports corresponding to the denominators in I. These supports are disjoint, and hence, for each j,

$$\sum_{I\in\mathcal{D}_j} \|f_I\|_2^2 \le \|f\|_2^2.$$

There are $O(1+s)$ possible block lengths, proving (6.22). The same argument applies after inserting the additional multipliers. □

Proposition 6.23. *Let $N_0 \ge 2$ and suppose that $\log Q \le c_0 (\log N_0)^{1/2}$. Then, for every dyadic $N \ge N_0$,*

$$\| \operatorname{High}_{Q,N} f \|_2 \ll Q^{-1/2} \exp\left((2+o(1)) \frac{\sqrt{\log Q}}{\log\log Q} \right) \|f\|_2, \quad (6.24)$$

$$\left\| \sup_{\substack{M\ge N_0\\ M\in 2^{\mathbb{N}}}} |\operatorname{High}_{Q,M} f| \right\|_2 \ll Q^{-1/2} \exp\left((2+o(1)) \frac{\sqrt{\log Q}}{\log\log Q} \right) \|f\|_2. \quad (6.25)$$

Here $o(1)$ is taken as $Q \to \infty$, uniformly in the allowed range of N_0. The bounded values of Q are absorbed into the implicit constant.

Proof. It is enough to prove (6.25), since (6.24) follows by taking $N_0 = N$. At scale M, write

$$R(M) = \max\{\exp((\log M)^{1/10}), Q^8\}.$$

The allowed range of Q gives $R(M) = M^{o(1)}$, uniformly for $M \ge N_0$. Hence every denominator $q \le R(M)$ satisfies $M \gtrsim q$. In particular, $M \gtrsim 2^s$ whenever $q \sim 2^s$, as required in Lemma 6.18.

Consider first the terms in (6.14) of the form

$$\mathfrak{s}(a,q) \widehat{\widetilde{M}}_M(\xi - a/q)\eta_s(\xi - a/q), \qquad q \sim 2^s, \quad q > Q.$$

At a fixed denominator scale, the condition $q \le R(M)$ introduces an upper cutoff in the sum over q, depending on M. Apply Corollary 6.21 with $K = \lfloor R(M) \rfloor$, setting the coefficients with $q \le Q$ equal to zero. Together with Lemma 3.21, this bounds the maximal norm of these terms by

$$\sum_{2^s \gtrsim Q} (1+s)^2 (2^s)^{-1/2} \exp\left((2+o(1)) \frac{\sqrt{\log(2^s)}}{\log\log(2^s)} \right) \|f\|_2.$$

The additional fixed logarithmic factor is absorbed into the $o(1)$ in the following estimate. The summands decrease geometrically, up to the displayed subexponential factor, so this is

$$\ll Q^{-1/2} \exp\left((2+o(1)) \frac{\sqrt{\log Q}}{\log\log Q} \right) \|f\|_2.$$

We next consider the terms containing

$$\mathfrak{s}(a,q) \widehat{\widetilde{M}}_M(\xi - a/q) \big(\eta_s - \eta_{m_q(M)}\big)(\xi - a/q), \qquad q \sim 2^s, \quad q \le Q,$$

where $16^{-m_q(M)} = \frac{Q}{qM^2}$. Fix q, a, put $\theta = \xi - a/q$, and let $M_1 \ge N_0$ be the first dyadic scale for which $(\eta_s - \eta_{m_q(M_1)})(\theta) \ne 0$. If no such scale exists, there is nothing to estimate. Otherwise,

$$|\theta| \gg \frac{Q}{qM_1^2}.$$

For every dyadic $M \ge M_1$, (6.3) gives

$$|\widehat{\widetilde{M}}_M(\theta)| \ll (M^2|\theta|)^{-1/2} + M^{-1} \ll \left(\frac{q}{Q}\right)^{1/2} \frac{M_1}{M}.$$

Here the M^{-1} term is absorbed because $|\theta| \le 1/2$. Consequently,

$$\sum_{\substack{M \ge M_1 \\ M \in 2^{\mathbb{N}}}} |\widehat{\widetilde{M}}_M(\theta)|^2 \ll \frac{q}{Q}.$$

At a fixed denominator scale, the rational frequency supports are disjoint. Bounding the supremum by the square function and applying Plancherel therefore gives

$$\begin{aligned} \left\| \sup_{\substack{M \ge N_0 \\ M \in 2^{\mathbb{N}}}} \left| \left(\sum_{q \sim 2^s} \sum_{a \in \mathbb{A}_q} \mathfrak{s}(a,q) \widehat{\widetilde{M}}_M(\xi - a/q)(\eta_s - \eta_{m_q(M)})(\xi - a/q) \widehat{f}(\xi) \right)^\vee \right| \right\|_2 \\ \ll \sup_{\substack{q \sim 2^s \\ a \in \mathbb{A}_q}} |\mathfrak{s}(a,q)| \left(\frac{q}{Q}\right)^{1/2} \|f\|_2 \ll Q^{-1/2} \exp\left((2+o(1)) \frac{\sqrt{\log(2^s)}}{\log\log(2^s)} \right) \|f\|_2. \end{aligned}$$

Summing over $2^s \lesssim Q$ gives

$$\ll Q^{-1/2}(1+\log Q) \exp\left((2+o(1)) \frac{\sqrt{\log Q}}{\log\log Q} \right) \|f\|_2,$$

which has the required form after absorbing the logarithmic factor into the $o(1)$.

It remains to estimate the exceptional terms. The exceptional character may depend on M. For every primitive real character χ that occurs at some dyadic scale $M \ge N_0$, let

$$\mathcal{M}(\chi) = \{M \ge N_0 : \chi \text{ is the exceptional character at scale } M\}.$$

Write $q_\chi, \beta_\chi, \sigma_\chi$, and $\mathcal{B}_\chi$ for the corresponding conductor, exceptional zero, and constants from Lemma 3.27, and put

$$D_\chi = \frac{Q}{\mathcal{B}_\chi}.$$

These quantities are fixed when M is restricted to $\mathcal{M}(\chi)$. Let $\mathrm{High}^{\mathrm{exc},\chi}_{Q,M}$ denote the sum of the two exceptional terms in (6.14) when $M \in \mathcal{M}(\chi)$, and set it equal to zero otherwise. We first consider the terms

$$-\mathfrak{s}_{\chi_{q_\chi d}}(a, q_\chi d)\widehat{\widetilde{M}}_{M,\beta_\chi}(\xi - a/(q_\chi d))\eta_s(\xi - a/(q_\chi d)), \qquad d > D_\chi.$$

For the fixed character χ, the condition $q_\chi d \le R(M)$ restricts the sum to $d \le R(M)/q_\chi$, with an upper limit depending on M. Apply Corollary 6.21 with $K = \lfloor R(M) \rfloor$, retaining only denominators $q = q_\chi d$ with $d > D_\chi$, and restrict the time supremum to $M \in \mathcal{M}(\chi)$. At the denominator scale $q_\chi d \sim 2^s$, the resulting factor is

$$(1+s)^2 \ll (1 + \log(q_\chi d))^2.$$

By Lemma 3.27,

$$\begin{aligned}(1 + \log(q_\chi d))^2 |\mathfrak{s}_{\chi_{q_\chi d}}(a, q_\chi d)| &\ll (1 + \log(q_\chi d))^2 \sigma_\chi d^{-1/2} \rho(d) \\ &= Q^{-1/2} \sigma_\chi \mathcal{B}_\chi^{1/2} (1 + \log(q_\chi d))^2 \rho(d) \left(\frac{D_\chi}{d}\right)^{1/2}.\end{aligned}$$

Grouping d into dyadic ranges and using the maximal-order estimate for ρ, the maximal norm of these terms is

$$\begin{aligned}&\ll Q^{-1/2} \sigma_\chi \mathcal{B}_\chi^{1/2} \sum_{2^k \gtrsim D_\chi} (1 + \log q_\chi + k)^2 \sup_{d \sim 2^k} \rho(d) \left(\frac{D_\chi}{2^k}\right)^{1/2} \|f\|_2 \\ &\ll Q^{-1/2} \sigma_\chi \mathcal{B}_\chi^{1/2} (1 + \log q_\chi + \log Q)^{O(1)} \exp\left((2 + o(1)) \frac{\sqrt{\log Q}}{\log\log Q}\right) \|f\|_2.\end{aligned}$$

Indeed, since $D_\chi \le Q$, the ranges $2^k \le Q$ contribute only fixed logarithmic powers. For $2^k > Q$, the factor $(D_\chi/2^k)^{1/2}$ gives geometric decay after including the displayed logarithmic and subexponential factors.

We finally consider the terms

$$-\mathfrak{s}_{\chi_{q_\chi d}}(a, q_\chi d)\widehat{\widetilde{M}}_{M,\beta_\chi}(\xi - a/(q_\chi d))\big(\eta_s - \eta_{\ell_{d,\chi}(M)}\big)(\xi - a/(q_\chi d)), \qquad d \le D_\chi,$$

where

$$16^{-\ell_{d,\chi}(M)} = \frac{Q}{\mathcal{B}_\chi d M^2}.$$

Fix d, a, put $\theta = \xi - a/(q_\chi d)$, and let $M_1 \in \mathcal{M}(\chi)$ be the first scale for which the cutoff difference is nonzero. If there is no such scale, the corresponding term vanishes. Otherwise,

$$|\theta| \gg \frac{Q}{\mathcal{B}_\chi d M_1^2}.$$

Using (6.3), $M^{\beta_\chi - 1} \le 1$, and $M_1 \ge N_0$, we obtain

$$\sum_{\substack{M \in \mathcal{M}(\chi) \\ M \ge M_1}} |\widehat{\widetilde{M}}_{M,\beta_\chi}(\theta)|^2 \le \sum_{\substack{M \ge M_1 \\ M \in 2^{\mathbb{N}}}} |\widehat{\widetilde{M}}_{M,\beta_\chi}(\theta)|^2 \ll \frac{\mathcal{B}_\chi d}{Q} + M_1^{-2} \ll \frac{\mathcal{B}_\chi d}{Q}.$$

In the last inequality we used $\mathcal{B}_\chi d \ge 1$ and $Q \ll N_0^2$.

The square-function and Plancherel argument therefore bounds the maximal norm at each dyadic d-scale by

$$\sup_{\substack{d\sim 2^k\\ a\in\mathbb{A}_{q_\chi d}}} |\mathfrak{s}_{\chi_{q_\chi d}}(a,q_\chi d)|\left(\frac{\mathcal{B}_\chi d}{Q}\right)^{1/2}\|f\|_2 \ll Q^{-1/2}\sigma_\chi \mathcal{B}_\chi^{1/2}\sup_{d\sim 2^k}\rho(d)\,\|f\|_2.$$

Since $d\le D_\chi\le Q$, summing the dyadic d-scales gives

$$\ll Q^{-1/2}\sigma_\chi\mathcal{B}_\chi^{1/2}(1+\log Q)\exp\left((2+o(1))\frac{\sqrt{\log Q}}{\log\log Q}\right)\|f\|_2.$$

Combining the two exceptional contributions, we have proved

$$\left\|\sup_{M\in\mathcal{M}(\chi)}|\operatorname{High}^{\mathrm{exc},\chi}_{Q,M}f|\right\|_2 \ll Q^{-1/2}\sigma_\chi\mathcal{B}_\chi^{1/2}(1+\log q_\chi+\log Q)^{O(1)}\exp\left((2+o(1))\frac{\sqrt{\log Q}}{\log\log Q}\right)\|f\|_2. \tag{6.26}$$

At each scale M, at most one exceptional character occurs. Consequently, pointwise,

$$\sup_{M\ge N_0}|\operatorname{High}^{\mathrm{exc}}_{Q,M}f|^2\le\sum_\chi\sup_{M\in\mathcal{M}(\chi)}|\operatorname{High}^{\mathrm{exc},\chi}_{Q,M}f|^2. \tag{6.27}$$

If $q_\chi\sim 2^j$, then, after decreasing the absolute constant in the definition of the Page region, the zero of χ lies in the exceptional region for the modulus bound 2^{j+1}. The Landau–Page theorem therefore permits at most one such character in each dyadic conductor range. By the stronger estimate in Lemma 3.27,

$$\sum_\chi\left(\sigma_\chi\mathcal{B}_\chi^{1/2}(1+\log q_\chi+\log Q)^{O(1)}\right)^2\ll\sum_{j\ge0}2^{-2j/25}(1+j+\log Q)^{O(1)}\ll(1+\log Q)^{O(1)}. \tag{6.28}$$

Combining (6.27) with (6.26), and then using (6.28) gives the required exceptional contribution. The remaining logarithmic factor is absorbed into the $o(1)$, proving (6.25). □

Proposition 6.16, Proposition 6.23, Lemma 6.10, and the exact identity (6.15) prove Proposition 4.19. This completes the proofs of Theorems 1.1 and 1.3.

7. Quantitative maximal estimates near L^1

This section proves the distributional estimate underlying Theorem 1.5(ii). For $C>0$ and $t\ge0$, define

$$\Psi_C(t)=\int_0^t\exp\left(C\sqrt{\log(e+u)\log(e+\log(e+u))}\right)du.$$

Then Ψ_C is an increasing convex function and

$$\Psi_C(t)\asymp_C t\exp\left(C\sqrt{\log(e+t)\log(e+\log(e+t))}\right). \tag{7.1}$$

In particular, $\Psi_C(2t)\ll_C\Psi_C(t)$ and $\Psi_C(t)\ge t$. For $N\ge2$, define the unweighted prime-square averages by

$$A_{N,\mathbb{P}^2}f(x)=\frac{1}{\pi(N)}\sum_{p\le N}f(x-p^2).$$

Theorem 7.2 (Orlicz maximal estimate)**.** *There are absolute constants $C, C_0 \geq 1$ such that the following holds. Choose A_N from $A_{N,\Lambda}$, $A_{N,\mathbb{N}^2}$, and $A_{N,\mathbb{P}^2}$, and put $A^*f = \sup_{N\geq 2} |A_N f|$. Then, for every finitely supported $f : \mathbb{Z} \to \mathbb{C}$ and every $\lambda > 0$,*

$$\big|\{x \in \mathbb{Z} : A^*f(x) > \lambda\}\big| \ll \sum_{\substack{x\in\mathbb{Z} \\ |f(x)|>\lambda/C_0}} \Psi_C\left(\frac{|f(x)|}{\lambda}\right). \tag{7.3}$$

Consequently,

$$\big|\{x \in \mathbb{Z} : A^*f(x) > \lambda\}\big| \ll \sum_{x\in\mathbb{Z}} \Psi_C\left(\frac{|f(x)|}{\lambda}\right). \tag{7.4}$$

In particular, for every finite $F \subset \mathbb{Z}$ and $t \geq 1$,

$$\big|\{x \in \mathbb{Z} : A^*\mathbf{1}_F(x) > t^{-1}\}\big| \ll |F|\Psi_C(t). \tag{7.5}$$

The truncated estimate gives the following quantitative $\ell^p(\mathbb{Z})$ bound.

Corollary 7.6. *For $1 < p \leq 2$ and $f \in \ell^p(\mathbb{Z})$,*

$$\|A^*f\|_{\ell^p} \leq \exp\left(\frac{C}{p-1}\log\frac{e}{p-1}\right)\|f\|_{\ell^p}, \tag{7.7}$$

after enlarging the absolute constant C.

Proof. By (7.3) and the layer-cake formula,

$$\begin{aligned}\|A^*f\|_{\ell^p}^p &= p\int_0^\infty \lambda^{p-1}\big|\{A^*f > \lambda\}\big|\,d\lambda \\ &\ll p\sum_x \int_0^{C_0|f(x)|} \lambda^{p-1}\Psi_C\left(\frac{|f(x)|}{\lambda}\right)\,d\lambda \\ &= p\|f\|_{\ell^p}^p \int_{1/C_0}^\infty \Psi_C(u)u^{-p-1}\,du.\end{aligned}$$

Put $\varepsilon = p-1$. By (7.1) and the substitution $u = e^v$,

$$\int_{1/C_0}^\infty \Psi_C(u)u^{-p-1}\,du \ll 1 + \int_0^\infty \exp\left(-\varepsilon v + C\sqrt{(1+v)\log(e+v)}\right)\,dv.$$

For $v \geq 0$,

$$C\sqrt{(1+v)\log(e+v)} \leq \frac{\varepsilon v}{2} + \frac{C_1}{\varepsilon}\log\frac{e}{\varepsilon},$$

with an absolute $C_1 > 0$. Hence

$$\int_{1/C_0}^\infty \Psi_C(u)u^{-p-1}\,du \ll \varepsilon^{-1}\exp\left(\frac{C_1}{\varepsilon}\log\frac{e}{\varepsilon}\right).$$

Taking p-th roots and enlarging C proves (7.7) for finitely supported f. Truncation and Fatou's lemma extend the estimate to every $f \in \ell^p(\mathbb{Z})$. □

Next, we bound the maximal operator associated with a Ionescu–Wainger model, then compare it with the annular arithmetic averages in $\ell^2(\mathbb{Z})$. Balancing these estimates gives the restricted bound.

Proposition 7.8. *Let $r, k \in \mathbb{N}$, with $r \geq 2$ and $k \geq 4$, and let $Q \geq 2^k$. Put*

$$\begin{aligned} p_r &= \frac{2r}{2r-1}, & B &= C_*^{k^2Q^{1/(k-1/2)}}, \\ \varepsilon &= (2^{20}rB^{2r+8})^{-1}, & C_{r,k} &= (C_*r\log(2+k))^{C_*k}, \end{aligned} \tag{7.9}$$

where $C_* > 1$ is a sufficiently large absolute constant. Then there exist an integer $P \le 3^Q$ and a Ionescu–Wainger set $\Sigma = \Sigma_{\le k} \subset P^{-1}\mathbb{Z}/\mathbb{Z}$ with the following properties.

(i) *Every rational point of denominator at most Q belongs to Σ:*

$$\bigcup_{1\le q\le Q} \left\{ \frac{a}{q} \pmod 1 : a \in \mathbb{Z} \right\} \subseteq \Sigma. \tag{7.10}$$

Every element of Σ, when written in reduced form, has denominator at most B. With the choice of ε in (7.9), the corresponding major-arc parameter set is $(r, \frac{1}{2})$-good.

(ii) *Let ν be a complex measure on $\mathbb{Z}_P$ with $\|\nu\|_{\mathrm{TV}} \le 1$, and let m be a multiplier on $\mathbb{R}$ supported in $[-\varepsilon, \varepsilon]$. Then, for $p_r \le p \le 2r$,*

$$\left\| T \sum_{\alpha\in\Sigma} \widehat{\nu}(\alpha) m(\cdot - \alpha) \right\|_{\ell^p(\mathbb{Z})\to\ell^p(\mathbb{Z})} \le C_{r,k} \|T_m\|_{L^p(\mathbb{R})\to L^p(\mathbb{R})}, \tag{7.11}$$

where the translates of m are periodized on $\mathbb{T}$.

(iii) *Put $\Omega = [-\varepsilon, \varepsilon] \times \Sigma \subset \mathbb{R} \times \mathbb{Q}/\mathbb{Z}$. If $f \in \ell^p(\mathbb{Z})$ has Fourier support in $\pi(\Omega)$, then*

$$C_{r,k}^{-1} \|f\|_{\ell^p(\mathbb{Z})} \le \|\mathcal{S}_\Omega^{-1} f\|_{L^p(\mathbb{A}_\mathbb{Z})} \le C_{r,k} \|f\|_{\ell^p(\mathbb{Z})} \tag{7.12}$$

for $p_r \le p \le 2r$.

(iv) *Let V be a finite-dimensional normed space and $0 < \delta \le (100P)^{-1}$. If $G : \mathbb{R} \times \mathbb{Z}_P \to V$ has real Fourier support in $[-\delta, \delta]$, then, for $1 \le p \le \infty$,*

$$\left\| \big(G(n, n \bmod P)\big)_{n\in\mathbb{Z}} \right\|_{\ell^p(\mathbb{Z};V)} \asymp \|G\|_{L^p(\mathbb{R}\times\mathbb{Z}_P;V)}, \tag{7.13}$$

with absolute implied constants.

Proof. Apply [49, Lemma 5.1] with $\rho = \frac{2}{k-\frac{1}{2}}$. Then

$$\left\lfloor \frac{2}{\rho} \right\rfloor + 1 = k, \qquad Q^{\rho/2} = Q^{1/(k-1/2)}.$$

The lemma produces a set of pairwise coprime integers defining $\Sigma_{\le k}$, together with an integer $P \le 3^Q$, such that (7.10) holds. Its constituent coprime integers are at most $B^{1/k}$, after enlarging C_*, and therefore every denominator occurring in $\Sigma_{\le k}$ is at most B. The choice of ε in (7.9) satisfies the goodness condition in [49, (5.1)]. This proves (i).

For (ii), define the adelic multiplier

$$M(\theta, \alpha) = m(\theta)\widehat{\nu}(\alpha), \qquad (\theta, \alpha) \in \mathbb{R} \times P^{-1}\mathbb{Z}/\mathbb{Z}.$$

Extend M by zero when $\alpha \notin P^{-1}\mathbb{Z}/\mathbb{Z}$. The associated operator factors into its real and finite-group parts. Indeed, if $F \in L^p(\mathbb{R} \times \mathbb{Z}_P)$, then

$$T_M F(x, z) = \int_{\mathbb{Z}_P} T_m\big(F(\cdot, z - y)\big)(x)\, d\nu(y).$$

Minkowski's inequality and translation invariance on $\mathbb{Z}_P$ therefore give

$$\|T_M F\|_{L^p(\mathbb{R}\times\mathbb{Z}_P)} \le \int_{\mathbb{Z}_P} \big\|T_m\big(F(\cdot, \cdot - y)\big)\big\|_{L^p(\mathbb{R}\times\mathbb{Z}_P)}\, d|\nu|(y) \le \|T_m\|_{L^p(\mathbb{R})\to L^p(\mathbb{R})} \|\nu\|_{\mathrm{TV}} \|F\|_{L^p(\mathbb{R}\times\mathbb{Z}_P)}. \tag{7.14}$$

For a general $F \in L^p(\mathbb{A}_\mathbb{Z})$, define

$$F_P(x, z) = \int_{\widehat{\mathbb{Z}}} F(x, z + Pu)\, du, \qquad z \in \mathbb{Z}_P,$$

where du is normalized Haar measure and any integer representative of z may be used. Jensen's inequality gives

$$\|F_P\|_{L^p(\mathbb{R}\times\mathbb{Z}_P)} \le \|F\|_{L^p(\mathbb{A}_\mathbb{Z})}.$$

This averaging preserves the Fourier coefficients at $\alpha \in \mathbb{T}[P]$ and removes the remaining arithmetic frequencies. Since M vanishes outside these frequencies, $T_M F$ is the lift of $T_M F_P$, and hence

$$\|T_M F\|_{L^p(\mathbb{A}_\mathbb{Z})} = \|T_M F_P\|_{L^p(\mathbb{R}\times\mathbb{Z}_P)}.$$

Using (7.14) and $\|\nu\|_{\mathrm{TV}} \le 1$, we obtain

$$\|T_M\|_{L^p(\mathbb{A}_\mathbb{Z})\to L^p(\mathbb{A}_\mathbb{Z})} \le \|T_m\|_{L^p(\mathbb{R})\to L^p(\mathbb{R})}. \tag{7.15}$$

The multiplier on $\mathbb{T}$ obtained from M and the rational frequency set Σ is

$$M_\Sigma^\mathbb{Z}(\xi) = \sum_{\alpha\in\Sigma} M(\xi-\alpha,\alpha) = \sum_{\alpha\in\Sigma} \widehat{\nu}(\alpha) m(\xi-\alpha).$$

Hence the adelic Ionescu–Wainger multiplier theorem [49, Theorem 1.7] and (7.15) give

$$\|T_{M_\Sigma^\mathbb{Z}}\|_{\ell^p(\mathbb{Z})\to\ell^p(\mathbb{Z})} \le (r\log(2+k))^{O(k)} \|T_M\|_{L^p(\mathbb{A}_\mathbb{Z})\to L^p(\mathbb{A}_\mathbb{Z})} \le C_{r,k}\|T_m\|_{L^p(\mathbb{R})\to L^p(\mathbb{R})}.$$

This is (7.11).

Estimate (7.12) is the major-arc sampling theorem [49, Theorem 1.6], again with its constant bounded by $C_{r,k}$. This proves (iii).

For (iv), expand G in the finite variable:

$$G(x,z) = \sum_{a\in\mathbb{Z}_P} G_a(x) e(az/P), \qquad \operatorname{supp}\widehat{G_a} \subseteq [-\delta,\delta].$$

Its restriction to the diagonal $n \mapsto (n, n \bmod P)$ is therefore

$$G(n, n \bmod P) = \sum_{a\in\mathbb{Z}_P} G_a(n) e(an/P).$$

Thus the Fourier transform of the a-th summand is supported in $\frac{a}{P} + [-\delta,\delta] \subset \mathbb{T}$. If $a \ne a'$, then

$$\left\| \frac{a}{P} - \frac{a'}{P} \right\|_\mathbb{T} \ge \frac{1}{P} > 2\delta.$$

Hence these intervals are pairwise disjoint. Equivalently, the map

$$\pi : [-\delta,\delta]\times\mathbb{T}[P] \longrightarrow \mathbb{T}, \qquad \pi(\theta,\alpha) = \theta+\alpha \pmod 1,$$

is injective. Put

$$\Omega_P = \left[-\frac{1}{100P}, \frac{1}{100P}\right] \times \mathbb{T}[P].$$

Since $\delta \le (100P)^{-1}$, the Fourier support of G is contained in Ω_P, and its sampled function is precisely $\mathcal{S}_{\Omega_P} G(n) = G(n, n \bmod P)$. The quantitative Shannon sampling theorem [49, Proposition 1.5], applied with $Q = P$, $c = 1/100$, and target space V, now gives

$$\left\| \big(G(n, n \bmod P)\big)_{n\in\mathbb{Z}} \right\|_{\ell^p(\mathbb{Z};V)} \asymp \|G\|_{L^p(\mathbb{R}\times\mathbb{Z}_P;V)},$$

with constants independent of P, δ, and G. This is (7.13). □

The following standard continuous estimates are included for completeness and to record their uniformity in β.

Lemma 7.16. *Uniformly in $0 < \beta \le 1$, $J \ge 1$, and $|\sigma_j| \le 1$, one has, for $1 < p \le 2$,*

$$\left\| T_{\sum_{j=1}^{J} \sigma_j (\widehat{\widetilde{M}}_{2^j,\beta} - \widehat{\widetilde{M}}_{2^{j-1},\beta})} f \right\|_{L^p(\mathbb{R})} \ll \frac{1}{p-1} \|f\|_{L^p(\mathbb{R})}, \tag{7.17}$$

$$\left\| \sup_{N \ge 2} |\widetilde{M}_{N,\beta} * f| \right\|_{L^p(\mathbb{R})} \ll \frac{1}{p-1} \|f\|_{L^p(\mathbb{R})}. \tag{7.18}$$

Here $\widetilde{M}_{N,\beta}$ is the continuous annular measure defined in (6.2). Since (7.18) holds with the supremum over all $N \ge 2$, it also holds when the supremum is restricted to dyadic N.

Proof. The change of variables $x = t^2$ in (6.2) gives

$$\widetilde{M}_{N,\beta}(x) = \frac{x^{\beta/2-1}}{2N} \mathbf{1}_{[N^2/4, N^2]}(x). \tag{7.19}$$

On the Fourier side, setting $t = 2^j v$ in (6.2) gives

$$\widehat{\widetilde{M}}_{2^j,\beta}(\theta) = 2^{j(\beta-1)} F_\beta(4^j\theta), \qquad F_\beta(u) = \int_{1/2}^{1} v^{\beta-1} e(-uv^2)\, dv.$$

Integration by parts, together with subtraction at $u = 0$, gives uniformly in β

$$|F_\beta(u)| \ll \min\{1, |u|^{-1}\}, \qquad |F_\beta(u) - F_\beta(u/4)| \ll \min\{|u|, |u|^{-1}\}.$$

For each j, adding and subtracting $2^{(j-1)(\beta-1)} F_\beta(4^j\theta)$ gives

$$\begin{aligned} &\left|2^{j(\beta-1)} F_\beta(4^j\theta) - 2^{(j-1)(\beta-1)} F_\beta(4^{j-1}\theta)\right| \\ &\le \left|2^{j(\beta-1)} - 2^{(j-1)(\beta-1)}\right| |F_\beta(4^j\theta)| + 2^{(j-1)(\beta-1)} \left|F_\beta(4^j\theta) - F_\beta(4^{j-1}\theta)\right|. \end{aligned}$$

Since $2^{(j-1)(\beta-1)} \le 1$, it follows that

$$\sum_{j=1}^{J} \left|\widehat{\widetilde{M}}_{2^j,\beta}(\theta) - \widehat{\widetilde{M}}_{2^{j-1},\beta}(\theta)\right| \ll \sum_{j=1}^{J} \left|2^{j(\beta-1)} - 2^{(j-1)(\beta-1)}\right| + \sum_{j=1}^{J} \min\{|4^j\theta|, |4^j\theta|^{-1}\} \ll 1. \tag{7.20}$$

Let K be the convolution kernel of the operator in (7.17). Estimate (7.20) and Plancherel's theorem give

$$\|K * f\|_{L^2(\mathbb{R})} \ll \|f\|_{L^2(\mathbb{R})}. \tag{7.21}$$

Collecting equal kernels, we have

$$K = -\sigma_1 \widetilde{M}_{1,\beta} + \sum_{j=1}^{J-1} (\sigma_j - \sigma_{j+1}) \widetilde{M}_{2^j,\beta} + \sigma_J \widetilde{M}_{2^J,\beta}.$$

Thus every $\widetilde{M}_{2^j,\beta}$ occurs with a coefficient of modulus at most 2. If $4^{j-1} < x < 4^j$, then (7.19) and $0 < \beta \le 1$ give

$$|K(x)| \ll \frac{x^{\beta/2-1}}{2^j} = \frac{1}{x} \frac{x^{\beta/2}}{2^j} \ll \frac{1}{x}, \qquad |K'(x)| \ll \frac{x^{\beta/2-2}}{2^j} = \frac{1}{x^2} \frac{x^{\beta/2}}{2^j} \ll \frac{1}{x^2},$$

At an endpoint $x = 4^j$, the jump of K is bounded by

$$\left|K(4^j+) - K(4^j-)\right| \ll \frac{(4^j)^{\beta/2-1}}{2^j} \ll 4^{-j}.$$

Consequently,

$$\int_{|x|>2|h|} |K(x-h) - K(x)|\, dx \ll |h| \int_{2|h|}^{\infty} \frac{dx}{x^2} + |h| \sum_{4^j \gtrsim |h|} 4^{-j} \ll 1.$$

Thus K is a Calderón–Zygmund kernel, with its L^2 and kernel constants bounded uniformly in J, β, and (σ_j). The Calderón–Zygmund weak endpoint estimate gives

$$\big|\{x \in \mathbb{R} : |K * f(x)| > \lambda\}\big| \ll \lambda^{-1}\|f\|_{L^1(\mathbb{R})}.$$

For $1 < p \le 3/2$, Marcinkiewicz interpolation with (7.21) gives

$$\|K * f\|_{L^p(\mathbb{R})} \ll (p-1)^{-1}\|f\|_{L^p(\mathbb{R})}.$$

For $3/2 \le p \le 2$, interpolation between the $L^{3/2}$ and L^2 estimates gives

$$\|K * f\|_{L^p(\mathbb{R})} \ll \|f\|_{L^p(\mathbb{R})} \ll (p-1)^{-1}\|f\|_{L^p(\mathbb{R})}.$$

This proves (7.17).

By (7.19), we have $0 \le \widetilde{M}_{N,\beta}(x) \ll N^{-2}\mathbf{1}_{[0,N^2]}(x)$, and hence

$$\sup_{N\ge 1} |\widetilde{M}_{N,\beta} * f(x)| \ll M^+_{\mathrm{HL}} f(x).$$

The Hardy–Littlewood estimate recalled in Section 2 proves (7.18). □

Proposition 7.22. *Let $r, k, Q, P, \Sigma_{\le k}, B, \varepsilon, p_r$, and $C_{r,k}$ be as in Proposition 7.8. Let ν be a complex measure on $\mathbb{Z}_P$ with $\|\nu\|_{\mathrm{TV}} \le 1$, and let $0 < \beta \le 1$. Define $L^{\nu,\beta}_N$ by*

$$\widehat{L^{\nu,\beta}_N}(\xi) = \sum_{\alpha \in \Sigma_{\le k}} \widehat{\nu}(\alpha)\widehat{\widetilde{M}}_{N,\beta}(\xi - \alpha)\eta\left(\frac{\xi - \alpha}{\varepsilon}\right).$$

Then

$$\Big\| \sup_{\substack{N\ge 1\\ N\in 2^{\mathbb{N}}}} |L^{\nu,\beta}_N f| \Big\|_{\ell^{p_r}(\mathbb{Z})} \ll rC^2_{r,k} \log\big(e + Q + r\log B\big)\|f\|_{\ell^{p_r}(\mathbb{Z})}.$$

The implied constant is independent of ν and β.

Proof. With ε as in (7.9), set

$$\delta = \min\left\{\frac{\varepsilon}{16}, \frac{1}{100P}\right\}, \qquad J_* = \left\lceil \frac{1}{2}\log_2 \delta^{-1} \right\rceil, \qquad N_* = 2^{J_*}.$$

Thus $1 \le N_*\sqrt{\delta} < 2$. For the scales $N \le N_*$, write $L_j = L^{\nu,\beta}_{2^j}$, $0 \le j \le J_*$. For arbitrary $|\sigma_j| \le 1$, Proposition 7.8, applied to the continuous multiplier

$$\sum_{j=1}^{J_*} \sigma_j \left(\widehat{\widetilde{M}}_{2^j,\beta} - \widehat{\widetilde{M}}_{2^{j-1},\beta}\right)\eta(\,\cdot\,/\varepsilon),$$

and Lemma 7.16 give

$$\Big\| \sum_{j=1}^{J_*} \sigma_j (L_j - L_{j-1}) f \Big\|_{\ell^{p_r}(\mathbb{Z})} \ll C_{r,k}(p_r - 1)^{-1}\|f\|_{\ell^{p_r}(\mathbb{Z})} \ll rC_{r,k}\|f\|_{\ell^{p_r}(\mathbb{Z})}. \tag{7.23}$$

Here we also used that the inverse Fourier transform of $\eta(\,\cdot\,/\varepsilon)$ has uniformly bounded $L^1(\mathbb{R})$-norm. The same multiplier estimate, without taking differences, gives

$$\|L_0 f\|_{\ell^{p_r}(\mathbb{Z})} \ll C_{r,k}\|f\|_{\ell^{p_r}(\mathbb{Z})}.$$

The Rademacher–Menshov inequality [41, (2.7)] gives

$$\Big\| \max_{1\le m\le J_*} \Big| \sum_{j=1}^{m} (L_j - L_{j-1}) f \Big| \Big\|_{\ell^{p_r}(\mathbb{Z})} \ll \sum_{\ell \le \lceil \log_2 J_* \rceil} \Big\| \Big(\sum_{I\in\mathcal{D}_\ell} \Big| \sum_{j\in I} (L_j - L_{j-1}) f \Big|^2 \Big)^{1/2} \Big\|_{\ell^{p_r}(\mathbb{Z})}.$$

Here $\mathcal{D}_\ell$ consists of the nonempty intersections of $\{1,\dots,J_*\}$ with the blocks $(m2^\ell,(m+1)2^\ell]$, $m\ge 0$. For fixed ℓ, let $(\epsilon_I)_{I\in\mathcal{D}_\ell}$ be independent Rademacher variables and set $\sigma_j=\epsilon_I$ for the unique $I\in\mathcal{D}_\ell$ containing j. Khintchine's inequality, Fubini's theorem, and (7.23) give

$$\begin{aligned}\left\|\left(\sum_{I\in\mathcal{D}_\ell}\left|\sum_{j\in I}(L_j-L_{j-1})f\right|^2\right)^{1/2}\right\|_{\ell^{p_r}(\mathbb{Z})} &\ll \left(\mathbb{E}_\epsilon\left\|\sum_{j=1}^{J_*}\sigma_j(L_j-L_{j-1})f\right\|_{\ell^{p_r}(\mathbb{Z})}^{p_r}\right)^{1/p_r}\\ &\ll rC_{r,k}\|f\|_{\ell^{p_r}(\mathbb{Z})}.\end{aligned}$$

Summing over ℓ and including the estimate for L_0 yields

$$\left\|\max_{0\le j\le J_*}|L_jf|\right\|_{\ell^{p_r}(\mathbb{Z})}\ll rC_{r,k}\log(2+J_*)\|f\|_{\ell^{p_r}(\mathbb{Z})}. \tag{7.24}$$

For $N\ge N_*$, introduce the narrower cutoff $\kappa(\theta)=\eta(\theta/\delta)$, and define

$$F(x,y)=\sum_{\alpha\in\Sigma_{\le k}}\int_{\mathbb{R}}\eta\left(\frac{\theta}{\varepsilon}\right)\widehat{f}(\alpha+\theta)e(x\theta+y\alpha)\,d\theta.$$

Estimate (7.12), followed by (7.11) with $m(\theta)=\eta(\theta/\varepsilon)$ and $\nu=\delta_0$, gives

$$\|F\|_{L^{p_r}(\mathbb{R}\times\mathbb{Z}_P)}\ll C_{r,k}\left\|\left(\sum_{\alpha\in\Sigma_{\le k}}\eta\left(\frac{\xi-\alpha}{\varepsilon}\right)\widehat{f}(\xi)\right)^\vee\right\|_{\ell^{p_r}(\mathbb{Z})}\ll C_{r,k}^2\|f\|_{\ell^{p_r}(\mathbb{Z})}. \tag{7.25}$$

For a finite set $\mathcal{N}$ of dyadic $N\ge N_*$, define

$$G_N=\big((\widetilde{M}_{N,\beta}*_{\mathbb{R}}\kappa^\vee)\otimes\nu\big)*_{\mathbb{R}\times\mathbb{Z}_P}F,\qquad N\in\mathcal{N}.$$

Equivalently, $\widehat{G}_N(\theta,\alpha)=\widehat{\widetilde{M}}_{N,\beta}(\theta)\kappa(\theta)\widehat{\nu}(\alpha)\widehat{F}(\theta,\alpha)$. Since convolution by $\kappa^\vee$ and by ν has uniformly bounded L^{p_r}-operator norm, Lemma 7.16, applied in the real variable, gives

$$\left\|\sup_{N\in\mathcal{N}}|G_N|\right\|_{L^{p_r}(\mathbb{R}\times\mathbb{Z}_P)}\ll r\|F\|_{L^{p_r}(\mathbb{R}\times\mathbb{Z}_P)}.$$

The real Fourier support of every G_N is contained in $[-\delta,\delta]$. Apply (7.13) to the $\ell^\infty(\mathcal{N})$-valued function $(G_N)_{N\in\mathcal{N}}$. Together with (7.25), this gives

$$\left\|\sup_{N\in\mathcal{N}}|G_N(n,n\bmod P)|\right\|_{\ell^{p_r}(\mathbb{Z})}\ll r\|F\|_{L^{p_r}(\mathbb{R}\times\mathbb{Z}_P)}\ll rC_{r,k}^2\|f\|_{\ell^{p_r}(\mathbb{Z})}. \tag{7.26}$$

Because $\delta\le\varepsilon/16$, $\eta(\theta/\varepsilon)=1$ on $\operatorname{supp}\kappa$. Consequently,

$$G_N(n,n\bmod P)=\left(\sum_{\alpha\in\Sigma_{\le k}}\widehat{\nu}(\alpha)\widehat{\widetilde{M}}_{N,\beta}(\xi-\alpha)\kappa(\xi-\alpha)\widehat{f}(\xi)\right)^\vee(n).$$

Thus (7.26) controls the part of $L_N^{\nu,\beta}f$ carrying the common cutoff κ. Letting $\mathcal{N}$ increase to all dyadic $N\ge N_*$ gives the same bound for the full supremum of these terms.

It remains to estimate the difference between the original cutoff and κ. Since $\eta(\theta/\varepsilon)\kappa(\theta)=\kappa(\theta)$, this difference has continuous multiplier

$$R_N(\theta)=\widehat{\widetilde{M}}_{N,\beta}(\theta)\eta\left(\frac{\theta}{\varepsilon}\right)\big(1-\kappa(\theta)\big).$$

The measures $\widetilde{M}_{N,\beta}$ have uniformly bounded mass, and the inverse Fourier transforms of the cutoff functions have uniformly bounded $L^1(\mathbb{R})$-norm. Hence

$$\|T_{R_N}\|_{L^1(\mathbb{R})\to L^1(\mathbb{R})} \ll 1.$$

Moreover, $1-\kappa(\theta)\neq 0$ implies $|\theta|\geq \delta/2$. The Fourier estimate (6.3) and Plancherel's theorem therefore give

$$\|T_{R_N}\|_{L^2(\mathbb{R})\to L^2(\mathbb{R})} \ll (N\sqrt{\delta})^{-1}.$$

Since $\frac{1}{p_r} = \left(1-\frac{1}{r}\right)+\frac{1}{2r}$, interpolation yields

$$\|T_{R_N}\|_{L^{p_r}(\mathbb{R})\to L^{p_r}(\mathbb{R})} \ll (N\sqrt{\delta})^{-1/r}.$$

Applying the Ionescu–Wainger multiplier estimate in Proposition 7.8, we obtain

$$\left\| \left(\sum_{\alpha\in\Sigma_{\leq k}} \widehat{\nu}(\alpha) R_N(\xi-\alpha)\widehat{f}(\xi) \right)^{\vee} \right\|_{\ell^{p_r}(\mathbb{Z})} \ll C_{r,k}(N\sqrt{\delta})^{-1/r}\|f\|_{\ell^{p_r}(\mathbb{Z})}.$$

Taking the supremum over N by summing these estimates over the dyadic scales and using $N_*\sqrt{\delta}\asymp 1$, we get

$$\left\| \sup_{\substack{N\geq N_*\\ N\in 2^{\mathbb{N}}}} \left| \left(\sum_{\alpha\in\Sigma_{\leq k}} \widehat{\nu}(\alpha) R_N(\xi-\alpha)\widehat{f}(\xi) \right)^{\vee} \right| \right\|_{\ell^{p_r}(\mathbb{Z})} \ll C_{r,k} \sum_{\ell\geq 0} \left(2^{\ell}N_*\sqrt{\delta}\right)^{-1/r}\|f\|_{\ell^{p_r}(\mathbb{Z})} \ll rC_{r,k}\|f\|_{\ell^{p_r}(\mathbb{Z})}. \tag{7.27}$$

Finally, $P\leq 3^Q$ and (7.9) imply

$$\log\delta^{-1} \ll Q + r\log B, \qquad \log(2+J_*) \ll \log\big(e+Q+r\log B\big).$$

Combining (7.24), (7.26), and (7.27) proves

$$\left\| \sup_{\substack{N\geq 1\\ N\in 2^{\mathbb{N}}}} |L_N^{\nu,\beta} f| \right\|_{\ell^{p_r}(\mathbb{Z})} \ll rC_{r,k}^2 \log\big(e+Q+r\log B\big)\|f\|_{\ell^{p_r}(\mathbb{Z})},$$

as required. □

We now compare the rational-frequency operator in Proposition 7.22 with the circle-method model V_N defined in (6.9). Let N_0 be the least dyadic integer such that

$$N_0 \geq \exp\left(C\big((\log Q)^2 + r\log B\big)\right), \tag{7.28}$$

where C is a sufficiently large absolute constant. For every dyadic $N\geq N_0$, the definition of $R=R(N)$ gives

$$R\leq N^{1/8}, \qquad N_0\geq 2B, \qquad \frac{\log(eQ)}{N_0\sqrt{\varepsilon}} \leq Q^{-1}. \tag{7.29}$$

In particular, the hypotheses of Lemma 6.10 hold throughout this range.

Proposition 7.30. *Using the assumptions and notation of Propositions 7.8 and 7.22, put*

$$D(r,k,Q) = rC_{r,k}^2\log(e+Q+r\log B). \tag{7.31}$$

For dyadic $N\geq N_0$, there are decompositions

$$\widetilde{A}_{N,\Lambda} = L_N + H_N, \qquad \widetilde{A}_{N,\mathbb{N}^2} = L_N^{\square} + H_N^{\square},$$

such that

$$\Big\| \sup_{N\ge N_0} |L_N f| \Big\|_{\ell^{pr}(\mathbb{Z})} + \Big\| \sup_{N\ge N_0} |L_N^{\square} f| \Big\|_{\ell^{pr}(\mathbb{Z})} \ll D(r,k,Q)\|f\|_{\ell^{pr}(\mathbb{Z})}, \tag{7.32}$$

$$\Big\| \sup_{N\ge N_0} |H_N f| \Big\|_{\ell^{2}(\mathbb{Z})} + \Big\| \sup_{N\ge N_0} |H_N^{\square} f| \Big\|_{\ell^{2}(\mathbb{Z})} \ll Q^{-1/4}\|f\|_{\ell^{2}(\mathbb{Z})}. \tag{7.33}$$

Proof. All suprema and sums in N below are over dyadic scales. All rational fractions are written in reduced form, and s is chosen so that $2^s \le q < 2^{s+1}$. We write $R = R(N)$. By Lemma 6.10, with $A = 2$,

$$\|\widetilde{A}_{N,\Lambda} - V_N\|_{\ell^2(\mathbb{Z})\to\ell^2(\mathbb{Z})} \ll Q^{-1/2}(1+\log N)^{-2}.$$

Summing over dyadic $N \ge N_0$, we obtain

$$\Big\| \sup_{N\ge N_0} |(\widetilde{A}_{N,\Lambda} - V_N)f| \Big\|_{\ell^2(\mathbb{Z})} \le \sum_{N\ge N_0} \|(\widetilde{A}_{N,\Lambda} - V_N)f\|_{\ell^2(\mathbb{Z})} \ll Q^{-1/2}\|f\|_{\ell^2(\mathbb{Z})}. \tag{7.34}$$

Recall from Section 6 that $\mathcal{M}(\chi)$ is the set of scales at which χ is exceptional, while q_χ and β_χ denote its conductor and exceptional zero. Since $R(N) \ge Q^8$, all exceptional zeros with $q_\chi \le Q$ arising for $N \ge N_0$ lie in a common Page region for the fixed conductor bound Q. Thus (2.5) permits at most one such character. Denote it, when it exists, by χ_*. By (7.10) and $\Sigma_{\le k} \subset P^{-1}\mathbb{Z}/\mathbb{Z}$, its conductor divides P.

For a primitive real character χ with $q_\chi \mid P$, put

$$\nu_P^\chi = \frac{1}{\phi(P)} \sum_{u\in\mathbb{A}_P} \chi(u)\delta_{u^2}.$$

This measure has total variation at most one. For $q \mid P$, reduction from $\mathbb{A}_P$ to $\mathbb{A}_q$ has equal-size fibers. If $q_\chi \mid q$, the character descends to the character χ_q modulo q; otherwise, it has mean zero on each fiber. Consequently,

$$\widehat{\nu_P^\chi}(a/q) = \begin{cases} \mathfrak{s}_{\chi_q}(a,q), & q_\chi \mid q, \\ 0, & q_\chi \nmid q. \end{cases} \tag{7.35}$$

In particular, $\widehat{\nu_P^1}(a/q) = \mathfrak{s}(a,q)$. Define

$$L_N = L_N^{\nu_P^1,1} - \mathbf{1}_{\mathcal{M}(\chi_*)}(N) L_N^{\nu_P^{\chi_*},\beta_{\chi_*}}, \qquad H_N = \widetilde{A}_{N,\Lambda} - L_N, \tag{7.36}$$

with the second term omitted when χ_* does not exist. Proposition 7.22, applied to the two measures above, gives

$$\Big\| \sup_{N\ge N_0} |L_N f| \Big\|_{\ell^{pr}(\mathbb{Z})} \ll D(r,k,Q)\|f\|_{\ell^{pr}(\mathbb{Z})}. \tag{7.37}$$

We now estimate H_N in $\ell^2(\mathbb{Z})$. The exceptional character may vary with N, but every character of conductor at most Q has already been included in (7.36). Let E_N^{exc} denote the contribution to V_N from exceptional characters of conductor greater than Q:

$$\widehat{E_N^{\mathrm{exc}} f}(\xi) = -\sum_{\substack{\chi \\ q_\chi > Q}} \mathbf{1}_{\mathcal{M}(\chi)}(N) \sum_{\substack{q\le R(N) \\ q_\chi \mid q}} \sum_{a\in\mathbb{A}_q} \mathfrak{s}_{\chi_q}(a,q)\widehat{\widetilde{M}}_{N,\beta_\chi}(\xi - a/q)\eta_s(\xi-a/q)\widehat{f}(\xi).$$

For each fixed N, at most one term in the outer sum is nonzero. Moreover, every denominator in this expression exceeds Q, since $q_\chi \mid q$.

To describe the remaining terms, put

$$\mathcal{C}_N(a,q;\theta)=\mathfrak{s}(a,q)\widetilde{\widetilde{M}}_N(\theta)-\mathbf{1}_{\mathcal{M}(\chi_*)}(N)\mathbf{1}_{\{q_{\chi_*}\mid q\}}\mathfrak{s}_{\chi_{*,q}}(a,q)\widetilde{\widetilde{M}}_{N,\beta_{\chi_*}}(\theta).$$

Here $\chi_{*,q}$ is induced by χ_* when $q_{\chi_*}\mid q$; the entire second term is taken to be zero if χ_* does not exist or $q_{\chi_*}\nmid q$. Define

$$\widehat{V_N^{>Q}f}(\xi)=\sum_{Q<q\le R(N)}\sum_{a\in\mathbb{A}_q}\mathcal{C}_N(a,q;\xi-a/q)\eta_s(\xi-a/q)\widehat{f}(\xi),$$

$$\widehat{L_N^{>Q}f}(\xi)=\sum_{\substack{a/q\in\Sigma_{\le k}\\ q>Q}}\mathcal{C}_N(a,q;\xi-a/q)\eta\left(\frac{\xi-a/q}{\varepsilon}\right)\widehat{f}(\xi),$$

$$\widehat{D_Nf}(\xi)=\sum_{q\le Q}\sum_{a\in\mathbb{A}_q}\mathcal{C}_N(a,q;\xi-a/q)\left(\eta_s(\xi-a/q)-\eta\left(\frac{\xi-a/q}{\varepsilon}\right)\right)\widehat{f}(\xi).$$

Thus $V_N^{>Q}$ is the part of $V_N-E_N^{\mathrm{exc}}$ with $q>Q$, and $L_N^{>Q}$ is the corresponding denominator restriction of L_N. For $q\le Q$, every fraction belongs to $\Sigma_{\le k}$ by (7.10), and the arithmetic coefficients agree by (7.35). Their difference is therefore precisely D_N. It follows from (6.9) and (7.36) that

$$H_N=(\widetilde{A}_{N,\Lambda}-V_N)+E_N^{\mathrm{exc}}+V_N^{>Q}-L_N^{>Q}+D_N. \tag{7.38}$$

By Lemmas 3.21 and 3.24,

$$|\mathfrak{s}(a,q)|+|\mathfrak{s}_{\chi_q}(a,q)|\ll q^{-3/8}. \tag{7.39}$$

Fix an exceptional character χ with $q_\chi>Q$. For $N\in\mathcal{M}(\chi)$, the operator E_N^{exc} consists of its contribution, whose coefficients vanish unless $q_\chi\mid q$. Corollary 6.21, together with (7.39), gives

$$\left\|\sup_{\substack{N\ge N_0\\ N\in\mathcal{M}(\chi)}}|E_N^{\mathrm{exc}}f|\right\|_{\ell^2(\mathbb{Z})}\ll\sum_{2^s\ge q_\chi/2}(1+s)^2\,2^{-3s/8}\|f\|_{\ell^2(\mathbb{Z})}\ll q_\chi^{-3/8}\log(eq_\chi)^2\|f\|_{\ell^2(\mathbb{Z})}.$$

At each N, at most one exceptional character occurs. The Landau–Page argument preceding (6.28) also permits at most one in each dyadic conductor range. Hence

$$\begin{aligned}\left\|\sup_{N\ge N_0}|E_N^{\mathrm{exc}}f|\right\|_{\ell^2(\mathbb{Z})}^2&\le\sum_{\substack{\chi\\ q_\chi>Q}}\left\|\sup_{\substack{N\ge N_0\\ N\in\mathcal{M}(\chi)}}|E_N^{\mathrm{exc}}f|\right\|_{\ell^2(\mathbb{Z})}^2\\&\ll\sum_{2^{j+1}>Q}2^{-3j/4}(1+j)^4\|f\|_{\ell^2(\mathbb{Z})}^2\\&\ll Q^{-3/4}\log(eQ)^4\|f\|_{\ell^2(\mathbb{Z})}^2\ll Q^{-1/2}\|f\|_{\ell^2(\mathbb{Z})}^2.\end{aligned} \tag{7.40}$$

Both $V_N^{>Q}$ and $L_N^{>Q}$ contain ordinary terms and, possibly, χ_*-terms with $q>Q$. For $V_N^{>Q}$, Corollary 6.21 allows the N-dependent upper bound $q\le R(N)$. For $L_N^{>Q}$, every denominator is at most B, and the choice of ε in (7.9) gives

$$\operatorname{supp}\eta(\,\cdot\,/\varepsilon)\subseteq\{\theta:\eta_s(\theta)=1\},\qquad q\sim 2^s,\quad q\le B.$$

Since $N_0\ge 2B$ by (7.29), the filtered form of the same corollary applies. For the ordinary terms use $\beta=1$; for the exceptional terms use $\beta=\beta_{\chi_*}$ and restrict N to $\mathcal{M}(\chi_*)$. The coefficient

bound (7.39) and

$$\sum_{2^s>Q/2} (1+s)^2\, 2^{-3s/8} \ll Q^{-3/8}\log(eQ)^2 \ll Q^{-1/4}$$

therefore give

$$\Big\| \sup_{N\ge N_0} |V_N^{>Q} f| \Big\|_{\ell^2(\mathbb{Z})} \ll Q^{-1/4}\|f\|_{\ell^2(\mathbb{Z})}, \tag{7.41}$$

$$\Big\| \sup_{N\ge N_0} |L_N^{>Q} f| \Big\|_{\ell^2(\mathbb{Z})} \ll Q^{-1/4}\|f\|_{\ell^2(\mathbb{Z})}. \tag{7.42}$$

It remains to estimate the cutoff difference D_N. By (6.3) and (7.39), uniformly for $q \le Q$ and $0 < |\theta| \le 1/2$,

$$|\mathcal{C}_N(a,q;\theta)| \ll \frac{1}{N\sqrt{|\theta|}}.$$

Consequently, summing the dyadic geometric series gives

$$\sum_{N\ge N_0} |\mathcal{C}_N(a,q;\theta)|^2 \ll \frac{1}{|\theta|}\sum_{N\ge N_0}\frac{1}{N^2} \ll \frac{1}{N_0^2|\theta|}.$$

The cutoff difference vanishes for $|\theta| < \varepsilon/2$, and its support is contained in that of η_s. Let $D_{N,s}$ denote the part of D_N with $q \sim 2^s$. The rational-frequency supports are disjoint at this scale. The pointwise bound of the supremum by the square function and Plancherel's theorem therefore give

$$\begin{aligned}
&\Big\| \sup_{N\ge N_0} |D_{N,s} f| \Big\|_{\ell^2(\mathbb{Z})}^2 \le \sum_{N\ge N_0} \|D_{N,s} f\|_{\ell^2(\mathbb{Z})}^2 \\
&= \int_{\mathbb{T}} \sum_{\substack{q\sim 2^s \\ q\le Q}} \sum_{a\in\mathbb{A}_q} \left| \eta_s(\xi - a/q) - \eta\left(\frac{\xi - a/q}{\varepsilon}\right)\right|^2 \sum_{N\ge N_0} |\mathcal{C}_N(a,q;\xi-a/q)|^2 |\widehat{f}(\xi)|^2\, d\xi \\
&\ll \frac{1}{N_0^2\varepsilon}\|f\|_{\ell^2(\mathbb{Z})}^2.
\end{aligned}$$

Taking square roots and summing over the $O(\log(eQ))$ denominator scales, we obtain

$$\Big\| \sup_{N\ge N_0} |D_N f| \Big\|_{\ell^2(\mathbb{Z})} \ll \frac{\log(eQ)}{N_0\sqrt{\varepsilon}}\|f\|_{\ell^2(\mathbb{Z})} \ll Q^{-1}\|f\|_{\ell^2(\mathbb{Z})}, \tag{7.43}$$

where the last inequality is (7.29). This estimate includes the possible χ_*-contribution through $\mathcal{C}_N$. Combining (7.38) with (7.34), (7.40), (7.41), (7.42), and (7.43), we conclude that

$$\Big\| \sup_{N\ge N_0} |H_N f| \Big\|_{\ell^2(\mathbb{Z})} \ll Q^{-1/4}\|f\|_{\ell^2(\mathbb{Z})}.$$

For the square averages, define

$$\nu_P^{\square} = \frac{1}{P}\sum_{u\in\mathbb{Z}_P} \delta_{u^2}, \qquad L_N^{\square} = L_N^{\nu_P^{\square},1}, \qquad H_N^{\square} = \widetilde{A}_{N,\mathbb{N}^2} - L_N^{\square}.$$

The measure $\nu_P^{\square}$ has mass one, so Proposition 7.22 gives

$$\Big\| \sup_{N\ge N_0} |L_N^{\square} f| \Big\|_{\ell^{p_r}(\mathbb{Z})} \ll D(r,k,Q)\|f\|_{\ell^{p_r}(\mathbb{Z})}.$$

For a reduced fraction a/q, with $q \mid P$, reduction modulo q gives

$$\widehat{\nu_P^{\square}}(a/q) = S(a,q) := \frac{1}{q}\sum_{u\in\mathbb{Z}_q} e(-au^2/q).$$

By the classical quadratic Gauss-sum bound [25, Section 3], we have $|S(a,q)| \le \sqrt{2/q}$. Define the square circle-method model by

$$\widehat{V_N^{\square}}(\xi) = \sum_{q\le R}\sum_{a\in\mathbb{A}_q} S(a,q)\widehat{\widetilde{M}}_N(\xi - a/q)\eta_s(\xi - a/q).$$

If $\theta = \xi - a/q$, with $q \le R$ and $|\theta| \le 2R/(qN^2)$, summation over residue classes modulo q, followed by partial summation, gives

$$\widehat{\widetilde{A}}_{N,\mathbb{N}^2}(\xi) = S(a,q)\widehat{\widetilde{M}}_N(\theta) + O\left(\frac{q}{N}(1+N^2|\theta|)\right) = S(a,q)\widehat{\widetilde{M}}_N(\theta) + O(R/N).$$

Outside these arcs, apply [21, Theorem 3.1] to the initial square averages at $M = N$ and $M = N/2$. Dirichlet's theorem supplies a reduced fraction a/q with $q \le 4M$ and $|2\xi - a/q| \le (4Mq)^{-1}$. If d is the reduced denominator of $a/(2q)$, then $q \le d \le 2q$. The Gauss factor and the continuous integral bound the main term in that theorem by

$$\ll d^{-1/2}\min\left\{1, \left(M^2|\xi - a/(2q)|\right)^{-1/2}\right\}.$$

If $d > R$, this is $O(R^{-1/2})$. If $d \le R$, being outside the arcs implies $|\xi - a/(2q)| > 2R/(dN^2)$, which gives the same bound since $M \ge N/2$. The approximation error is $O(M^{-1}\sqrt{q}) = O(N^{-1/2})$. Using $\widetilde{A}_{N,\mathbb{N}^2} = A_{N,\mathbb{N}^2} - \frac{1}{2}A_{N/2,\mathbb{N}^2}$, we therefore obtain

$$|\widehat{\widetilde{A}}_{N,\mathbb{N}^2}(\xi)| \ll R^{-1/2} + N^{-1/2}$$

outside the arcs.

By (7.29), $R \le N^{1/8}$. Hence the arcs above are disjoint and their corresponding cutoffs η_s equal one. Every term of $V_N^{\square}$ outside its corresponding arc satisfies, by (6.3),

$$|S(a,q)\widehat{\widetilde{M}}_N(\theta)\eta_s(\theta)| \ll q^{-1/2}\left(\frac{1}{N\sqrt{|\theta|}} + \frac{1}{N}\right) \ll R^{-1/2}.$$

At each ξ, at most one rational frequency contributes at each dyadic denominator scale, so there are $O(\log(eR))$ such terms. Combining the major-arc approximation with the bounds outside the arcs and applying Plancherel's theorem gives

$$\begin{aligned}\|\widetilde{A}_{N,\mathbb{N}^2} - V_N^{\square}\|_{\ell^2(\mathbb{Z})\to\ell^2(\mathbb{Z})} &\ll R/N + R^{-1/2}\log(eR) + N^{-1/2}\\ &\ll R^{-1/3} \ll Q^{-1/2}(1+\log N)^{-2}.\end{aligned}$$

The last two inequalities follow from (6.4) and (7.29). Summing over dyadic $N \ge N_0$, as in (7.34), yields

$$\left\|\sup_{N\ge N_0}|(\widetilde{A}_{N,\mathbb{N}^2} - V_N^{\square})f|\right\|_{\ell^2(\mathbb{Z})} \ll Q^{-1/2}\|f\|_{\ell^2(\mathbb{Z})}.$$

Finally, compare $V_N^{\square}$ with $L_N^{\square}$. For the terms with $q > Q$, Corollary 6.21 and $|S(a,q)| \le \sqrt{2/q}$ give, for each of the two sums, maximal $\ell^2(\mathbb{Z})$-operator norm at most

$$\ll \sum_{2^s > Q/2}(1+s)^2\,2^{-s/2} \ll Q^{-1/2}\log(eQ)^2 \ll Q^{-1/4}.$$

For the circle-method sum, allow the upper limit $q \le R(N)$ to depend on N. For the Ionescu–Wainger sum, use the filtered form: its denominators are at most B, its cutoff is supported where $\eta_s = 1$, and $N_0 \ge 2B$.

For $q \le Q$, the arithmetic coefficients agree, and their difference has multiplier

$$\sum_{q\le Q}\sum_{a\in\mathbb{A}_q} S(a,q)\widehat{\widetilde{M}}_N(\xi-a/q)\left(\eta_s(\xi-a/q)-\eta\left(\frac{\xi-a/q}{\varepsilon}\right)\right).$$

By (6.3), the coefficients $S(a,q)\widehat{\widetilde{M}}_N(\theta)$ satisfy the same dyadic square-sum bound used above for $\mathcal{C}_N$. The cutoff vanishes for $|\theta| < \varepsilon/2$, and the supports are disjoint at each denominator scale. The preceding Plancherel calculation therefore bounds the maximal $\ell^2(\mathbb{Z})$-operator norm of this difference by

$$\ll \frac{\log(eQ)}{N_0\sqrt{\varepsilon}} \ll Q^{-1},$$

using (7.29).

Combining these three bounds for

$$H_N^{\square} = (\widetilde{A}_{N,\mathbb{N}^2} - V_N^{\square}) + (V_N^{\square} - L_N^{\square})$$

gives

$$\Big\| \sup_{N\ge N_0} |H_N^{\square} f| \Big\|_{\ell^2(\mathbb{Z})} \ll Q^{-1/4}\|f\|_{\ell^2(\mathbb{Z})}.$$

Together with (7.37) and the low estimate for $L_N^{\square}$, this proves (7.32) and (7.33). □

We now optimize the parameters in Proposition 7.30 and deduce the endpoint estimates.

Proof of Theorem 7.2. Let $\widetilde{A}^*$ denote either of the dyadic annular maximal operators

$$f \longmapsto \sup_{N\in 2^{\mathbb{N}}} |\widetilde{A}_{N,\Lambda} f|, \qquad f \longmapsto \sup_{N\in 2^{\mathbb{N}}} |\widetilde{A}_{N,\mathbb{N}^2} f|.$$

For either choice of $\widetilde{A}^*$, write $\widetilde{A}_N = L_N + H_N$ for the corresponding decomposition in Proposition 7.30. We first work with dyadic scales. Chebyshev's estimate $\sum_{n\le N}\Lambda(n) \ll N$ gives $\|\widetilde{A}_N \mathbf{1}_F\|_{\ell^1(\mathbb{Z})} \ll |F|$ for prime squares; the same bound is immediate for squares. There are $O(1+\log N_0)$ scales below N_0.

For $N \ge N_0$, if $|\widetilde{A}_N \mathbf{1}_F| > t^{-1}$, then either $|L_N \mathbf{1}_F| > (2t)^{-1}$ or $|H_N \mathbf{1}_F| > (2t)^{-1}$. The union bound and Chebyshev's inequality, with exponents 1, p_r, and 2, followed by (7.32) and (7.33), therefore give, for $t \ge 1$,

$$\begin{aligned}
|\{\widetilde{A}^*\mathbf{1}_F > t^{-1}\}| &\le t\sum_{N<N_0}\|\widetilde{A}_N\mathbf{1}_F\|_{\ell^1(\mathbb{Z})} + (2t)^{p_r}\Big\|\sup_{N\ge N_0}|L_N\mathbf{1}_F|\Big\|_{\ell^{p_r}(\mathbb{Z})}^{p_r} + 4t^2\Big\|\sup_{N\ge N_0}|H_N\mathbf{1}_F|\Big\|_{\ell^2(\mathbb{Z})}^2 \\
&\ll |F|\left(t(1+\log N_0) + t^{p_r}D(r,k,Q)^{p_r} + t^2Q^{-1/2}\right) \\
&= t|F|\left(1+\log N_0 + t^{1/(2r-1)}D(r,k,Q)^{p_r} + tQ^{-1/2}\right). \qquad (7.44)
\end{aligned}$$

Here we used $\|\mathbf{1}_F\|_{\ell^p(\mathbb{Z})}^p = |F|$ and $p_r - 1 = (2r-1)^{-1}$. For sufficiently large t, put

$$v = \log(e+t), \qquad Q = 2^{\lceil 2\log_2(e+t)\rceil}, \qquad r = \lceil v\rceil, \qquad k = \left\lfloor \sqrt{\frac{v}{\log(e+v)}} \right\rfloor.$$

Then $k \ge 4$, $Q \ge 2^k$, $t^{1/(2r-1)} \ll 1$, and $tQ^{-1/2} \ll 1$. Moreover, the definitions of B and $C_{r,k}$ in (7.9) give

$$\log B \ll k^2 \exp\left(\frac{Cv}{k}\right), \qquad \log C_{r,k} \ll k\log(e+v).$$

Together with the definitions of N_0 and $D(r,k,Q)$ in (7.28) and (7.31), these bounds give

$$\log(1+\log N_0)+\log D(r,k,Q) \ll k\log(e+v)+\frac{v}{k} \ll \sqrt{v\log(e+v)}.$$

Thus the expression in parentheses in (7.44) is at most $\exp\left(C\sqrt{v\log(e+v)}\right)$. For bounded t, the same conclusion follows by taking fixed admissible values of r,k,Q. By (7.1),

$$|\{\widetilde{A}^*\mathbf{1}_F > t^{-1}\}| \ll |F|\Psi_C(t). \tag{7.45}$$

By (4.18) and its square analogue,

$$\sup_{\substack{N\in 2^{\mathbb{N}}\\ N\geq 2}} |A_N f| \leq 2\widetilde{A}^* f + \frac{1}{2}|f(\cdot-1)|.$$

The last term is needed only for squares and satisfies the required restricted estimate, since $\Psi_C(t)\geq t\geq 1$. If $N\leq N'<2N$ and N' is dyadic, positivity gives $|A_N f|\leq 2A_{N'}|f|$. Together with (7.45), these bounds prove (7.5) for the square and von Mangoldt averages.

It remains to pass to the unweighted prime-square averages. For $f\geq 0$ and $2^m\leq N<2^{m+1}$, the prime number theorem gives

$$A_{N,\mathbb{P}^2} f \ll \frac{m+1}{2^m}\sum_{j=0}^{m}\frac{1}{j+1}\sum_{\substack{2^j\leq p<2^{j+1}\\ p\in\mathbb{P}}}(\log p) f(\cdot-p^2) \ll A_\Lambda^* f\,(m+1)\sum_{j=0}^{m}\frac{2^{j-m}}{j+1} \ll A_\Lambda^* f.$$

Thus (7.5) holds for all three maximal operators.

We pass to general functions by the standard dyadic layer-cake decomposition; see [28, Chapter 1, §2]. Let $C_\infty\geq 1$ bound the masses of the three families of kernels. For finitely supported f, the part with $|f|\leq \lambda/(2C_\infty)$ contributes at most $\lambda/2$. Put

$$E_j=\left\{\frac{2^j\lambda}{2C_\infty}<|f|\leq\frac{2^{j+1}\lambda}{2C_\infty}\right\},\qquad j\geq 0.$$

Since $\sum_{j\geq 0}(j+1)^{-2}<2$, positivity and (7.5) give

$$\begin{aligned}|\{A^*f>\lambda\}| &\leq \sum_{j\geq 0}\left|\left\{A^*\mathbf{1}_{E_j}>\frac{2^{-j}}{4(j+1)^2}\right\}\right| \ll \sum_{j\geq 0}|E_j|\Psi_{C_1}\left(4\cdot 2^j(j+1)^2\right)\\ &\ll \sum_{j\geq 0}|E_j|\Psi_{C_2}\left(\frac{2^j}{2C_\infty}\right) \leq \sum_{\substack{x\in\mathbb{Z}\\ |f(x)|>\lambda/(2C_\infty)}}\Psi_{C_2}\left(\frac{|f(x)|}{\lambda}\right).\end{aligned}$$

Here C_1 is the constant in (7.5). The third inequality follows from (7.1), with $C_2>C_1$ sufficiently large, since $\log(j+1)=o(\sqrt{j\log(e+j)})$. The last inequality uses monotonicity of Ψ_{C_2}. Renaming C_2 as C proves (7.3) with $C_0=2C_\infty$; enlarging the sum gives (7.4). □

Let (X,μ,T) be an invertible measure-preserving system. For any of the three families in Theorem 7.2, use the ergodic notation of Section 2.9.

Corollary 7.46. *For $1<p\leq 2$,*

$$\|A_T^* f\|_{L^p(X)} \leq \exp\left(\frac{C}{p-1}\log\frac{e}{p-1}\right)\|f\|_{L^p(X)}. \tag{7.47}$$

Moreover,

$$\mu\{A_T^* f>\lambda\} \ll \int_X \Psi_C\left(\frac{|f|}{\lambda}\right)d\mu,\qquad \lambda>0. \tag{7.48}$$

If $\int_X \Psi_C(|f|/a)\,d\mu < \infty$ *for some* $a > 0$*, then the corresponding averages converge almost everywhere and in* $L^1(X)$*.*

Proof. Calderón's transference principle [8, Theorem 1] gives (7.47) from (7.7). The modular transference argument in Section 2.9, applied to (7.4), gives (7.48).

For bounded g, [39, Corollary 2(ii)], applied to the polynomial n^2, gives almost everywhere convergence of the square averages and the logarithmically weighted prime-square averages. The latter are normalized by $\sum_{p\le N}\log p$; the prime number theorem permits normalization by N, and Abel summation gives convergence of the unweighted prime averages. The additional prime powers in $A_{N,\Lambda,T}g$ contribute at most

$$\frac{\|g\|_{L^\infty(X)}}{N}\sum_{\substack{p^j\le N\\ j\ge 2}}\log p \ll \|g\|_{L^\infty(X)}N^{-1/2}\log^2(2N) = o(1).$$

Thus all three families converge almost everywhere for bounded functions.

Suppose that $\int_X \Psi_C(|f|/a)\,d\mu < \infty$ for some $a > 0$, and set $g_m = f\mathbf{1}_{\{|f|\le m\}}$. The doubling property from (7.1) gives $\Psi_C(|f|/\lambda) \ll_{a,\lambda,C} \Psi_C(|f|/a)$ for every $\lambda > 0$. Since the averages of each g_m converge almost everywhere, (7.48) yields

$$\mu\left\{\limsup_{N,M\to\infty}|A_{N,T}f - A_{M,T}f| > 2\lambda\right\} \ll \mu\{A_T^*(f-g_m) > \lambda\} \ll \int_X \Psi_C\left(\frac{|f-g_m|}{\lambda}\right)d\mu.$$

Using dominated convergence, pointwise convergence $f - g_m \to 0$, and $|f - g_m| \le |f|$, we conclude that

$$\mu\left\{\limsup_{N,M\to\infty}|A_{N,T}f - A_{M,T}f| > 2\lambda\right\} = 0.$$

Taking λ through the positive rationals proves almost everywhere convergence.

Convergence in $L^1(X)$ follows first for bounded functions by dominated convergence, and then for all $L^1(X)$ functions by density and the uniform $L^1(X)$-operator bound. □

Proof of Theorem 1.5(ii). For the unweighted prime-square averages,

$$A_T^*f = \sup_{m\ge 1}\left|m^{-1}\sum_{k=1}^{m} f\circ T^{p_k^2}\right|,$$

so (7.47) gives the claim. □

8. Persistent universal L^1-badness of prime squares

We now study the endpoint L^1 behaviour of averages along the prime squares. We use LaVictoire's criterion [31], which reduces the ergodic conclusion to distributional properties of subsets of finite cyclic groups $\mathbb{Z}/q\mathbb{Z}$. For each modulus q, we take Λ_q to be the set of squares of units modulo q. The Chinese remainder theorem gives its multiplicative structure. We quote Granville–Kurlberg [16] for the Poisson-spacing conclusion, but verify the two spacing conditions needed for our purpose directly. Dirichlet's theorem shows that every element of Λ_q is represented by a prime square, and the prime number theorem in arithmetic progressions gives the required limiting estimate.

Throughout this section, $p_1 < p_2 < \cdots$ denotes the sequence of primes and $n_k = p_k^2$. We prove that (n_k) is persistently universally L^1-bad, in the standard sense that for every nonatomic ergodic measure-preserving system and every infinite $\mathcal{S} \subset \mathbb{N}$, there is an $f \in L^1$ for which the

averages

$$N^{-1}\sum_{k\le N} T^{n_k} f, \qquad N \in \mathcal{S},$$

diverge on a set of positive measure. We use the following criterion of LaVictoire. For $\Lambda \subset \mathbb{Z}/q\mathbb{Z}$, write

$$\mathbb{P}_q(\Lambda) = \frac{|\Lambda|}{q}, \qquad s_q(\Lambda) = \mathbb{P}_q(\Lambda)^{-1}, \qquad \Lambda^\gamma = \Lambda + (0, \gamma s_q(\Lambda)) \subset \mathbb{Z}/q\mathbb{Z},$$

where intervals and distances are understood cyclically. When $\Lambda = \Lambda_q$, we write simply $s_q = s_q(\Lambda_q)$.

The following form of LaVictoire's criterion combines [31, Theorem 3.1] with the Conze transference principle and Sawyer's Banach principle, as explained in [31, Section 1].

Theorem 8.1. *Let (n_k) be an increasing sequence of natural numbers. Suppose that, for every $q > 1$, there is a set $\Lambda_q \subset \{n_k \bmod q : k \ge 1\}$ which is multiplicative under the Chinese remainder theorem: if $(q, r) = 1$, then Λ_{qr} corresponds to $\Lambda_q \times \Lambda_r$. Suppose also that there are two sequences (u_j) and (q_j) of odd integers whose combined collection is pairwise coprime and such that*

$$\mathbb{P}_{q_j}(\Lambda_{q_j}) \longrightarrow 0, \qquad \inf_j \mathbb{P}_{u_j}(\Lambda_{u_j}) > 0, \tag{8.2}$$

$$\gamma - \liminf_{j\to\infty} \mathbb{P}_{q_j}(\Lambda_{q_j}^\gamma) = o(\gamma) \quad (\gamma \downarrow 0), \tag{8.3}$$

and that there is an $\alpha > 0$ such that, for every sufficiently small $\gamma > 0$,

$$\liminf_{j\to\infty} \frac{1}{|\Lambda_{q_j}|^2} \big|\{(u, v) \in \Lambda_{q_j}^2 : \operatorname{dist}_{q_j}(u - v, \Lambda_{q_j}) > \gamma s_{q_j}\}\big| > 5\alpha. \tag{8.4}$$

Finally, suppose that some $\beta > 0$ has the following property. If Q is any product of distinct terms from the two sequences (u_j) and (q_j), then for every $a \in \Lambda_Q$,

$$\liminf_{N\to\infty} \frac{1}{N} \big|\{1 \le k \le N : n_k \equiv a \pmod{Q}\}\big| > \frac{\beta}{|\Lambda_Q|}. \tag{8.5}$$

Then (n_k) is persistently universally L^1-bad.

For comparison, we record the following special case of the theorem of Granville and Kurlberg [16, Theorem 1].

Theorem 8.6 (Granville–Kurlberg [16])**.** *For each odd prime p, let $\Omega_p \subset \mathbb{Z}/p\mathbb{Z}$, put $r_p = |\Omega_p|/p$ and $s_p = r_p^{-1}$, and define Ω_q from the Ω_p, for squarefree q, by the Chinese remainder theorem. Suppose that, for every fixed $k \ge 1$ and all distinct $0, h_1, \dots, h_{k-1} \pmod p$,*

$$N_k(\mathbf{h}, \Omega_p) := \big|\{t \in \mathbb{Z}/p\mathbb{Z} : t, t + h_1, \dots, t + h_{k-1} \in \Omega_p\}\big| = r_p^k p\big(1 + O_k((1 - r_p)p^{-c})\big) \tag{8.7}$$

for some $c > 0$, and suppose that $s_p = p^{o(1)}$. Then the normalized spacings of Ω_q are Poisson distributed as $s_q = q/|\Omega_q| \to \infty$. In particular, if the elements of Ω_q are listed cyclically and $F_q(\theta)$ is the proportion of normalized gaps exceeding θ, then $F_q(\theta) \to e^{-\theta}$ for every $\theta > 0$.

We verify the correlation hypothesis for the nonzero quadratic residues.

Lemma 8.8. *Let p be an odd prime and let*

$$\Lambda_p = \{x^2 \pmod p : x \not\equiv 0 \pmod p\}.$$

For every fixed $k \ge 1$ and distinct $0, h_1, \dots, h_{k-1} \pmod p$,

$$N_k(\mathbf{h}, \Lambda_p) = \frac{p}{2^k} + O_k(p^{1/2}) = \left(\frac{p-1}{2p}\right)^k p\big(1 + O_k(p^{-1/2})\big).$$

Consequently (8.7) *holds, for example with* $c = 1/3$.

Proof. Let χ_p be the quadratic character modulo p, extended by $\chi_p(0) = 0$, and put $h_0 = 0$. For every $x \pmod p$,

$$\mathbf{1}_{\Lambda_p}(x) = \frac{1}{2}\big(1 + \chi_p(x) - \mathbf{1}_{\{0\}}(x)\big).$$

Therefore

$$N_k(\mathbf{h}, \Lambda_p) = 2^{-k} \sum_{t \bmod p} \prod_{i=0}^{k-1} \big(1 + \chi_p(t + h_i) - \mathbf{1}_{\{0\}}(t + h_i)\big).$$

The term obtained by choosing 1 from every factor is $p/2^k$. Every term containing at least one factor $\mathbf{1}_{\{0\}}(t + h_i)$ is $O_k(1)$: one such factor fixes $t = -h_i$, and there are at most 3^k terms in the expansion. The remaining nonconstant terms have the form

$$\sum_{t \bmod p} \chi_p\left(\prod_{i \in I}(t + h_i)\right), \qquad \varnothing \neq I \subset \{0, \dots, k-1\}.$$

Because the h_i are distinct, the polynomial inside the character is squarefree. For $|I| = 1$ the sum is 0, and for $|I| \geq 2$ Weil's bound for multiplicative character sums gives absolute value at most $(|I| - 1)p^{1/2}$, see [25, Section 11.1]. Summing over the $2^k - 1$ nonempty sets I proves

$$N_k(\mathbf{h}, \Lambda_p) = 2^{-k}p + O_k(p^{1/2}).$$

Since

$$\left(\frac{p-1}{2p}\right)^k p = 2^{-k}p + O_k(1),$$

the second formula follows. Finally $1 - r_p = (p+1)/(2p) \sim 1$, so the relative error $O_k(p^{-1/2})$ is $O_k((1 - r_p)p^{-1/3})$. □

Proof of Theorem 1.5(i). For every $q \geq 2$, define

$$\Lambda_q = \{b^2 \pmod q : (b, q) = 1\}.$$

If $(q, r) = 1$, the Chinese remainder theorem identifies Λ_{qr} with $\Lambda_q \times \Lambda_r$. Moreover,

$$\Lambda_q \subseteq \{p_k^2 \pmod q : k \geq 1\}.$$

Indeed, if $a \in \Lambda_q$, choose a unit $b \pmod q$ such that $b^2 \equiv a \pmod q$. Dirichlet's theorem gives a prime $p \equiv b \pmod q$, and hence $p^2 \equiv a \pmod q$.

Choose distinct odd primes u_j and pairwise disjoint sets B_j of odd primes, with $|B_j| = j$, such that $u_i \notin B_j$ for all $i, j \geq 1$ and

$$p > 3^j \quad (p \in B_j), \qquad \sum_{p \in B_j} \frac{64}{\sqrt{p}} \leq 1.$$

This can be done successively, since arbitrarily large unused primes are available at each stage. Put $q_j = \prod_{p \in B_j} p$. The combined collection $(u_j), (q_j)$ is odd and pairwise coprime.

For every odd squarefree q, the Chinese remainder theorem and the local square-root count give

$$|\Lambda_q| = \prod_{p \mid q} \frac{p-1}{2} = \frac{\phi(q)}{2^{\omega(q)}}, \qquad s_q = \frac{q}{|\Lambda_q|} = \prod_{p \mid q} \frac{2p}{p-1}. \tag{8.9}$$

Consequently

$$\mathbb{P}_{u_j}(\Lambda_{u_j}) \geq \frac{1}{3}, \qquad 3^{-j} \leq \mathbb{P}_{q_j}(\Lambda_{q_j}) \leq 2^{-j}, \qquad 2^j \leq s_{q_j} \leq 3^j.$$

This proves (8.2) and $s_{q_j} \to \infty$.

We verify the two remaining spacing conditions directly by finite moment estimates. This avoids using the strict finite-fraction inequality in [31, Lemma 3.2]. For $\Lambda = \Lambda_q$ and $m = |\Lambda|$, define

$$R_q(t) = |\{a \in \Lambda : a + t \in \Lambda\}|, \qquad T_q(t) = |\{(a, b) \in \Lambda^2 : a - b - t \in \Lambda\}|.$$

For an odd prime p, write $m_p = (p-1)/2$. The quadratic-character estimate used in Lemma 8.8, with two distinct shifts, gives the following convenient uniform bound:

$$R_p(t) \le \left(1 + \frac{54}{\sqrt{p}}\right) \frac{m_p^2}{p} \qquad (t \not\equiv 0 \pmod p). \tag{8.10}$$

Indeed the absolute error is at most $6\sqrt{p}$, while $m_p \ge p/3$. Also $R_p(0) = m_p$ and

$$T_p(t) = \sum_{b \in \Lambda_p} R_p(-b - t).$$

At most one term in this sum has zero shift. Applying (8.10) to the other terms and using $m_p \le (9/\sqrt{p}) m_p^3/p$ gives

$$T_p(t) \le \left(1 + \frac{63}{\sqrt{p}}\right) \frac{m_p^3}{p} \qquad (t \in \mathbb{Z}/p\mathbb{Z}).$$

Both counts multiply under the Chinese remainder theorem. For $q = q_j$, our choice of B_j gives

$$\prod_{p|q} (1 + 64/\sqrt{p}) \le \exp\Big(\sum_{p|q} 64/\sqrt{p}\Big) \le e < 3.$$

Estimate (8.10) therefore yields

$$R_q(t) \le 3\frac{m^2}{q} \quad \text{if } t \not\equiv 0 \pmod p \text{ for every } p \mid q, \qquad T_q(t) \le 3\frac{m^3}{q} \quad \text{for every } t. \tag{8.11}$$

Fix $0 < \gamma \le 1$. There are between $\gamma q/m - 1$ and $\gamma q/m$ integers d with $0 < d < \gamma q/m$. For such distinct pairs d, e, we have $0 < |d - e| < \gamma q/m \le 3^j < p$ for every $p \in B_j$. The bound for R_q therefore gives $|(\Lambda + d) \cap (\Lambda + e)| \le 3m^2/q$. The union bound and the first two terms of inclusion–exclusion give

$$\left(\frac{\gamma q}{m} - 1\right) m - \frac{3}{2} \left(\frac{\gamma q}{m}\right)^2 \frac{m^2}{q} \le |\Lambda^\gamma| \le \frac{\gamma q}{m}\, m.$$

Dividing by q, we obtain

$$\gamma - \frac{m}{q} - \frac{3}{2}\gamma^2 \le \mathbb{P}_{q_j}(\Lambda^\gamma_{q_j}) \le \gamma.$$

Since $m/q \le 2^{-j} \to 0$, it follows that

$$0 \le \gamma - \liminf_{j \to \infty} \mathbb{P}_{q_j}(\Lambda^\gamma_{q_j}) \le \frac{3}{2}\gamma^2.$$

This proves (8.3).

If $\operatorname{dist}_q(a - b, \Lambda) \le \gamma q/m$, then $a - b - w \equiv t \pmod q$ for some $w \in \Lambda$ and some integer t with $|t| \le \gamma q/m$. Summing (8.11) over these at most $2\gamma q/m + 1$ integers gives

$$\frac{1}{m^2} |\{(a, b) \in \Lambda^2 : \operatorname{dist}_q(a - b, \Lambda) \le \gamma q/m\}| \le \frac{1}{m^2} \left(\frac{2\gamma q}{m} + 1\right) \frac{3m^3}{q} = 6\gamma + \frac{3m}{q}.$$

Since $m/q \le 2^{-j} \to 0$, taking complements and then the lower limit yields

$$\liminf_{j \to \infty} \frac{1}{|\Lambda_{q_j}|^2} |\{(a, b) \in \Lambda_{q_j}^2 : \operatorname{dist}_{q_j}(a - b, \Lambda_{q_j}) > \gamma s_{q_j}\}| \ge 1 - 6\gamma.$$

For $0 < \gamma < 1/16$, we have $1 - 6\gamma > 5/8 > 5(1/160)$. Thus (8.4) holds with $\gamma_0 = 1/16$ and $\alpha = 1/160$.

Finally, let Q be a product of distinct terms from (u_j) and (q_j), and let $a \in \Lambda_Q$. The integer Q is odd and square-free. By the Chinese remainder theorem, the congruence $b^2 \equiv a \pmod{Q}$ has exactly $2^{\omega(Q)}$ unit solutions $b \pmod{Q}$. For each such b, the prime number theorem in the fixed arithmetic progression $b \pmod{Q}$ gives

$$|\{k \le N : p_k \equiv b \pmod{Q}\}| = \frac{N}{\phi(Q)} + o_Q(N).$$

Distinct roots give disjoint progressions, apart from the finitely many primes dividing Q, and hence

$$\lim_{N\to\infty} \frac{1}{N}|\{k \le N : p_k^2 \equiv a \pmod{Q}\}| = \frac{2^{\omega(Q)}}{\phi(Q)} = \frac{1}{|\Lambda_Q|},$$

where the last equality is (8.9). Therefore (8.5) holds with any fixed $0 < \beta < 1$, for example $\beta = 1/2$. All hypotheses of Theorem 8.1 are now verified, and (p_k^2) is persistently universally L^1-bad. □

9. Quantitative Goldbach-type estimate: reductions

We now prove that almost every integer $n \equiv 12 \pmod{24}$ is the sum of a prime square and a prime of the form $x^2 + y^2 + 1$, with $(x, y) = 1$. The proof combines an analytic estimate for prime squares with a lower-bound sieve for the primitive sum-of-two-squares condition and an upper-bound sieve for the remaining large prime factors congruent to $3 \pmod 4$.

With $\mathcal{P}$ as defined in (2.3), put

$$\mathcal{N} = \{n < N^2 : n \equiv 12 \pmod{24},\ n \in \mathcal{P} + \mathbb{P}^2\}.$$

To prove Theorem 1.6, it is enough to show that, for every $A > 0$,

$$|\{n < N^2 : n \equiv 12 \pmod{24}\} \setminus \mathcal{N}| \ll_A N^2(\log N)^{-A}. \tag{9.1}$$

We may restrict throughout to $n \gg_A N^2(\log N)^{-A}$, since the number of smaller integers is already of the required order. For such n, let

$$\mathbb{P}_{3,n} = \{p \in \mathbb{P} : p \equiv 3 \pmod 4,\ p \nmid n - 1\},$$
$$\mathcal{A}_n = \{r - 1 < N^2 : r \equiv 11 \pmod{24},\ r \in \mathbb{P},\ n - r \in \mathbb{P}^2\}.$$

We use the sieve notation $S(\mathcal{F}, \mathfrak{P}, z)$ introduced in (2.12).

Lemma 9.2. *For every $\varepsilon > 0$,*

$$|\{r \in \mathcal{P} : n - r \in \mathbb{P}^2\}| \ge S(\mathcal{A}_n, \mathbb{P}_{3,n}, N^2) - O_\varepsilon(N^\varepsilon).$$

Proof. Let $r - 1$ be counted by $S(\mathcal{A}_n, \mathbb{P}_{3,n}, N^2)$ and write $n - r = q^2$ with q prime. If no prime $p \equiv 3 \pmod 4$ divides $r - 1$, then $r - 1 \equiv 2 \pmod 4$ and the primitive sum-of-two-squares criterion shows that $r - 1 = x^2 + y^2$ with $(x, y) = 1$. Hence $r \in \mathcal{P}$.

Otherwise, let $p \equiv 3 \pmod 4$ divide $r - 1$. Since $p \notin \mathbb{P}_{3,n}$, we have $p \mid n - 1$. Therefore

$$p \mid (n - 1) - (r - 1) = q^2,$$

so $p = q$. Thus every exceptional r is of the form $r = n - q^2$, where q is a prime divisor of $n - 1$. There are at most $\omega(n - 1) \ll_\varepsilon N^\varepsilon$ such primes. □

We now truncate the lower sieve. Fix

$$1 < \alpha < 2, \qquad z = N^{1/\alpha}.$$

Lemma 9.3. *For every $\varepsilon > 0$,*

$$S(\mathcal{A}_n, \mathbb{P}_{3,n}, z) - |\{r \in \mathcal{P} : n - r \in \mathbb{P}^2\}| = |T_n| + O_\varepsilon(N^\varepsilon), \tag{9.4}$$

where

$$T_n = \left\{ r - 1 = 2up_1p_2 : \begin{array}{l} r \in \mathbb{P}, \quad n - r \in \mathbb{P}^2, \\ z \le p_2 \le p_1, \quad p_i \in \mathbb{P}_{3,n}, \\ p \mid u \Longrightarrow p \equiv 1 \pmod 4 \end{array} \right\}. \tag{9.5}$$

Proof. Let $r - 1$ be counted by the difference on the left of (9.4). Then $r - 1$ has no prime divisor $p < z$ belonging to $\mathbb{P}_{3,n}$, but $r \notin \mathcal{P}$. By the primitive sum-of-two-squares criterion, $r - 1$ has a prime divisor congruent to 3 (mod 4).

Write $n - r = q^2$, with $q \in \mathbb{P}$. If a prime $p \equiv 3 \pmod 4$ divides both $r - 1$ and $n - 1$, then

$$p \mid (n - 1) - (r - 1) = q^2,$$

and hence $p = q$. Thus $q \mid n - 1$, and q uniquely determines $r = n - q^2$. The number of these representations is therefore at most $\omega(n-1) \ll_\varepsilon N^\varepsilon$. After removing them, every prime divisor of $r - 1$ congruent to 3 (mod 4) belongs to $\mathbb{P}_{3,n}$, and hence is at least z.

Since $r \equiv 11 \pmod{24}$, one has $r - 1 \equiv 2 \pmod 8$. The number of prime factors of $r - 1$ congruent to 3 (mod 4), counted with multiplicity, is even. There are therefore at least two such factors. There cannot be four, since

$$z^4 = N^{4/\alpha} > N^2 > r - 1.$$

Consequently,

$$r - 1 = 2up_1p_2,$$

where $z \le p_2 \le p_1$, $p_i \in \mathbb{P}_{3,n}$, and every prime divisor of u is congruent to 1 (mod 4). This gives T_n. The converse implication is immediate from the primitive sum-of-two-squares criterion, completing the proof. □

For the factorization in (9.5), put $\ell = 2up_2$. Since $p_1p_2u < N^2/2$ and $p_2 \le p_1$,

$$u < \frac{N^2}{2z^2}, \qquad z \le p_2 < \left(\frac{N^2}{2u}\right)^{1/2}.$$

Accordingly, define

$$L = \left\{ 2up_2 : \begin{array}{l} u \in \mathbb{N}, \quad u < \dfrac{N^2}{2z^2}, \\ p_2 \in \mathbb{P}, \quad p_2 \equiv 3 \pmod 4, \quad z \le p_2 < \left(\dfrac{N^2}{2u}\right)^{1/2}, \\ p \mid u \Longrightarrow p \equiv 1 \pmod 4 \quad \text{for every prime } p \end{array} \right\}, \tag{9.6}$$

and put

$$L_n = \{\ell \in L : (\ell, n - 1) = 1\}.$$

Thus, if $\ell = 2up_2 \in L_n$, then $p_2 \nmid n - 1$, and hence $p_2 \in \mathbb{P}_{3,n}$. The ordering $p_1 \ge p_2$ makes the choice of $\ell = 2up_2$ canonical. For every $\ell \in L$, define

$$M_n(\ell) = \{\ell p + 1 \colon \ell p + 1 < N^2 : p \equiv 3 \pmod 4,\ p \in \mathbb{P},\ n - 1 - \ell p = m^2 \text{ for some } m \in \mathbb{P},\ m > 3\}.$$

and let

$$W_{\ell,n} = \{p \in \mathbb{P} : p \nmid 12\ell n\}, \qquad z_\ell = (N^2/\ell)^{1/4}.$$

Lemma 9.7. *For every $\varepsilon > 0$,*

$$|T_n| \le \sum_{\ell \in L_n} S(M_n(\ell), W_{\ell,n}, z_\ell) + O_\varepsilon(N^\varepsilon). \tag{9.8}$$

Proof. Let $r - 1 \in T_n$. By (9.5), it has the canonical factorization $r - 1 = 2up_1p_2 = \ell p_1$ and $\ell = 2up_2 \in L$, where $z \le p_2 \le p_1$. Write $n - r = q^2$, with $q \in \mathbb{P}$. Since $n \equiv 12 \pmod{24}$ and $r \equiv 11 \pmod{24}$, we have $q^2 = n - r \equiv 1 \pmod{24}$, so $q > 3$. Moreover,

$$n - 1 - \ell p_1 = n - r = q^2.$$

We first remove the cases in which $\ell \notin L_n$. If $(\ell, n-1) > 1$, choose a prime $s \mid (\ell, n-1)$. So $s \mid q^2$, and hence $s = q$. In particular, $q \mid n - 1$, and q determines $r = n - q^2$. The number of such r is therefore at most $\omega(n-1) \ll_\varepsilon N^\varepsilon$.

For every remaining $r - 1 \in T_n$, we have $\ell \in L_n$. The relations

$$r = \ell p_1 + 1 < N^2, \qquad n - 1 - \ell p_1 = q^2, \qquad q > 3,$$

show that $r \in M_n(\ell)$. Furthermore,

$$z_\ell = \left(\frac{N^2}{\ell}\right)^{1/4} \le N^{1/2} < z \le p_1 < r.$$

Since r is prime, no prime below z_ℓ divides r. Consequently, r is counted by $S(M_n(\ell), W_{\ell,n}, z_\ell)$.

Thus, apart from $O_\varepsilon(N^\varepsilon)$ elements, every $r - 1 \in T_n$ gives an $\ell \in L_n$ for which r is counted by the corresponding sifted set. Summing over $\ell \in L_n$ proves (9.8). □

Combining Lemmas 9.2, 9.3, and 9.7, we obtain the reduction used below.

Corollary 9.9. *For every $\varepsilon > 0$,*

$$|\{r \in \mathcal{P} : n - r \in \mathbb{P}^2\}| \ge S(\mathcal{A}_n, \mathbb{P}_{3,n}, z) - \sum_{\ell \in L_n} S(M_n(\ell), W_{\ell,n}, z_\ell) - O_\varepsilon(N^\varepsilon).$$

It remains to obtain a lower bound for $S(\mathcal{A}_n, \mathbb{P}_{3,n}, z)$ and an upper bound for

$$\sum_{\ell \in L_n} S(M_n(\ell), W_{\ell,n}, z_\ell).$$

The common analytic input for these two sieve estimates is the following Bombieri–Vinogradov approximation, proved in Section 10. Fix a nonzero real function $\psi \in C_c^\infty((-1/8, 1/8))$, and put

$$\eta = \frac{\psi * \widetilde{\psi}}{(\psi * \widetilde{\psi})(0)}, \qquad \widetilde{\psi}(t) = \psi(-t).$$

Thus $\eta(0) = 1$, $\operatorname{supp} \eta \subset [-1/4, 1/4]$, and $\eta^\vee \ge 0$.

Lemma 9.10 (Bombieri–Vinogradov approximation)**.** *Let $A, \delta > 0$ and $L_0 < N^{2-\delta}$. There are sufficiently large constants $B = B(A, \delta)$ and $C = C(A, \delta)$ such that the following holds. Put*

$$R = (\log N)^C, \qquad \eta_N(\theta) = \eta\left(\frac{N^2\theta}{R}\right),$$

and suppose that $V_\ell \le (N^2/\ell)^{1/2}(\log N)^{-B}$ for $\ell < L_0$. Let $|d_{k,\ell}| \le 1$ and $(a_{k,\ell}, k) = 1$. Define

$$G_R(\ell, k, n) := \mathcal{T}_2(\ell, k, n; a_{k,\ell}) \prod_{2 < p \le R} \frac{p\phi((p,k))}{(p-1)^2} \left| \left\{ (u, v) \in \mathbb{A}_p^2 : \begin{array}{ll} u^2 + \ell v \equiv n & \pmod p, \\ v \equiv a_{k,\ell} & \pmod{(p,k)} \end{array} \right\} \right|, \tag{9.11}$$

where $\mathcal{T}_2(\ell,k,n;a_{k,\ell})$ *is defined in* (3.18) *and the second congruence is understood to be trivial when* $p\nmid k$. *Also define*

$$H_{\ell,R}(n):=\ell\int_{\mathbb{R}}\eta_N(\theta)^2\left(\int_2^N\frac{e(\theta u^2)}{\log u}\,du\right)\left(\int_{\substack{2\le v\le N^2\\ \ell v<N^2}}\frac{e(\ell\theta v)}{\log v}\,dv\right)e(-n\theta)\,d\theta.$$

Then

$$\sum_{n<N^2}\left|\sum_{\ell<L_0}\sum_{k<V_\ell}d_{k,\ell}r_{k,\ell}(n)\right|^2\ll_{A,\delta}N^4(\log N)^{-A},\tag{9.12}$$

where

$$r_{k,\ell}(n)=\sum_{\substack{m^2+\ell t=n\\ t\equiv a_{k,\ell}\pmod k}}\mathbf{1}_{\mathbb{P}}(m)\mathbf{1}_{\mathbb{P}}(t)-\frac{G_R(\ell,k,n)}{\ell\phi(k)}H_{\ell,R}(n).$$

We now specify the residue classes used in Lemma 9.10. In the lower sieve, for square-free d supported on primes greater than 3, let $a_{24d,1}\pmod{24d}$ be determined by

$$a_{24d,1}\equiv 11\pmod{24},\qquad a_{24d,1}\equiv 1\pmod d,\tag{9.13}$$

Then $(a_{24d,1},24d)=1$.

For the upper sieve, write $\ell=2v$, where $(v,6)=1$. If d is square-free and $(d,12\ell)=1$, the Chinese remainder theorem determines $a_{12d,\ell}\pmod{12d}$ by

$$va_{12d,\ell}\equiv 5\pmod{12},\qquad \ell a_{12d,\ell}\equiv -1\pmod d.\tag{9.14}$$

Since v is invertible modulo 12 and ℓ is invertible modulo d, one has $(a_{12d,\ell},12d)=1$. Moreover,

$$\ell a_{12d,\ell}\equiv 10\pmod{12},\qquad \ell a_{12d,\ell}\equiv -1\pmod d.$$

As $(12,d)=1$, these two congruences determine a unique class $b_{12d}\pmod{12d}$, depending only on d, such that

$$\ell a_{12d,\ell}\equiv b_{12d}\pmod{12d}.$$

In particular, $(b_{12d},12d)=2$.

Fix $A_0,K>0$. Choose $C=C(A_0,K)$ sufficiently large for Lemma 9.10 with logarithmic saving K, and put

$$\alpha=\frac{9}{8},\qquad z=N^{1/\alpha}=N^{8/9},\qquad R=(\log N)^C.\tag{9.15}$$

To simplify notation, we suppress the dependence on R and set

$$X_n=\frac{G_R(1,24,n)}{\phi(24)}H_{1,R}(n),$$
$$\frac{\omega_n(d)}{d}=\frac{\phi(24)G_R(1,24d,n)}{\phi(24d)G_R(1,24,n)},\qquad \Theta_n(z)=\prod_{\substack{3<p<z\\ p\equiv 3\pmod 4}}\left(1-\frac{\omega_n(p)}{p}\right).\tag{9.16}$$

The following propositions are proved in Section 11 by combining Lemma 9.10 with the lower and upper Rosser–Iwaniec sieves. We use the sieve functions f_κ and F_κ with the normalization fixed in Section 2. Their estimates are uniform for

$$n\equiv 12\pmod{24},\qquad N^2(\log N)^{-A_0}\le n<N^2.$$

Proposition 9.17 (Lower sieve). *For* $N^2(\log N)^{-A_0}\le n<N^2$, $n\equiv 12\pmod{24}$,

$$S(\mathcal{A}_n,\mathbb{P}_{3,n},z)\ge X_n\Theta_n(z)\big(f_{1/2}(9/8)+o(1)\big)+\mathcal{E}_1(n),$$

where

$$\sum_{n<N^2} |\mathcal{E}_1(n)|^2 \ll_{A_0,K} N^4(\log N)^{-K}. \tag{9.18}$$

For $\alpha > 1$, define

$$\mathcal{W}(\alpha) := \frac{\alpha}{4\sqrt{2}} \int_2^{2\alpha} \frac{t-2+(t-1)\log(t-1)}{t^2(t-1)\sqrt{1-t/(2\alpha)}}\, dt. \tag{9.19}$$

The integral is finite, since the singularity at $t = 2\alpha$ is integrable.

Proposition 9.20 (Upper sieve)**.** *For $N^2(\log N)^{-A_0} \le n < N^2$, $n \equiv 12 \pmod{24}$,*

$$\sum_{\ell \in L_n} S(M_n(\ell), W_{\ell,n}, z_\ell) \le \big(c_{9/8}F_1(2)\mathcal{W}(9/8) + o(1)\big)X_n\Theta_n(z) + \mathcal{E}_2(n),$$

where

$$c_\alpha = \frac{16e^{-\gamma/2}}{\sqrt{\pi\alpha}}$$

and

$$\sum_{n<N^2} |\mathcal{E}_2(n)|^2 \ll_{A_0,K} N^4(\log N)^{-K}. \tag{9.21}$$

Corollary 9.9, Proposition 9.17, and Proposition 9.20 reduce the problem to comparing the two sieve constants. Put

$$\Delta = f_{1/2}(9/8) - c_{9/8}F_1(2)\mathcal{W}(9/8).$$

We shall also need the following two lemmas. Their proofs are given in Section 11.

Lemma 9.22. *Let $A_0 > 0$. There is $C_{A_0} > 0$ such that, for every fixed sufficiently large C, with $R = (\log N)^C$, one has*

$$X_n\Theta_n(z) \gg_{A_0,C} N(\log N)^{-C_{A_0}} \tag{9.23}$$

whenever $n \equiv 12 \pmod{24}$ and $N^2(\log N)^{-A_0} \le n < N^2$.

Lemma 9.24. *With the normalization of* [24, (1.8)–(1.10)]*,*

$$\Delta > 0.$$

We now complete the proof of Theorem 1.6.

Proof of Theorem 1.6. Fix $A_0 > 0$, and let C_{A_0} be given by Lemma 9.22. Set $K = A_0 + 2C_{A_0} + 10$. Choose the constants in Lemma 9.10 and in the two sieve propositions for this value of K, enlarging C, if necessary, so that Lemma 9.22 also applies.

By Lemma 9.24, $\Delta > 0$. Hence, for all sufficiently large N, Corollary 9.9 and Propositions 9.17 and 9.20 give

$$|\{r \in \mathcal{P} : n - r \in \mathbb{P}^2\}| \ge \frac{\Delta}{2}X_n\Theta_n(z) + \mathcal{E}_1(n) - \mathcal{E}_2(n) - O_\varepsilon(N^\varepsilon). \tag{9.25}$$

By (9.23), there is $c_{A_0} > 0$ such that

$$\frac{\Delta}{2}X_n\Theta_n(z) \ge 4c_{A_0}N(\log N)^{-C_{A_0}}$$

throughout the stated range of n. Taking $\varepsilon = 1/2$, we also have, for sufficiently large N,

$$O_\varepsilon(N^\varepsilon) \le c_{A_0}N(\log N)^{-C_{A_0}}.$$

Discard the integers $n < N^2$ for which

$$|\mathcal{E}_1(n)| + |\mathcal{E}_2(n)| > c_{A_0} N (\log N)^{-C_{A_0}}.$$

By Cauchy–Schwarz, Chebyshev's inequality, (9.18), and (9.21), their number is at most

$$\frac{N^4 (\log N)^{-K}}{N^2 (\log N)^{-2C_{A_0}}} \ll_{A_0} N^2 (\log N)^{-A_0}.$$

For every remaining n, the right-hand side of (9.25) is at least

$$2c_{A_0} N (\log N)^{-C_{A_0}} > 0.$$

Thus $n \in \mathcal{P} + \mathbb{P}^2$.

Finally, the integers below $N^2(\log N)^{-A_0}$, together with the discarded exceptional set, contribute $O_{A_0}(N^2(\log N)^{-A_0})$. This proves (9.1). □

10. Quantitative Goldbach-type estimate: the circle method

This section proves Lemma 9.10. Fourier inversion reduces the problem to estimates for a prime-square exponential sum and a prime sum in an arithmetic progression. Let

$$P_U(\theta) = \sum_{m \le U} \mathbf{1}_{\mathbb{P}}(m) e(m^2\theta),$$

$$F_{k,\ell,W}(\theta) = \sum_{\substack{t \le W \\ \ell t < N^2 \\ t \equiv a_{k,\ell} \pmod k}} \mathbf{1}_{\mathbb{P}}(t) e(\ell t \theta),$$

where $2 \le U \le N$ and $2 \le W \le N^2$. The restriction $\ell t < N^2$ does not change any Fourier coefficient indexed by $n < N^2$.

Choose $C = C(A, \delta)$ sufficiently large, put $R = (\log N)^C$, and define

$$\mathfrak{M} = \bigcup_{q \le R} \bigcup_{a \in \mathbb{A}_q} \left\{ \theta \in \mathbb{T} : \left\| \theta - \frac{a}{q} \right\|_{\mathbb{T}} \le \frac{R}{N^2} \right\}, \qquad \mathfrak{m} = \mathbb{T} \setminus \mathfrak{M}.$$

Distinct reduced fractions with denominators at most R are separated by at least R^{-2}. Hence the arcs in $\mathfrak{M}$ are disjoint for sufficiently large N. Moreover, they contain the usual major arcs

$$\left\{ \theta : \left\| \theta - \frac{a}{q} \right\|_{\mathbb{T}} \le \frac{R}{qN^2} \right\}, \qquad q \le R.$$

Thus $\mathfrak{m}$ is contained in the minor-arc set occurring in [44, Theorem 1]. Partial summation therefore gives, for every $D > 0$,

$$\sup_{\theta \in \mathfrak{m}} |P_U(\theta)| \ll_D \frac{U}{\log U} (\log N)^{-D} \tag{10.1}$$

whenever $U \ge N(\log N)^{-2D}$. Below this range we use $|P_U(\theta)| \le U$.

Lemma 10.2. *Under the hypotheses of Lemma 9.10, uniformly for $2 \le U \le N$ and $2 \le W \le N^2$,*

$$\sum_{n < N^2} \left| \int_{\mathfrak{m}} P_U(\theta) \sum_{\ell < L_0} \sum_{k < V_\ell} d_{k,\ell} F_{k,\ell,W}(\theta) e(-n\theta)\, d\theta \right|^2 \ll_{A,\delta} N^4 (\log N)^{-A}. \tag{10.3}$$

Proof. Bessel's inequality gives

$$\sum_{n<N^2}\left|\int_{\mathfrak{m}} P_U(\theta)\sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}F_{k,\ell,W}(\theta)e(-n\theta)\,d\theta\right|^2 \le \sup_{\theta\in\mathfrak{m}}|P_U(\theta)|^2\int_{\mathbb{T}}\left|\sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}F_{k,\ell,W}(\theta)\right|^2 d\theta. \tag{10.4}$$

Opening the final integral and using orthogonality, we obtain

$$\int_{\mathbb{T}}\left|\sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}F_{k,\ell,W}(\theta)\right|^2 d\theta = \sum_{\ell_1,\ell_2<L_0}\sum_{\substack{k_1<V_{\ell_1}\\ k_2<V_{\ell_2}}} d_{k_1,\ell_1}\overline{d_{k_2,\ell_2}} \sum_{\substack{t_i\in\mathbb{P},\ t_i\le W\\ \ell_i t_i<N^2\\ t_i\equiv a_{k_i,\ell_i}\,(\mathrm{mod}\,k_i)\\ \ell_1t_1=\ell_2t_2}} 1.$$

We estimate separately the terms with $t_1=t_2$ and $t_1\neq t_2$.

If $t_1=t_2$, then $\ell_1=\ell_2=\ell$. When the two congruences are compatible, they restrict t_1 to one residue class modulo $[k_1,k_2]$. Since $|d_{k,\ell}|\le 1$, the contribution of these terms is at most

$$\sum_{\ell<L_0}\sum_{k_1,k_2<V_\ell}\left(\frac{N^2}{\ell[k_1,k_2]}+1\right)\ll N^2(\log N)^4, \tag{10.5}$$

where we used

$$\sum_{k_1,k_2\le V}\frac{1}{[k_1,k_2]}\ll(\log(2V))^3, \qquad V_\ell^2\le\frac{N^2}{\ell}.$$

Suppose now that $t_1\neq t_2$. Since t_1 and t_2 are distinct primes, the relation $\ell_1t_1=\ell_2t_2$ implies

$$\ell_1=st_2, \qquad \ell_2=st_1$$

for some $s\ge 1$. Put

$$w_\ell(t)=\sum_{k<V_\ell}\mathbf{1}_{t\equiv a_{k,\ell}\,(\mathrm{mod}\,k)}.$$

Using again $|d_{k,\ell}|\le 1$, the off-diagonal contribution is bounded by

$$\sum_{\substack{s\ge1,\ t_1,t_2\in\mathbb{P}\\ t_1\neq t_2,\ t_i\le W\\ st_1,st_2<L_0,\ st_1t_2<N^2}} w_{st_2}(t_1)w_{st_1}(t_2).$$

For every dyadic interval $t\sim T$, expanding the square gives

$$\sum_{t\sim T}w_\ell(t)^2\le\sum_{k_1,k_2<V_\ell}\left(\frac{T}{[k_1,k_2]}+1\right)\ll T(\log N)^3+V_\ell^2. \tag{10.6}$$

Indeed, for each k_1,k_2, the two congruences are either incompatible or determine one residue class modulo $[k_1,k_2]$.

Restrict s,t_1,t_2 to dyadic intervals $s\sim S$ and $t_i\sim T_i$. For every nonempty such range, $ST_1T_2\ll N^2$. After dropping the primality conditions, Cauchy–Schwarz gives

$$\sum_{\substack{s\sim S,\ t_i\sim T_i\\ st_1,st_2<L_0,\ st_1t_2<N^2}} w_{st_2}(t_1)w_{st_1}(t_2)\le\left(\sum_{\substack{s\sim S,\ t_2\sim T_2\\ st_2<L_0}}\sum_{t_1\sim T_1} w_{st_2}(t_1)^2\right)^{1/2}\left(\sum_{\substack{s\sim S,\ t_1\sim T_1\\ st_1<L_0}}\sum_{t_2\sim T_2} w_{st_1}(t_2)^2\right)^{1/2}. \tag{10.7}$$

By (10.6) and $V_\ell^2 \le N^2/\ell$,

$$\sum_{\substack{s\sim S,\ t_2\sim T_2\\ st_2<L_0}} \sum_{t_1\sim T_1} w_{st_2}(t_1)^2 \ll ST_1T_2(\log N)^3 + N^2 \sum_{s\sim S}\frac{1}{s}\sum_{t_2\sim T_2}\frac{1}{t_2} \ll N^2(\log N)^3.$$

The same estimate holds with t_1 and t_2 interchanged. Therefore (10.7) is $O(N^2(\log N)^3)$.

There are $O((\log N)^3)$ choices for S, T_1, T_2. Combining (10.5) and (10.7), we obtain

$$\int_{\mathbb{T}} \left| \sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}F_{k,\ell,W}(\theta)\right|^2 d\theta \ll N^2(\log N)^4 + \sum_{S,T_1,T_2} N^2(\log N)^3 \ll N^2(\log N)^6. \quad (10.8)$$

If $U \ge N(\log N)^{-2D}$, then (10.1), (10.4), and (10.8) give $O(N^4(\log N)^{4-2D})$. For smaller U, the bound $|P_U| \le U$ gives $O(N^4(\log N)^{6-4D})$. Taking D sufficiently large proves (10.3). □

We now rescale the function η fixed before Lemma 9.10. On $\mathbb{R}$, put

$$\eta_N(\theta) = \eta\left(\frac{N^2\theta}{R}\right),$$

and use the same notation for its periodization on $\mathbb{T}$. Moreover, $|\eta_N| \le 1$ and the Euclidean inverse Fourier transform of η_N is nonnegative.

Since η is smooth and even, $\eta'(0) = 0$, and hence $\eta(t) = 1 + O(t^2)$ for $|t| \le 1$. Together with $|\eta| \le 1$, this gives

$$|1 - \eta_N(\theta)^2| \ll \min\left\{1, \left(\frac{N^2|\theta|}{R}\right)^2\right\}. \quad (10.9)$$

For $2 \le U \le N$ and $2 \le W \le N^2$, put

$$I_U(\gamma) = \int_2^U \frac{e(\gamma u^2)}{\log u}\,du, \qquad J_{\ell,W}(\gamma) = \int_{\substack{2\le v\le W\\ \ell v<N^2}} \frac{e(\ell\gamma v)}{\log v}\,dv,$$

and define

$$H_{\ell,R}(n;U,W) := \ell \int_{\mathbb{R}} \eta_N(\gamma)^2 I_U(\gamma) J_{\ell,W}(\gamma) e(-n\gamma)\,d\gamma. \quad (10.10)$$

Thus

$$H_{\ell,R}(n) = H_{\ell,R}(n;N,N^2).$$

Lemma 10.11. *Uniformly in ℓ, U, W and $n < N^2$,*

$$H_{\ell,R}(n;U,W) \ge 0, \qquad H_{\ell,R}(n;U,W) = H_\ell(n;U,W) + O(NR^{-1/2}),$$

where

$$H_\ell(n;U,W) = \int_{\substack{2\le u\le U\\ 2\le (n-u^2)/\ell\le W}} \frac{du}{\log u \log((n-u^2)/\ell)}.$$

Proof. Fourier inversion gives

$$\int_{\mathbb{R}} I_U(\gamma) J_{\ell,W}(\gamma) e(-n\gamma)\,d\gamma = \frac{1}{\ell} H_\ell(n;U,W).$$

The inverse Fourier transforms of I_U and $J_{\ell,W}$ are nonnegative measures. Moreover,

$$(\eta_N^2)^\vee = \eta_N^\vee * \eta_N^\vee \ge 0.$$

Hence the inverse Fourier transform of $\eta_N^2 I_U J_{\ell,W}$ is nonnegative, proving $H_{\ell,R}(n;U,W) \ge 0$.

Van der Corput's estimate and integration by parts give

$$|I_U(\gamma)| \ll \min\{N, |\gamma|^{-1/2}\}, \qquad |J_{\ell,W}(\gamma)| \ll \ell^{-1}\min\{N^2, |\gamma|^{-1}\}. \tag{10.12}$$

By the definitions of $H_{\ell,R}$ and H_ℓ,

$$H_{\ell,R}(n;U,W) - H_\ell(n;U,W) = \ell\int_{\mathbb{R}} \big(\eta_N(\gamma)^2 - 1\big) I_U(\gamma) J_{\ell,W}(\gamma) e(-n\gamma)\, d\gamma.$$

Using (10.9), (10.12), and splitting the integral at N^{-2} and R/N^2, we obtain

$$\begin{aligned} |H_{\ell,R}(n;U,W) - H_\ell(n;U,W)| &\ll N^3 \int_0^{N^{-2}} \left(\frac{N^2\gamma}{R}\right)^2 d\gamma \\ &\quad + \int_{N^{-2}}^{R/N^2} \left(\frac{N^2\gamma}{R}\right)^2 \gamma^{-3/2}\, d\gamma + \int_{R/N^2}^{\infty} \gamma^{-3/2}\, d\gamma \\ &\ll \frac{N}{R^2} + \frac{N}{R^{1/2}} \ll NR^{-1/2}. \end{aligned} \tag{10.13}$$

□

We next approximate the two exponential sums on $\mathfrak{M}$. If $\theta = a/q + \gamma \in \mathfrak{M}$, define

$$\begin{aligned} P_U^{(0)}(\theta) &= \mathfrak{s}(-a,q) I_U(\gamma), \\ F_{k,\ell,W}^{(0)}(\theta) &= \frac{\Upsilon(-a\ell, q, a_{k,\ell}, k)}{\phi(k)} J_{\ell,W}(\gamma). \end{aligned}$$

Lemma 10.14. *Uniformly for $2 \le U \le N$ and $2 \le W \le N^2$,*

$$\sum_{n<N^2} \left| \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} \int_{\mathfrak{M}} \big(P_U F_{k,\ell,W} - P_U^{(0)} F_{k,\ell,W}^{(0)}\big)(\theta) e(-n\theta)\, d\theta \right|^2 \ll_{A,\delta} N^4 (\log N)^{-A}.$$

Proof. Fix $\theta = a/q + \gamma \in \mathfrak{M}$, put $X_\ell = N^2/\ell$, and write $h = [k,q]$. Define

$$\Delta(X;h) = \sup_{\substack{(b,h)=1 \\ 2\le y\le X}} \left|\pi(y;h,b) - \frac{\mathrm{Li}(y)}{\phi(h)}\right|, \qquad E(y;h,b) = \pi(y;h,b) - \frac{\mathrm{Li}(y)}{\phi(h)}.$$

Both $F_{k,\ell,W}$ and $J_{\ell,W}$ are unchanged when W is replaced by $\min\{W, X_\ell\}$, so assume $W \le X_\ell$. Every prime in the progression is coprime to k. Those not coprime to h must therefore divide q, and their total contribution has absolute value at most q. For the remaining primes, split into the reduced classes

$$C_{k,q} = \{b \in \mathbb{A}_h : b \equiv a_{k,\ell} \pmod k\}.$$

Their contribution is

$$\sum_{b\in C_{k,q}} e(a\ell b/q) \sum_{\substack{t\le W,\ \ell t<N^2 \\ t\equiv b \pmod h}} \mathbf{1}_{\mathbb{P}}(t) e(\ell\gamma t).$$

Since $W \le X_\ell$, the strict cutoff can remove only the term $t = W = X_\ell$. Thus

$$F_{k,\ell,W}(\theta) = \sum_{\substack{t\le W \\ t\equiv a_{k,\ell} \pmod k}} \mathbf{1}_{\mathbb{P}}(t) e(\ell\theta t) + O(1), \qquad J_{\ell,W}(\gamma) = \int_2^W \frac{e(\ell\gamma v)}{\log v}\, dv.$$

The $O(1)$ correction is zero unless $W = X_\ell$ is a prime in the progression, in which case it equals $-e(\ell\theta X_\ell)$. For each $b \in C_{k,q}$, Stieltjes integration gives

$$\sum_{\substack{t\le W\\ t\equiv b \pmod h}} \mathbf{1}_{\mathbb{P}}(t)e(\ell\gamma t) = \frac{1}{\phi(h)}\int_2^W \frac{e(\ell\gamma v)}{\log v}\,dv + \int_{2^-}^W e(\ell\gamma v)\,dE(v;h,b).$$

Partial summation gives

$$\begin{aligned}\left|\int_{2^-}^W e(\ell\gamma v)\,dE(v;h,b)\right| &\le |E(W;h,b)| + 2\pi|\gamma|\ell\int_2^W |E(v;h,b)|\,dv\\ &\le \big(1+2\pi|\gamma|\ell(W-2)\big)\Delta(X_\ell;h) \ll \big(1+|\gamma|\ell X_\ell\big)\Delta(X_\ell;h).\end{aligned}$$

Since $|\gamma| \le R/N^2$, $\ell X_\ell = N^2$, and $|C_{k,q}| \le q \le R$, summing these errors gives $O(R^2\Delta(X_\ell;[k,q]))$. Also,

$$\frac{1}{\phi([k,q])}\sum_{b\in C_{k,q}} e(a\ell b/q) = \frac{\Upsilon(-a\ell,q,a_{k,\ell},k)}{\phi(k)}.$$

Restoring both the strict endpoint and the primes dividing q, we obtain an estimate for the full original sum:

$$\begin{aligned}F_{k,\ell,W}(\theta) &= F^{(0)}_{k,\ell,W}(\theta) + E_{k,\ell,W}(\theta),\\ \sup_{2\le W'\le X_\ell}|E_{k,\ell,W'}(\theta)| &\ll R^2\Delta(X_\ell;[k,q]) + q + 1.\end{aligned} \tag{10.15}$$

We use the Bombieri–Vinogradov theorem: for every $T > 0$, there is $L = L(T)$ such that

$$\sum_{h\le X^{1/2}(\log X)^{-L}} \Delta(X;h) \ll_T X(\log X)^{-T}. \tag{10.16}$$

For fixed q and h, every positive integer k satisfying $[k,q] = h$ has $h/k \in \{1,\dots,q\}$. The map $k \mapsto h/k$ is injective on this set, so each h has at most q preimages. We obtain

$$\sum_{1\le k<V_\ell}\sup_{2\le W'\le X_\ell}|E_{k,\ell,W'}(a/q+\gamma)| \ll R^3\sum_{h\le RV_\ell}\Delta(X_\ell;h) + (q+1)V_\ell.$$

Here $[k,q] \le kq \le RV_\ell$. Since $N^\delta < X_\ell \le N^2$, we have $\delta\log N < \log X_\ell \le 2\log N$. For sufficiently large $B = B(T,C,\delta)$ and N,

$$RV_\ell \le X_\ell^{1/2}(\log N)^{-B+C} \le X_\ell^{1/2}(\log X_\ell)^{-L}.$$

Thus (10.16), with a sufficiently strong logarithmic saving, controls the first term uniformly in the denominator and all cutoffs. Taking $T = K+3C+1$ in (10.16) and choosing $B = B(K,C,\delta)$ sufficiently large, we obtain

$$\begin{aligned}R^3\sum_{1\le\ell<L_0}\sum_{h\le RV_\ell}\Delta(X_\ell;h) &\ll_{K,C,\delta} (\log N)^{-K-1}\sum_{1\le\ell<L_0}X_\ell\\ &= N^2(\log N)^{-K-1}\sum_{1\le\ell<L_0}\frac{1}{\ell} \ll N^2(\log N)^{-K}.\end{aligned}$$

The finite error in (10.15) satisfies, uniformly for $q \le R$,

$$\begin{aligned}(q+1)\sum_{1\le\ell<L_0}\sum_{1\le k<V_\ell}1 &\le 2RN\sum_{1\le\ell<L_0}\ell^{-1/2}\\ &\ll RN\sqrt{L_0} \ll N^{2-\delta/2}(\log N)^C \ll_{K,C,\delta} N^2(\log N)^{-K},\end{aligned}$$

where we used $V_\ell \le N/\sqrt{\ell}$, $L_0 < N^{2-\delta}$, and $R = (\log N)^C$. Together with (10.15) and the estimate obtained from (10.16), this proves

$$\sup_{\theta\in\mathfrak{M}} \sum_{1\le \ell<L_0} \sum_{1\le k<V_\ell} \sup_{2\le W'\le X_\ell} |E_{k,\ell,W'}(\theta)| \ll_{K,C,\delta} N^2(\log N)^{-K}. \tag{10.17}$$

We next approximate P_U. If $U < N(\log N)^{-2D}$, then

$$|P_U(\theta)| + |P_U^{(0)}(\theta)| \ll U \ll N(\log N)^{-D},$$

so (10.18) follows trivially. We may therefore assume that $U \ge N(\log N)^{-2D}$, and hence $\log U \sim \log N$. Separating the primes dividing q and then splitting the remaining primes into reduced residue classes modulo q, we have

$$P_U(\theta) = \sum_{r\in\mathbb{A}_q} e(ar^2/q) \sum_{\substack{m\le U\\ m\equiv r \pmod q}} \mathbf{1}_{\mathbb{P}}(m) e(\gamma m^2) + O(\log q).$$

Since $q \le R = (\log N)^C$, the Siegel–Walfisz theorem gives, with an arbitrarily large logarithmic saving,

$$\max_{\substack{r\in\mathbb{A}_q\\ 2\le y\le U}} \left|\pi(y;q,r) - \frac{\mathrm{Li}(y)}{\phi(q)}\right| \ll_D N(\log N)^{-D-2C-3}.$$

Partial summation and $|\gamma|U^2 \le R$ therefore give

$$\sum_{\substack{m\le U\\ m\equiv r \pmod q}} \mathbf{1}_{\mathbb{P}}(m) e(\gamma m^2) = \frac{1}{\phi(q)} \int_2^U \frac{e(\gamma u^2)}{\log u}\, du + O_D\big(NR(\log N)^{-D-2C-3}\big).$$

Summing this formula over $r \in \mathbb{A}_q$, using $q \le R$, and absorbing the primes dividing q, we obtain

$$\begin{aligned} P_U(a/q+\gamma) &= \frac{1}{\phi(q)} \sum_{r\in\mathbb{A}_q} e(ar^2/q) \int_2^U \frac{e(\gamma u^2)}{\log u}\, du + O_D\big(N(\log N)^{-D}\big) \\ &= P_U^{(0)}(a/q+\gamma) + O_D\big(N(\log N)^{-D}\big). \end{aligned} \tag{10.18}$$

Since the arcs are disjoint, and since there are $\ll R^2$ reduced fractions with denominator at most R, we conclude that $\mathrm{meas}(\mathfrak{M}) \ll \frac{R^3}{N^2}$. We have

$$P_U F_{k,\ell,W} - P_U^{(0)} F_{k,\ell,W}^{(0)} = P_U^{(0)} E_{k,\ell,W} + (P_U - P_U^{(0)}) F_{k,\ell,W}.$$

For the first term, Bessel's inequality gives

$$\begin{aligned} \sum_{n<N^2} \left| \int_{\mathfrak{M}} P_U^{(0)}(\theta) \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} E_{k,\ell,W}(\theta) e(-n\theta)\, d\theta \right|^2 &\le \int_{\mathfrak{M}} |P_U^{(0)}(\theta)|^2 \left| \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} E_{k,\ell,W}(\theta) \right|^2 d\theta \\ &\ll N^2 \frac{R^3}{N^2} \big(N^2(\log N)^{-K}\big)^2 = N^4 R^3 (\log N)^{-2K}, \end{aligned}$$

where we used (10.17) and $|P_U^{(0)}| \ll N$. For the second term, Bessel's inequality, (10.18), and (10.8) give

$$\begin{aligned} &\sum_{n<N^2} \left| \int_{\mathfrak{M}} (P_U - P_U^{(0)})(\theta) \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} F_{k,\ell,W}(\theta) e(-n\theta)\, d\theta \right|^2 \\ &\quad \le \|P_U - P_U^{(0)}\|_{L^\infty(\mathfrak{M})}^2 \int_{\mathbb{T}} \left| \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} F_{k,\ell,W}(\theta) \right|^2 d\theta \ll N^2(\log N)^{-2D}\, N^2(\log N)^6. \end{aligned}$$

Choosing K and D so that $2K > A + 3C$ and $2D > A + 6$ completes the proof. □

We identify the arithmetic factor. For $q \geq 1$, put

$$\mathfrak{G}_q(\ell, k, n) = \sum_{a \in \mathbb{A}_q} \Upsilon(a\ell, q, a_{k,\ell}, k)\mathfrak{s}(a, q)e(an/q).$$

Lemma 10.19. *The series*

$$\sum_{\substack{q \geq 1 \\ P^+(q) \leq R}} \mathfrak{G}_q(\ell, k, n)$$

converges absolutely and satisfies

$$\sum_{\substack{q \geq 1 \\ P^+(q) \leq R}} \mathfrak{G}_q(\ell, k, n) = \prod_{p \leq R} \left(1 + \sum_{\nu \geq 1} \mathfrak{G}_{p^\nu}(\ell, k, n)\right) = G_R(\ell, k, n), \tag{10.20}$$

where G_R is defined in (9.11). Moreover, for some absolute $C_1 > 0$,

$$\begin{aligned} |\mathfrak{s}(a, q)| &\ll q^{-1/2}\tau(q)^{C_1}, \\ |\Upsilon(a\ell, q, a_{k,\ell}, k)| &\ll q^{-1}(q, k\ell)\tau(q)^{C_1}, \\ |\mathfrak{G}_q(\ell, k, n)| &\ll q^{-1}(q, k\ell)\tau(q)^{C_1}, \end{aligned} \tag{10.21}$$

and

$$\sum_{\ell < L_0} \sum_{k < V_\ell} \frac{(q, k\ell)}{\ell\phi(k)} \ll \tau(q)^{C_1}(\log N)^2. \tag{10.22}$$

Proof. The first two estimates in (10.21) follow from (3.26) and [14, Lemma 3.8]. Equation (3.19), with $b = a_{k,\ell}$, shows that $\mathfrak{G}_q(\ell, k, n)$ is multiplicative in q. Comparing the Euler products in (3.17), we see that, for every odd prime p,

$$\mathfrak{G}_{p^\nu}(\ell, k, n) = 0 \quad (\nu \geq 2), \qquad |\mathfrak{G}_p(\ell, k, n)| \ll p^{-1}(p, k\ell).$$

At $p = 2$, only finitely many exponents contribute and their total is bounded absolutely. It follows by multiplicativity that

$$|\mathfrak{G}_q(\ell, k, n)| \ll q^{-1}(q, k\ell)\tau(q)^{C_1}.$$

This proves the third estimate in (10.21). Moreover, $\mathfrak{G}_{p^\nu}(\ell, k, n) = 0$ for every odd p and $\nu \geq 2$, while at $p = 2$ only finitely many exponents contribute. Thus, among the integers q satisfying $P^+(q) \leq R$, only finitely many have $\mathfrak{G}_q(\ell, k, n) \neq 0$. Consequently,

$$\sum_{\substack{q \geq 1 \\ P^+(q) \leq R}} |\mathfrak{G}_q(\ell, k, n)| < \infty.$$

Choose $Q > R$ so that the interval (R, Q) contains no prime. Then $p < Q$ is equivalent to $p \leq R$. Taking $b = a_{k,\ell}$ in Corollary 3.16 therefore gives (10.20).

For (10.22), we use

$$(q, k\ell) \leq \sum_{\substack{d_1 d_2 | q \\ d_1 | k,\ d_2 | \ell}} \phi(d_1 d_2).$$

Consequently,

$$\sum_{\ell < L_0} \sum_{k < V_\ell} \frac{(q, k\ell)}{\ell\phi(k)} \leq \sum_{d_1 d_2 | q} \phi(d_1 d_2) \sum_{\substack{\ell < L_0 \\ d_2 | \ell}} \frac{1}{\ell} \sum_{\substack{k < V_\ell \\ d_1 | k}} \frac{1}{\phi(k)}$$

$$\ll (\log N)^2 \sum_{d_1 d_2 | q} \frac{\phi(d_1 d_2)}{d_2 \phi(d_1)}$$
$$\le (\log N)^2 \sum_{d_1 d_2 | q} 1 \le \tau(q)^2 (\log N)^2,$$

where we used $\phi(d_1 d_2) \le d_2 \phi(d_1)$. After enlarging C_1, if necessary, this proves (10.22). □

We complete the major arcs.

Lemma 10.23. *Under the hypotheses of Lemma 9.10, uniformly for $2 \le U \le N$ and $2 \le W \le N^2$,*

$$\sum_{n<N^2} \left| \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} \left(\int_{\mathfrak{M}} P_U^{(0)}(\theta) F_{k,\ell,W}^{(0)}(\theta) e(-n\theta)\, d\theta - \frac{G_R(\ell,k,n)}{\ell \phi(k)} H_{\ell,R}(n;U,W) \right) \right|^2 \tag{10.24}$$
$$\ll_{A,\delta} N^4 (\log N)^{-A}.$$

Proof. After replacing a by $-a$, the integral in (10.24) equals

$$\frac{1}{\phi(k)} \sum_{q \le R} \sum_{a \in \mathbb{A}_q} \Upsilon(a\ell, q, a_{k,\ell}, k) \mathfrak{s}(a,q) e(an/q) \int_{|\gamma| \le R/N^2} I_U(\gamma) J_{\ell,W}(\gamma) e(-n\gamma)\, d\gamma. \tag{10.25}$$

We first replace the sharp cutoff in (10.25) by $\eta_N(\gamma)^2$. At the denominator scale $q \sim 2^s$, Plancherel and disjointness of the corresponding frequency intervals give

$$\left\| \left(\sum_{q \sim 2^s} \sum_{a \in \mathbb{A}_q} \big(\mathbf{1}_{|\gamma| \le R/N^2} - \eta_N(\gamma)^2 \big) \mathfrak{s}(a,q) I_U(\gamma) \sum_{\ell<L_0} \sum_{k<V_\ell} \frac{d_{k,\ell}}{\phi(k)} \Upsilon(a\ell, q, a_{k,\ell}, k) J_{\ell,W}(\gamma) \right)^\vee \right\|_2^2$$
$$\ll \sum_{q \sim 2^s} \sum_{a \in \mathbb{A}_q} q^{-3} \tau(q)^{O(1)} (\log N)^{O(1)} \int_{\mathbb{R}} \big| \mathbf{1}_{|\gamma| \le R/N^2} - \eta_N(\gamma)^2 \big|^2 \min\{N^2, |\gamma|^{-1}\} \min\{N^4, |\gamma|^{-2}\}\, d\gamma$$
$$\ll \sum_{q \sim 2^s} \sum_{a \in \mathbb{A}_q} q^{-3} \tau(q)^{O(1)} (\log N)^{O(1)} \left(\frac{N^{14}}{R^4} \int_0^{N^{-2}} \gamma^4\, d\gamma + \frac{N^8}{R^4} \int_{N^{-2}}^{R/N^2} \gamma\, d\gamma \right)$$
$$\ll N^4 R^{-2} (\log N)^{O(1)} \sum_{q \sim 2^s} \phi(q) q^{-3} \tau(q)^{O(1)}$$
$$\ll N^4 R^{-2} 2^{-s+o(s)} (\log N)^{O(1)}.$$

Here $\gamma = \theta - a/q$; in the first inequality we used (10.21), (10.22), and the bounds for I_U and $J_{\ell,W}$. Since the frequency supports corresponding to distinct reduced fractions a/q, $q \le R$, are disjoint, we may sum the preceding squared estimates over $2^s \le R$. Their total is

$$\ll N^4 R^{-2} (\log N)^{O(1)} \sum_{2^s \le R} 2^{-s+o(s)} \ll N^4 R^{-2} (\log N)^{O(1)} \ll_A N^4 (\log N)^{-A},$$

provided C is sufficiently large.

Put $Y = \frac{N}{100 R^{1/2}}$. Distinct reduced fractions with denominators at most Y are separated by at least Y^{-2}, whereas η_N is supported in an interval of width $O(R/N^2)$. Thus all the translates $\eta_N(\cdot - a/q)$, $q \le Y$, have disjoint supports. We may therefore extend the major-arc sum to the additional R-smooth moduli

$$R < q \le Y, \qquad P^+(q) \le R,$$

provided that we estimate the L^2-contribution of the added terms. This contribution is negligible by the following argument.

For $q \sim 2^s$, the Fourier transform in n of the added terms is

$$\sum_{\substack{q\sim 2^s\\ P^+(q)\le R}} \sum_{a\in\mathbb{A}_q} \eta_N(\gamma)^2 \mathfrak{s}(a,q) I_U(\gamma) \sum_{\ell<L_0} \sum_{k<V_\ell} \frac{d_{k,\ell}}{\phi(k)} \Upsilon(a\ell, q, a_{k,\ell}, k) J_{\ell,W}(\gamma), \qquad \gamma = \theta - a/q.$$

The translated supports are disjoint at this denominator scale. Therefore, by Plancherel, (10.21), and (10.22), its squared ℓ^2-norm is

$$\begin{aligned} &\ll \sum_{\substack{q\sim 2^s\\ P^+(q)\le R}} \sum_{a\in\mathbb{A}_q} q^{-3}\tau(q)^{O(1)} (\log N)^{O(1)} \int_{\mathbb{R}} |\eta_N(\gamma)|^4 \min\{N^2, |\gamma|^{-1}\} \min\{N^4, |\gamma|^{-2}\}\, d\gamma \\ &\ll N^4 (\log N)^{O(1)} \sum_{q\sim 2^s} \phi(q) q^{-3} \tau(q)^{O(1)} \ll N^4\, 2^{-s+o(s)} (\log N)^{O(1)}. \end{aligned}$$

Taking square roots, summing over the dyadic scales $2^s > R$, and then squaring gives

$$N^4 (\log N)^{O(1)} \left(\sum_{2^s > R} 2^{-s/2+o(s)} \right)^2 \ll N^4 R^{-1+o(1)} (\log N)^{O(1)},$$

which shows the contribution of these terms are negligible, as claimed above. After replacing $\mathbf{1}_{\{|\gamma|\le R/N^2\}}$ by η_N^2 and extending the sum from $q \le R$ to $q \le Y$ and $P^+(q) \le R$. Fourier inversion and (10.10) give

$$\begin{aligned} &\sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} \int_{\mathfrak{M}} P_U^{(0)}(\theta) F_{k,\ell,W}^{(0)}(\theta) e(-n\theta)\, d\theta \\ &\qquad = \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} \frac{H_{\ell,R}(n;U,W)}{\ell\phi(k)} \sum_{\substack{q\le Y\\ P^+(q)\le R}} \mathfrak{G}_q(\ell,k,n) + \mathcal{E}(n), \end{aligned} \tag{10.26}$$

where

$$\sum_{n<N^2} |\mathcal{E}(n)|^2 \ll_{A,\delta} N^4 (\log N)^{-A}.$$

For the remaining terms with $q > Y$ and $P^+(q) \le R$, (10.21), (10.22), and $|H_{\ell,R}(n;U,W)| \ll N$ give

$$\begin{aligned} \left| \sum_{\ell<L_0} \sum_{k<V_\ell} d_{k,\ell} \frac{H_{\ell,R}(n;U,W)}{\ell\phi(k)} \sum_{\substack{q>Y\\ P^+(q)\le R}} \mathfrak{G}_q(\ell,k,n) \right| &\le \sum_{\substack{q>Y\\ P^+(q)\le R}} \sum_{\ell<L_0} \sum_{k<V_\ell} \frac{|H_{\ell,R}(n;U,W)|}{\ell\phi(k)} |\mathfrak{G}_q(\ell,k,n)| \\ &\ll N (\log N)^{O(1)} \sum_{\substack{q>Y\\ P^+(q)\le R}} \frac{\tau(q)^{O(1)}}{q}. \end{aligned}$$

Rankin's argument gives

$$\begin{aligned} \sum_{\substack{q>Y\\ P^+(q)\le R}} \frac{\tau(q)^{O(1)}}{q} &\le Y^{-1/\log(2R)} \prod_{p\le R} \left(\sum_{\nu\ge 0} \frac{(\nu+1)^{O(1)}}{p^{\nu(1-1/\log(2R))}} \right) \\ &\ll Y^{-1/\log(2R)} \exp\left(O\left(\sum_{p\le R} p^{-1+1/\log(2R)} \right) \right) \end{aligned} \tag{10.27}$$

$$\ll Y^{-1/\log(2R)} \exp\left(O\left(\sum_{p\le R}\frac{1}{p}\right)\right)$$
$$\ll Y^{-1/\log(2R)}(\log R)^{O(1)} \ll_A (\log N)^{-A}.$$

After increasing C, its square summed over $n < N^2$ is $O_A(N^4(\log N)^{-A})$. The completed arithmetic sum is now exactly $G_R(\ell,k,n)$, by (10.20), (10.26), and (10.27). This proves (10.24). □

We finish the proof of the stated approximation.

Proof of Lemma 9.10. For $n < N^2$, Fourier inversion gives

$$\sum_{\substack{m\le U,\ t\le W\\ m^2+\ell t=n\\ t\equiv a_{k,\ell} \pmod k}} \mathbf{1}_{\mathbb{P}}(m)\mathbf{1}_{\mathbb{P}}(t) = \int_{\mathbb{T}} P_U(\theta)F_{k,\ell,W}(\theta)e(-n\theta)\,d\theta.$$

After multiplying by $d_{k,\ell}$ and summing over ℓ and k, introduce $E_1(n), E_2(n), E_3(n)$ through the following successive equalities:

$$\sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell} \sum_{\substack{m\le U,\ t\le W\\ m^2+\ell t=n\\ t\equiv a_{k,\ell} \pmod k}} \mathbf{1}_{\mathbb{P}}(m)\mathbf{1}_{\mathbb{P}}(t) \tag{10.28}$$
$$= \sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}\int_{\mathfrak{M}} P_U(\theta)F_{k,\ell,W}(\theta)e(-n\theta)\,d\theta + E_1(n), \qquad \text{by Lemma 10.2,}$$
$$= \sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}\int_{\mathfrak{M}} P_U^{(0)}(\theta)F_{k,\ell,W}^{(0)}(\theta)e(-n\theta)\,d\theta + E_1(n) + E_2(n), \qquad \text{by Lemma 10.14,}$$
$$= \sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}\frac{G_R(\ell,k,n)}{\ell\phi(k)}H_{\ell,R}(n;U,W) + E_1(n) + E_2(n) + E_3(n), \qquad \text{by Lemma 10.23.}$$

The three cited lemmas give, uniformly in U and W,

$$\sum_{n<N^2}\left(|E_1(n)|^2 + |E_2(n)|^2 + |E_3(n)|^2\right) \ll_{A,\delta} N^4(\log N)^{-A}.$$

Estimate (10.28) therefore implies

$$\sum_{n<N^2}\left|\sum_{\ell<L_0}\sum_{k<V_\ell} d_{k,\ell}\left(\sum_{\substack{m\le U,\ t\le W\\ m^2+\ell t=n\\ t\equiv a_{k,\ell} \pmod k}} \mathbf{1}_{\mathbb{P}}(m)\mathbf{1}_{\mathbb{P}}(t) - \frac{G_R(\ell,k,n)}{\ell\phi(k)}H_{\ell,R}(n;U,W)\right)\right|^2 \ll_{A,\delta} N^4(\log N)^{-A}.$$

Taking $U = N$ and $W = N^2$ proves (9.12). Indeed, when $n < N^2$, the equation $m^2 + \ell t = n$ forces $m < N$ and $\ell t < N^2$, while $H_{\ell,R}(n;N,N^2) = H_{\ell,R}(n)$. □

11. Quantitative Goldbach-type estimate: sieve arguments

We prove Propositions 9.17 and 9.20. Recall that $A_0, K > 0$ and the parameters α, z, R were fixed in (9.15), with C chosen so that Lemma 9.10 gives the logarithmic saving K. Throughout this section,

$$n \equiv 12 \pmod{24}, \qquad N^2(\log N)^{-A_0} \le n < N^2, \tag{11.1}$$

and all constants implicit in $o(1)$ are uniform in this range. Put

$$\mathbb{P}_3 = \{p \in \mathbb{P} : p > 3,\ p \equiv 3 \pmod 4\}, \qquad W_\ell = \{p \in \mathbb{P} : p \nmid 12\ell\}.$$

The following observation replaces the n-dependent sets of sieve primes by fixed sets, allowing Lemma 9.10 to be applied with coefficients independent of n.

Lemma 11.2. *For every $\varepsilon > 0$,*

$$S(\mathcal{A}_n, \mathbb{P}_{3,n}, z) \geq S(\mathcal{A}_n, \mathbb{P}_3, z), \tag{11.3}$$

$$\sum_{\ell \in L_n} S(M_n(\ell), W_{\ell,n}, z_\ell) \leq \sum_{\ell \in L} S(M_n(\ell), W_\ell, z_\ell) + O_\varepsilon(N^\varepsilon). \tag{11.4}$$

Proof. Since $r - 1 \equiv 10 \pmod{24}$, the prime 3 never divides an element of $\mathcal{A}_n$. Thus the first inequality follows from $\mathbb{P}_{3,n} \setminus \{3\} \subseteq \mathbb{P}_3$. For the second, suppose that an integer $r = \ell t + 1 \in M_n(\ell)$ is counted after sifting by $W_{\ell,n}$, but is removed after sifting by W_ℓ. Then some prime $p < z_\ell$ divides both n and r. Since $n - r = m^2$ for a prime m, it follows that $m = p$, and hence

$$r = n - p^2.$$

For each $p \mid n$, the number of representations $r - 1 = \ell t$, with $\ell \in L$ and t prime, is at most $\tau(r-1)$. Consequently, the total number of such pairs (ℓ, r) is

$$\ll \sum_{\substack{p \mid n \\ n - p^2 > 1}} \tau(n - p^2 - 1) \ll_\varepsilon N^\varepsilon.$$

Enlarging the remaining sum from L_n to L can only increase it, which proves (11.4). □

For an odd prime p, set

$$\rho_p(a) = |\{x \in \mathbb{F}_p^\times : x^2 = a\}|.$$

Thus $\rho_p(a) = 1 + (\frac{a}{p})$ when $p \nmid a$, and $\rho_p(a) = 0$ when $p \mid a$.

We now apply the residue classes fixed in (9.13) and (9.14) to the two sieve sequences. For the lower sieve, let d be square-free and supported on $\mathbb{P}_3$. Since $(d, 24) = 1$, the congruences in (9.13) give

$$d \mid r - 1, \quad r \equiv 11 \pmod{24} \quad \Longleftrightarrow \quad r \equiv a_{24d,1} \pmod{24d}.$$

Consequently,

$$|\mathcal{A}_{n,d}| = \sum_{\substack{m^2 + t = n \\ t \equiv a_{24d,1} \pmod{24d}}} \mathbf{1}_{\mathbb{P}}(m)\mathbf{1}_{\mathbb{P}}(t) = \frac{G_R(1, 24d, n)}{\phi(24d)} H_{1,R}(n) + r_{24d,1}(n),$$

where the second equality follows from the definition of the remainder in Lemma 9.10. Hence, recalling the definitions in (9.16),

$$|\mathcal{A}_{n,d}| = X_n \frac{\omega_n(d)}{d} + r_{24d,1}(n). \tag{11.5}$$

Indeed,

$$X_n \frac{\omega_n(d)}{d} = \frac{G_R(1, 24, n)}{\phi(24)} H_{1,R}(n) \frac{\phi(24) G_R(1, 24d, n)}{\phi(24d) G_R(1, 24, n)} = \frac{G_R(1, 24d, n)}{\phi(24d)} H_{1,R}(n).$$

Lemma 11.6. *For $p > 3$, $p \equiv 3 \pmod 4$,*

$$\frac{\omega_n(p)}{p} = \begin{cases} \dfrac{\rho_p(n-1)}{p - 1 - \rho_p(n)}, & p \leq R, \\ \dfrac{1}{p-1}, & p > R. \end{cases}$$

In particular, for $p \geq 7$, $0 \leq \omega_n(p)/p \leq 2/(p-3) < 1$.

Proof. For $p \leq R$, the unrestricted local equation is $x^2+t = n$, with $x, t \in \mathbb{F}_p^\times$. It has $p-1-\rho_p(n)$ solutions. The additional condition $p \mid t-1$ fixes $t = 1$ and leaves $\rho_p(n-1)$ choices for x. Taking the ratio of these two local factors proves the first case. If $p > R$, the prime p does not occur in G_R, and hence

$$\frac{\phi(24)G_R(1,24p,n)}{\phi(24p)G_R(1,24,n)} = \frac{1}{p-1}.$$

The final claim follows from $\rho_p(a) \leq 2$. □

For the upper sieve, write $\ell = 2v$ and recall the class $a_{12d,\ell}$ from (9.14). Since the prime m is greater than 3, one has $m^2 \equiv 1 \pmod{24}$. Thus

$$m^2 + \ell t = n - 1, \qquad n \equiv 12 \pmod{24}$$

implies

$$\ell t \equiv 10 \pmod{24}, \qquad \text{or equivalently} \qquad vt \equiv 5 \pmod{12}.$$

Moreover,

$$d \mid \ell t + 1 \iff \ell t \equiv -1 \pmod{d}.$$

By (9.14), these two conditions are equivalent to

$$t \equiv a_{12d,\ell} \pmod{12d}.$$

It follows that

$$|M_n(\ell)_d| = \sum_{\substack{m^2+\ell t = n-1 \\ t \equiv a_{12d,\ell} \pmod{12d}}} \mathbf{1}_{\mathbb{P}}(m)\mathbf{1}_{\mathbb{P}}(t). \tag{11.7}$$

Lemma 11.8. *There is a multiplicative function* $\omega_n(\ell, \cdot)$, *defined on square-free integers supported on* W_ℓ, *such that*

$$\frac{G_R(\ell, 12d, n-1)}{\phi(12d)} = \frac{G_R(\ell, 12, n-1)}{\phi(12)} \frac{\omega_n(\ell, d)}{d}. \tag{11.9}$$

For $p \nmid 12\ell$, *its factors are*

$$\frac{\omega_n(\ell,p)}{p} = \begin{cases} \dfrac{\rho_p(n)}{p-1-\rho_p(n-1)}, & p \leq R, \\ \dfrac{1}{p-1}, & p > R. \end{cases} \tag{11.10}$$

In particular, $0 \leq \omega_n(\ell,p)/p < 1$.

Proof. Suppose first that $p \leq R$. The unrestricted local equation

$$x^2 + \ell t \equiv n - 1 \pmod{p}, \qquad x, t \in \mathbb{F}_p^\times,$$

has $p - 1 - \rho_p(n-1)$ solutions. The condition $p \mid \ell t + 1$ fixes $t = -\ell^{-1}$, and the equation then becomes $x^2 \equiv n \pmod{p}$, which has $\rho_p(n)$ solutions. The ratio of these local factors is therefore

$$\frac{\rho_p(n)}{p-1-\rho_p(n-1)}.$$

For $p > R$, the prime p does not occur in the product defining G_R, so the factor p to the modulus contributes only

$$\frac{\phi(12)}{\phi(12p)} = \frac{1}{p-1}.$$

Multiplicativity gives (11.9). If the local factor of $G_R(\ell, 12, n-1)$ at some prime $p \le R$ vanishes, then the corresponding local factor of $G_R(\ell, 12d, n-1)$ also vanishes for every square-free d supported on W_ℓ. In this case we define $\omega_n(\ell, p)$ by (11.10); the identity (11.9) is then examined locally, and both sides vanish whenever the common obstructing factor appears.

For $p \ge 7$, (11.10) is at most $2/(p-3) < 1$. The case $p = 5$ follows directly from the possible values of $\rho_5(n)$ and $\rho_5(n-1)$, and gives a bound of $1/2$. □

Put

$$X_n(\ell) = \frac{G_R(\ell, 12, n-1)}{\ell\phi(12)} H_{\ell,R}(n-1), \qquad \Theta_{n,\ell}(y) = \prod_{\substack{p<y \\ p\in W_\ell}} \left(1 - \frac{\omega_n(\ell,p)}{p}\right).$$

Applying Lemma 9.10 to (11.7) and using (11.9), we obtain

$$|M_n(\ell)_d| = X_n(\ell)\frac{\omega_n(\ell,d)}{d} + r_{12d,\ell}(n-1). \tag{11.11}$$

For $p > R$, both local densities equal $1/(p-1)$, independently of n. The uniform interval forms of Mertens' theorem give, for some fixed $K_0 \ge 2$ and all $2 \le w < y$,

$$\prod_{\substack{w\le p<y \\ p>3,\ p\equiv 3 \pmod 4}} \left(1 - \frac{1}{p-1}\right)^{-1} \le \left(1 + \frac{K_0}{\log w}\right)\left(\frac{\log y}{\log w}\right)^{1/2},$$

$$\prod_{\substack{w\le p<y \\ p\in W_\ell}} \left(1 - \frac{1}{p-1}\right)^{-1} \le \left(1 + \frac{K_0}{\log w}\right)\frac{\log y}{\log w}. \tag{11.12}$$

The second bound is uniform in ℓ, since restricting the product over odd primes only decreases it. Restricting both products to $p > R$ gives outer dimensions $1/2$ and 1, respectively. We retain the factors at $p \le R$ by pre-sieving.

Lemma 11.13. *Suppose that*

$$|\mathcal{F}_d| = Xg(d) + r_d,$$

where $X \ge 0$, and g is multiplicative on the square-free integers supported on a fixed set of primes $\mathfrak{P}$, with $g(1) = 1$ and $0 \le g(p) < 1$. Fix $C, A > 0$, set $R = (\log N)^C$, $D_0 = \exp((\log\log N)^3)$, and $D_1 = D/D_0$, and assume $R + 1 \le z \le D_1 \le N^A$ for sufficiently large N. Suppose that fixed constants $\kappa_0, \kappa > 0$ and $K_0 \ge 2$ satisfy, for every $2 \le w < y \le z$,

$$\prod_{\substack{w\le p<y \\ p\in\mathfrak{P},\ p\le R}} (1 - g(p))^{-1} \le \left(1 + \frac{K_0}{\log w}\right)\left(\frac{\log y}{\log w}\right)^{\kappa_0},$$

$$\prod_{\substack{w\le p<y \\ p\in\mathfrak{P},\ p>R}} (1 - g(p))^{-1} \le \left(1 + \frac{K_0}{\log w}\right)\left(\frac{\log y}{\log w}\right)^{\kappa}. \tag{11.14}$$

All $o(1)$ terms below are uniform for fixed $C, A, \kappa_0, \kappa, K_0$. Then

$$S(\mathcal{F}, \mathfrak{P}, z) \ge XV(z)\big(f_\kappa(s) + o(1)\big) + \sum_{m<D} c_m^- r_m,$$

$$S(\mathcal{F}, \mathfrak{P}, z) \le XV(z)\big(F_\kappa(s) + o(1)\big) + \sum_{m<D} c_m^+ r_m, \tag{11.15}$$

where

$$V(z)=\prod_{\substack{p<z\\ p\in\mathfrak{P}}}(1-g(p)),\qquad s=\frac{\log D_1}{\log z},\qquad |c_m^{\pm}|\le 1.$$

The coefficients $c_m^{\pm}$ depend only on the fixed prime sets and the sieve levels, and not on the values of g. The dimension κ_0 is used only in the preliminary sieve over $p\le R$; the sieve functions in (11.15) are determined by the outer dimension κ.

Proof. Put

$$V_0(R)=\prod_{\substack{p\le R\\ p\in\mathfrak{P}}}(1-g(p)),\quad s_0=\frac{\log D_0}{\log(R+1)},\quad \mathfrak{P}_0=\{p\in\mathfrak{P}:p\le R\},\quad \mathfrak{P}_1=\{p\in\mathfrak{P}:R<p<z\}.$$

Choose Rosser–Iwaniec weights satisfying

$$\operatorname{supp}\lambda_0^{\pm}\subseteq\{e<D_0:\mu^2(e)=1,\ p\mid e\Rightarrow p\in\mathfrak{P}_0\},$$
$$\operatorname{supp}\lambda_1^{\pm}\subseteq\{d<D_1:\mu^2(d)=1,\ p\mid d\Rightarrow p\in\mathfrak{P}_1\}.$$

The preliminary Rosser–Iwaniec weights are chosen independently of g, with $|\lambda_{0,e}^{\pm}|\le 1$ and the usual pointwise sieve inequalities. Apply the coefficient form of the fundamental lemma [24, (3.4)–(3.5)] to $\mathfrak{P}_0$, level D_0, and sifting limit $R+1$. This includes every prime in $\mathfrak{P}_0$, and the corresponding sieve product is $V_0(R)$. We obtain

$$V_0(R)\big(f_{\kappa_0}(s_0)-O(Q_{\kappa_0}(s_0)(\log D_0)^{O(1)})\big)\le\sum_e\lambda_{0,e}^{-}g(e)\le V_0(R),$$
$$V_0(R)\le\sum_e\lambda_{0,e}^{+}g(e)\le V_0(R)\big(F_{\kappa_0}(s_0)+O(Q_{\kappa_0}(s_0)(\log D_0)^{O(1)})\big).$$

The sieve functions satisfy

$$|F_{\kappa_0}(s)-1|+|f_{\kappa_0}(s)-1|\ll e^{-s},$$

while [24, (1.6)] gives

$$Q_{\kappa_0}(s)\le\exp\{-s\log s+s\log\log(3s)+O_{\kappa_0}(s)\}.$$

Since $D_0=\exp((\log\log N)^3)$ and $R=(\log N)^C$, we have $s_0=\log D_0/\log(R+1)\gg_C(\log\log N)^2$. Consequently, for every fixed $K>0$,

$$|F_{\kappa_0}(s_0)-1|+|f_{\kappa_0}(s_0)-1|+Q_{\kappa_0}(s_0)(\log D_0)^{O(1)}\ll e^{-s_0}\ll_K(\log N)^{-K}.$$

Thus, for either sign,

$$\sum_e\lambda_{0,e}^{\pm}g(e)=V_0(R)\big(1+O_K((\log N)^{-K})\big).\tag{11.16}$$

For the lower bound, write

$$P_0=\prod_{p\in\mathfrak{P}_0}p,\qquad P_1=\prod_{p\in\mathfrak{P}_1}p.$$

The outer lower weights satisfy the pointwise Rosser inequality

$$\sum_{d\mid a}\lambda_{1,d}^{-}\le 1_{(a,P_1)=1}.$$

Hence

$$S(\mathcal{F},\mathfrak{P},z)=\sum_{a\in\mathcal{F}}1_{(a,P_0)=1}1_{(a,P_1)=1}\geq\sum_{a\in\mathcal{F}}1_{(a,P_0)=1}\sum_{d\mid a}\lambda^-_{1,d}$$
$$=\sum_d\lambda^-_{1,d}\sum_{\substack{a\in\mathcal{F}\\ d\mid a}}1_{(a,P_0)=1}=\sum_d\lambda^-_{1,d}S(\mathcal{F}_d,\mathfrak{P}_0,2R).$$

Here the parameter $2R$ is harmless because $\mathfrak{P}_0$ already contains only primes $p\leq R$.

For the lower bound, if $\lambda^-_{1,d}\geq 0$ we insert the preliminary lower weights into $S(\mathcal{F}_d,\mathfrak{P}_0,2R)$, while if $\lambda^-_{1,d}<0$ we insert the preliminary upper weights. In either case, the remainders that occur are those in the approximation

$$|(\mathcal{F}_d)_e|=|\mathcal{F}_{ed}|=Xg(ed)+r_{ed},$$

where $\mathcal{F}_{ed}:=\{a\in\mathcal{F}:ed\mid a\}$. Thus the coefficient of r_{ed} is

$$c^-_{ed}=\begin{cases}\lambda^-_{1,d}\lambda^-_{0,e}, & \lambda^-_{1,d}\geq 0,\\ \lambda^-_{1,d}\lambda^+_{0,e}, & \lambda^-_{1,d}<0.\end{cases}$$

For the upper bound the choices are reversed, and

$$c^+_{ed}=\begin{cases}\lambda^+_{1,d}\lambda^+_{0,e}, & \lambda^+_{1,d}\geq 0,\\ \lambda^+_{1,d}\lambda^-_{0,e}, & \lambda^+_{1,d}<0.\end{cases}$$

The two prime ranges are disjoint, so every $m=ed$ has a unique such factorization with e supported on $\mathfrak{P}_0$ and d supported on $\mathfrak{P}_1$. Defining $c^\pm_m:=c^\pm_{ed}$, we therefore have

$$|c^\pm_m|\leq 1,\qquad m=ed<D_0D_1=D.$$

Put $V_1(z)=\prod_{p\in\mathfrak{P}_1}(1-g(p))$, so that $V(z)=V_0(R)V_1(z)$. By (11.16), the preliminary error is

$$\ll_K XV_0(R)(\log N)^{-K}\sum_d|\lambda^\pm_{1,d}|g(d).$$

The weights satisfy $|\lambda^\pm_{1,d}|\leq 1$ and are supported on square-free integers whose prime factors lie in $\mathfrak{P}_1$. Hence

$$\sum_d|\lambda^\pm_{1,d}|g(d)\leq\sum_{\substack{d\text{ square-free}\\ p\mid d\Rightarrow p\in\mathfrak{P}_1}}g(d)=\prod_{p\in\mathfrak{P}_1}(1+g(p)).$$

The equality follows by expanding the product: choosing $g(p)$ from the factors corresponding to the primes dividing d produces $g(d)$, by multiplicativity, and each such square-free integer occurs exactly once. Since $0\leq g(p)<1$, we also have $1+g(p)\leq(1-g(p))^{-1}$. Thus the preliminary error is

$$\ll_K XV_0(R)(\log N)^{-K}V_1(z)^{-1}=XV(z)(\log N)^{-K}V_1(z)^{-2}.$$

Taking $y=z$ and letting $w\downarrow R$ in (11.14) gives

$$V_1(z)^{-1}\ll\left(\frac{\log z}{\log R}\right)^\kappa\ll(\log N)^\kappa,$$

where we used $z\leq N^A$ and $R\geq 2$. The preliminary error is therefore $O_K(XV(z)(\log N)^{-K+2\kappa})$, which is $o(XV(z))$ upon choosing $K>2\kappa$. By the dimension-κ hypothesis (11.14), the Rosser–Iwaniec coefficient bounds for the outer weights give

$$\sum_d\lambda^-_{1,d}g(d)\geq V_1(z)\big(f_\kappa(s)+o(1)\big),$$

$$\sum_d \lambda^+_{1,d} g(d) \le V_1(z)\big(F_\kappa(s)+o(1)\big), \qquad s=\frac{\log D_1}{\log z}.$$

For the lower bound, the contribution of the main terms is

$$\begin{aligned} &X\sum_{\lambda^-_{1,d}\ge 0}\lambda^-_{1,d}g(d)\sum_e \lambda^-_{0,e}g(e) + X\sum_{\lambda^-_{1,d}<0}\lambda^-_{1,d}g(d)\sum_e\lambda^+_{0,e}g(e)\\ &\quad = XV_0(R)\sum_d \lambda^-_{1,d}g(d)+o(XV(z))\\ &\quad \ge XV_0(R)V_1(z)\big(f_\kappa(s)+o(1)\big) = XV(z)\big(f_\kappa(s)+o(1)\big), \end{aligned}$$

where the first equality follows from (11.16) and the preceding estimate for the preliminary error.

Similarly, the contribution of the main terms to the upper bound is

$$\begin{aligned} &X\sum_{\lambda^+_{1,d}\ge 0}\lambda^+_{1,d}g(d)\sum_e \lambda^+_{0,e}g(e) + X\sum_{\lambda^+_{1,d}<0}\lambda^+_{1,d}g(d)\sum_e\lambda^-_{0,e}g(e)\\ &\quad = XV_0(R)\sum_d \lambda^+_{1,d}g(d)+o(XV(z))\\ &\quad \le XV_0(R)V_1(z)\big(F_\kappa(s)+o(1)\big) = XV(z)\big(F_\kappa(s)+o(1)\big). \end{aligned}$$

Combining these estimates with the remainder coefficients $c^\pm_m$ defined above proves (11.15). □

The preliminary dimension bound (11.14) holds uniformly in both applications with $\kappa_0=2$. Indeed, Lemmas 11.6 and 11.8 give $g(p)\le 2/(p-3)=2/p+O(p^{-2})$ for $p\ge 7$, while $g(5)\le 1/2$ when that prime occurs. Mertens' uniform interval estimate supplies the required bound. This preliminary dimension affects only the error; the final sieve functions retain the outer dimensions $1/2$ and 1.

The separation of the small primes from the remaining sieve primes is similar to the two-stage sieve in [36, Section 2, especially (14)–(16)], where preliminary beta-sieve weights are combined with linear-sieve weights on the complementary range of primes.

Proof of Proposition 9.17. By (11.3), it is enough to bound $S(\mathcal{A}_n,\mathbb{P}_3,z)$ from below. Apply Lemma 11.13 to (11.5), with

$$D=\frac{N}{24(\log N)^B}.$$

The moduli $24d$, $d<D$, lie in the range of Lemma 9.10. Since $z=N^{8/9}$ and $D/D_0=N^{1-o(1)}$, the level conditions in Lemma 11.13 hold with $A=2$. By (11.12), the outer dimension is $1/2$, and

$$\frac{\log(D/D_0)}{\log z}=\frac{9}{8}+o(1).$$

Consequently,

$$S(\mathcal{A}_n,\mathbb{P}_{3,n},z)\ge X_n\Theta_n(z)\big(f_{1/2}(9/8)+o(1)\big)+\mathcal{E}_1(n).$$

Here $\Theta_n(z)$ is defined in (9.16), and

$$\mathcal{E}_1(n)=\sum_{d<D} c^-_d r_{24d,1}(n),$$

where Lemma 11.13 supplies coefficients c_d^- independent of n, with $|c_d^-| \le 1$. Lemma 9.10, with $\ell = 1$, gives

$$\sum_{n<N^2} |\mathcal{E}_1(n)|^2 \ll_{A_0,K} N^4 (\log N)^{-K}.$$

This proves the proposition. □

We next estimate the sum of the upper-sieve main terms. The calculation at primes $p \le R$ is exact, while all factors at $p > R$ are independent of n. Recall that H_1 denotes the $\ell = 1$ case of the archimedean main term from Lemma 10.11, namely

$$H_1(n) = H_1(n; N, N^2) = \int_{\substack{2\le y\le N \\ 2\le n-y^2\le N^2}} \frac{dy}{\log y \, \log(n-y^2)}.$$

For n satisfying (11.1), the interval $\sqrt{n}/4 \le y \le \sqrt{n}/2$ is contained in the domain of integration. On this interval we have $y \sim \sqrt{n}$ and $n - y^2 \sim n$, and hence

$$\log y \ll \log N, \qquad \log(n-y^2) \ll \log N.$$

Therefore

$$H_1(n) \ge \int_{\sqrt{n}/4}^{\sqrt{n}/2} \frac{dy}{\log y \, \log(n-y^2)} \gg \frac{\sqrt{n}}{(\log N)^2} \gg_{A_0} N(\log N)^{-A_0/2-2}, \tag{11.17}$$

uniformly for n in the range (11.1).

Lemma 11.18. *Let $\alpha = 9/8$. Uniformly in the range (11.1),*

$$\sum_{\ell\in L} X_n(\ell)\Theta_{n,\ell}(z_\ell) \le \big(c_\alpha \mathcal{W}(\alpha) + o(1)\big) X_n \Theta_n(z), \qquad c_\alpha = \frac{16 e^{-\gamma/2}}{\sqrt{\pi\alpha}}. \tag{11.19}$$

Here $\mathcal{W}(\alpha)$ is the integral defined in (9.19).

Proof. The proof contains a few steps.

Step 1: The parameters attached to ℓ. Write $\ell = 2up_2$ as in (9.6). Thus every prime divisor of u is $1 \pmod 4$, while $p_2 \equiv 3 \pmod 4$, and

$$u < \frac{N^2}{2z^2}, \qquad z \le p_2 < \left(\frac{N^2}{2u}\right)^{1/2}.$$

Recall also that the sifting level attached to $M_n(\ell)$ is

$$z_\ell = \left(\frac{N^2}{\ell}\right)^{1/4} = \left(\frac{N^2}{2up_2}\right)^{1/4}.$$

Since $z = N^{1/\alpha}$ and $\alpha = 9/8$, we have $z = N^{8/9}$, and hence $u < N^2/(2z^2) = N^{2/9}/2$. Using the upper bound for p_2, we get

$$2u^5 p_2 < 2u^5 \left(\frac{N^2}{2u}\right)^{1/2} = \sqrt{2}\, N u^{9/2} < N^2.$$

This is equivalent to $u^4 < N^2/(2up_2) = z_\ell^4$, and therefore $u < z_\ell$. In particular, every prime divisor of u lies below z_ℓ. Finally, since $p_2 \ge z$ and $\ell = 2up_2 \ge p_2$, we have

$$z_\ell = \left(\frac{N^2}{\ell}\right)^{1/4} \le \left(\frac{N^2}{p_2}\right)^{1/4} \le \left(\frac{N^2}{z}\right)^{1/4} = N^{5/18} < N^{8/9} = z.$$

Thus $p_2 \ge z > z_\ell$, and, since $R = (\log N)^C$, also $p_2 > R$ for all sufficiently large N.

Step 2: Factors at primes $p \le R$. We compare the factors at primes $p \le R$ in

$$G_R(1,24,n)\Theta_n(z) \quad \text{and} \quad G_R(\ell,12,n-1)\Theta_{n,\ell}(z_\ell), \qquad \ell = 2up_2.$$

For $3 < p \le R$, put

$$U_p = \frac{p\big(p-1-\rho_p(n)-\rho_p(n-1)\big)}{(p-1)^2}.$$

We compute the factors in the two sieve products using Lemmas 11.6 and 11.8.

Suppose first that $p \nmid \ell$. Since y is uniquely determined by x in

$$x^2 + \ell y \equiv n-1 \pmod p,$$

and $y = 0$ precisely when $x^2 \equiv n-1 \pmod p$, we have

$$\big|\{(x,y) \in \mathbb{A}_p^2 : x^2 + \ell y \equiv n-1 \pmod p\}\big| = p - 1 - \rho_p(n-1).$$

Thus the factor of $G_R(\ell,12,n-1)$ at p is

$$\frac{p\big(p-1-\rho_p(n-1)\big)}{(p-1)^2}.$$

Lemma 11.8 gives

$$\frac{\omega_n(\ell,p)}{p} = \frac{\rho_p(n)}{p-1-\rho_p(n-1)}.$$

Consequently, the factor of $G_R(\ell,12,n-1)\Theta_{n,\ell}(z_\ell)$ at p is

$$\frac{p\big(p-1-\rho_p(n-1)\big)}{(p-1)^2}\left(1 - \frac{\rho_p(n)}{p-1-\rho_p(n-1)}\right) = \frac{p\big(p-1-\rho_p(n)-\rho_p(n-1)\big)}{(p-1)^2} = U_p.$$

If $p \mid \ell$, then $p \le R < z \le p_2$, so $p \mid u$ and hence $p \equiv 1 \pmod 4$. In this case

$$x^2 + \ell y \equiv n-1 \pmod p \quad \Longleftrightarrow \quad x^2 \equiv n-1 \pmod p,$$

and therefore

$$\big|\{(x,y) \in \mathbb{A}_p^2 : x^2 + \ell y \equiv n-1 \pmod p\}\big| = (p-1)\rho_p(n-1).$$

Since $p \notin W_\ell$, there is no factor from $\Theta_{n,\ell}(z_\ell)$. The factor contributed by p is therefore

$$\frac{p}{(p-1)^2}(p-1)\rho_p(n-1) = \frac{p\rho_p(n-1)}{p-1}.$$

For the lower-sieve product, the congruence

$$x^2 + y \equiv n \pmod p$$

has $p - 1 - \rho_p(n)$ solutions in $\mathbb{A}_p^2$. Hence the factor of $G_R(1,24,n)$ at p is

$$\frac{p\big(p-1-\rho_p(n)\big)}{(p-1)^2}.$$

If $p \equiv 1 \pmod 4$, then p is not a lower-sieve prime, so this is also its factor in $G_R(1,24,n)\Theta_n(z)$. If $p \equiv 3 \pmod 4$, Lemma 11.6 gives

$$\frac{\omega_n(p)}{p} = \frac{\rho_p(n-1)}{p-1-\rho_p(n)}.$$

The factor in the full lower-sieve product is therefore

$$\frac{p\big(p-1-\rho_p(n)\big)}{(p-1)^2}\left(1 - \frac{\rho_p(n-1)}{p-1-\rho_p(n)}\right) = \frac{p\big(p-1-\rho_p(n)-\rho_p(n-1)\big)}{(p-1)^2} = U_p.$$

Thus the factor contributed by p to $G_R(1,24,n)\Theta_n(z)$ is

$$\begin{cases} \dfrac{p\big(p-1-\rho_p(n)\big)}{(p-1)^2}, & p\equiv 1 \pmod 4,\\ U_p, & p\equiv 3 \pmod 4. \end{cases}$$

To compare these products, define the non-negative multiplicative function h_R, supported on integers composed of primes $1 \pmod 4$, by

$$h_R(p^j)=\begin{cases} \dfrac{\rho_p(n-1)(p-1)}{p-1-\rho_p(n)-\rho_p(n-1)}, & p\le R,\\ \dfrac{p-1}{p-2}, & p>R, \end{cases} \qquad j\ge 1. \tag{11.20}$$

The second case will account for the factors with $p>R$ when the sum over u is estimated below. The denominator in the first case is positive. Indeed, $\rho_p(n),\rho_p(n-1)\in\{0,2\}$, so it is at least $p-5>0$ for $p\ge 7$. For $p=5$, two consecutive residues cannot both be non-zero quadratic residues, and hence $\rho_5(n)+\rho_5(n-1)\le 2$.

For $p\le R$, $p\equiv 1 \pmod 4$, we have

$$U_p\left(1+\frac{h_R(p)}{p-1}\right)=\frac{p\big(p-1-\rho_p(n)\big)}{(p-1)^2}, \qquad U_p h_R(p)=\frac{p\rho_p(n-1)}{p-1}.$$

Thus the first identity gives the factor in $G_R(1,24,n)\Theta_n(z)$, while the second gives the factor in $G_R(\ell,12,n-1)\Theta_{n,\ell}(z_\ell)$ when $p\mid u$. Set

$$B_R(n)=\prod_{3<p\le R}U_p, \qquad C_R(n)=B_R(n)\prod_{\substack{3<p\le R\\ p\equiv 1 \pmod 4}}\left(1+\frac{h_R(p)}{p-1}\right). \tag{11.21}$$

A direct calculation at 2 and 3 gives the same factor $8\cdot 3=24$ in $G_R(1,24,n)$ and $G_R(\ell,12,n-1)$. This and Lemma 11.6 imply that the product of the factors at $p\le R$ in $G_R(1,24,n)\Theta_n(z)$ is

$$24\prod_{\substack{3<p\le R\\ p\equiv 3 \pmod 4}}U_p\prod_{\substack{3<p\le R\\ p\equiv 1 \pmod 4}}\frac{p\big(p-1-\rho_p(n)\big)}{(p-1)^2}=24C_R(n), \tag{11.22}$$

Similarly, Lemma 11.8 gives the product of the factors at $p\le R$ in $G_R(\ell,12,n-1)\Theta_{n,\ell}(z_\ell)$ to be

$$24\prod_{\substack{3<p\le R\\ p\nmid u}}U_p\prod_{\substack{3<p\le R\\ p\mid u}}\frac{p\rho_p(n-1)}{p-1}=24B_R(n)\prod_{\substack{p\mid u\\ p\le R}}h_R(p).$$

Thus $C_R(n)$ is precisely the small-prime part of the lower-sieve main term, while $B_R(n)\prod_{p\mid u,\,p\le R}h_R(p)$ is the small-prime part of the upper-sieve term indexed by $\ell=2up_2$.

For primes beyond the pre-sieving range, let $q>R$ be an outer sieve prime. Since $q\nmid\ell$, the condition that q divide the sifted integer $\ell y+1$ is equivalent to $y\equiv -\ell^{-1} \pmod q$. Thus it selects one reduced residue class for the prime variable y modulo q. Hence the corresponding local density is $\frac{1}{\varphi(q)}=\frac{1}{q-1}$. This factor is independent of n. Mertens' theorem gives, uniformly for $y\ge N^{\varepsilon}$,

$$\prod_{R<p<y}\left(1-\frac{1}{p-1}\right)=(1+o(1))\frac{\log R}{\log y},$$

$$\prod_{\substack{R<p<y\\ p\equiv 3 \pmod 4}} \left(1-\frac{1}{p-1}\right) = (1+o(1))\left(\frac{\log R}{\log y}\right)^{1/2}. \tag{11.23}$$

We now compute the lower normalization. By definition,

$$X_n\Theta_n(z) = \frac{G_R(1,24,n)}{\varphi(24)} H_{1,R}(n) \prod_{\substack{3<p<z\\ p\equiv 3 \pmod 4}} \left(1-\frac{\omega_n(p)}{p}\right).$$

We split the product at R. The factors with $p \le R$ are part of the pre-sieved local computation. By (11.22), the factors at $p \le R$ give

$$\frac{G_R(1,24,n)}{\phi(24)} \prod_{\substack{3<p\le R\\ p\equiv 3 \pmod 4}} \left(1-\frac{\omega_n(p)}{p}\right) = \frac{24}{\phi(24)} C_R(n) = 3C_R(n).$$

Combining this with (11.23) for $p > R$, we obtain

$$X_n\Theta_n(z) = \big(3+o(1)\big) C_R(n) H_{1,R}(n) \left(\frac{\log R}{\log z}\right)^{1/2}. \tag{11.24}$$

For $q > R$, Lemma 11.8 gives $\frac{\omega_n(\ell,q)}{q} = \frac{1}{q-1}$. Since $p_2 \ge z > z_\ell$, the primes $q < z_\ell$ that divide $\ell = 2up_2$ are precisely the prime divisors of u. Hence

$$\prod_{\substack{R<q<z_\ell\\ q\in W_\ell}} \left(1-\frac{\omega_n(\ell,q)}{q}\right) = \prod_{R<q<z_\ell} \left(1-\frac{1}{q-1}\right) \prod_{\substack{q\mid u\\ q>R}} \left(1-\frac{1}{q-1}\right)^{-1},$$

where we also used $u < z_\ell$, proved above. For $q > R$,

$$\left(1-\frac{1}{q-1}\right)^{-1} = \frac{q-1}{q-2} = h_R(q).$$

Moreover, $h_R(q^j) = h_R(q)$ for every $j \ge 1$, and therefore

$$h_R(u) = \prod_{q\mid u} h_R(q).$$

Consequently,

$$\prod_{\substack{q\mid u\\ q\le R}} h_R(q) \prod_{\substack{q\mid u\\ q>R}} \left(1-\frac{1}{q-1}\right)^{-1} = h_R(u).$$

Combining this identity with (11.23) gives, uniformly for $\ell = 2up_2 \in L$,

$$X_n(\ell)\Theta_{n,\ell}(z_\ell) = \big(6+o(1)\big) B_R(n) h_R(u) \frac{H_{\ell,R}(n-1)}{\ell} \frac{\log R}{\log z_\ell}. \tag{11.25}$$

Step 3: The mean value of h_R. We prove that, uniformly in n,

$$\sum_{u\le y} h_R(u) = \left(\frac{e^{-\gamma/2}}{\sqrt{\pi}} + o(1)\right) \frac{C_R(n)}{B_R(n)} \frac{y}{\sqrt{\log R \log y}} \tag{11.26}$$

whenever $\log y \ge \frac{\log N}{\log\log N}$. We shall also prove the uniform upper bound

$$\sum_{u\le y} h_R(u) \ll \frac{C_R(n)}{B_R(n)} \frac{y}{\sqrt{\log R \log y}} \qquad (y \ge R^2). \tag{11.27}$$

We first establish the corresponding asymptotic for a fixed multiplicative function. Let h_∞ be supported on integers composed of primes congruent to 1 (mod 4), and define

$$h_\infty(p^j) = \frac{p-1}{p-2} \qquad (p \equiv 1 \pmod 4,\ j \geq 1).$$

Then

$$\sum_{p\leq x} h_\infty(p)\log p = \sum_{\substack{p\leq x\\ p\equiv 1 \pmod 4}} \log p + O\left(\sum_{p\leq x}\frac{\log p}{p}\right) = \frac{x}{2} + O_K\big(x(\log x)^{-K}\big)$$

for every fixed $K > 0$. Moreover,

$$h_\infty(p^j) = \frac{p-1}{p-2} \leq j+1 = \tau_2(p^j),$$

and hence $h_\infty(n) \leq \tau_2(n)$. Since the prime-sum estimate holds with an arbitrarily large logarithmic saving, [15, Theorem 1], applied with $\alpha = 1/2$ and $k = 2$, gives

$$\sum_{u\leq y} h_\infty(u) = \left(\frac{c_\infty}{\sqrt{\pi}} + O\left(\frac{1}{\log y}\right)\right)\frac{y}{\sqrt{\log y}}, \tag{11.28}$$

where

$$c_\infty = \prod_p \left(1 + \sum_{j\geq 1}\frac{h_\infty(p^j)}{p^j}\right)\left(1-\frac{1}{p}\right)^{1/2}. \tag{11.29}$$

We next compare h_R with h_∞, so we can replace h_∞ in (11.28) with h_R. For this purpose, we claim that there is a multiplicative function $a_{R,n}$, supported on integers composed of primes $p \leq R$, $p \equiv 1 \pmod 4$, such that

$$h_R = h_\infty * a_{R,n}, \tag{11.30}$$

and, for every fixed $J \geq 0$,

$$\sum_{d\geq 1}\frac{|a_{R,n}(d)|}{d}(1+\log d)^J \ll_J (\log R)^{O_J(1)}. \tag{11.31}$$

Here $*$ is the Dirichlet's convolution. We shall also need the estimate

$$\sum_{d\geq 1}\frac{|a_{R,n}(d)|}{d^{1-1/\log(2R)}} \ll (\log R)^{O(1)}. \tag{11.32}$$

To prove these claims, define $a_{R,n}$ by

$$\sum_{d\geq 1}\frac{a_{R,n}(d)}{d^s} = \prod_{\substack{p\leq R\\ p\equiv 1 \pmod 4}} \frac{1+\sum_{j\geq 1} h_R(p^j)p^{-js}}{1+\sum_{j\geq 1} h_\infty(p^j)p^{-js}}, \qquad \Re s > 1. \tag{11.33}$$

This immediately gives (11.30). Since both h_R and h_∞ are constant on the positive powers of a prime, the local quotient in (11.33) is $\frac{1+(h_R(p)-1)p^{-s}}{1+(h_\infty(p)-1)p^{-s}}$. It follows that

$$a_{R,n}(p^j) = \big(h_R(p)-h_\infty(p)\big)\big(1-h_\infty(p)\big)^{j-1} = \big(h_R(p)-h_\infty(p)\big)\left(-\frac{1}{p-2}\right)^{j-1} \tag{11.34}$$

for $p \leq R$, $p \equiv 1 \pmod 4$, and $j \geq 1$. By (11.20), one has $0 \leq h_R(p) \leq 6$: if $\rho_p(n-1) = 0$, then $h_R(p) = 0$, while otherwise this follows from the lower bounds for the denominator established immediately after (11.20). Since $h_\infty(p) = (p-1)/(p-2) \leq 4/3$, (11.34) gives

$$|a_{R,n}(p^j)| \ll (p-2)^{1-j}. \tag{11.35}$$

We now prove (11.32). Uniformly for $1-\frac{1}{\log(2R)}\le\sigma\le 1$, equation (11.35) gives

$$\sum_{j\ge1}\frac{|a_{R,n}(p^j)|}{p^{j\sigma}}\ll p^{-\sigma}.$$

Therefore

$$\sum_{d\ge1}\frac{|a_{R,n}(d)|}{d^\sigma}\le\prod_{\substack{p\le R\\ p\equiv1\ (\mathrm{mod}\ 4)}}\bigl(1+O(p^{-\sigma})\bigr)\ll\exp\left(O\left(\sum_{p\le R}p^{-\sigma}\right)\right)\ll(\log R)^{O(1)},$$

because $p^{1/\log(2R)}\ll1$ for $p\le R$. Taking $\sigma=1-1/\log(2R)$ proves (11.32).

We next prove the logarithmic-moment estimate (11.31). Put

$$A^*_{R,n}(s):=\sum_{d\ge1}\frac{|a_{R,n}(d)|}{d^s}=\prod_{\substack{p\le R\\ p\equiv1\ (\mathrm{mod}\ 4)}}\left(1+\sum_{\nu\ge1}\frac{|a_{R,n}(p^\nu)|}{p^{\nu s}}\right).$$

By (11.35), $A^*_{R,n}(1)\ll(\log R)^{O(1)}$. Moreover, for every fixed $j\ge1$, logarithmic differentiation of the Euler product gives

$$\left|\left(\frac{d}{ds}\right)^j\log A^*_{R,n}(s)\Big|_{s=1}\right|\ll_j\sum_{p\le R}\sum_{\nu\ge1}\frac{|a_{R,n}(p^\nu)|(\nu\log p)^j}{p^\nu}\ll_j\sum_{p\le R}\frac{(\log p)^j}{p}\ll_j(\log R)^j,$$

where the sum over ν is convergent uniformly by (11.35). Repeated differentiation of $A^*_{R,n}=\exp(\log A^*_{R,n})$ therefore gives

$$\left|\left(\frac{d}{ds}\right)^j A^*_{R,n}(s)\Big|_{s=1}\right|\ll_j(\log R)^{O_j(1)}.$$

On the other hand,

$$(-1)^j\left(\frac{d}{ds}\right)^j A^*_{R,n}(s)\Big|_{s=1}=\sum_{d\ge1}\frac{|a_{R,n}(d)|}{d}(\log d)^j.$$

It follows that

$$\sum_{d\ge1}\frac{|a_{R,n}(d)|}{d}(\log d)^j\ll_j(\log R)^{O_j(1)}\qquad(j\ge0).$$

Expanding $(1+\log d)^J$ and summing these estimates for $0\le j\le J$ proves (11.31).

We next claim that the contribution of large d in (11.30) is negligible. Put $D_y=\exp(\sqrt{\log y})$. Rankin's inequality and (11.32) give

$$\begin{aligned}\sum_{d>D_y}\frac{|a_{R,n}(d)|}{d}&\le D_y^{-1/\log(2R)}\sum_{d\ge1}\frac{|a_{R,n}(d)|}{d^{1-1/\log(2R)}}\\&\ll\exp\left(-\frac{\sqrt{\log y}}{\log(2R)}\right)(\log R)^{O(1)}\ll_K(\log N)^{-K}\end{aligned}\tag{11.36}$$

for every fixed $K>0$. The last estimate is uniform when $\log y\ge\log N/\log\log N$.

We can now transfer (11.28) to h_R. By (11.30),

$$\sum_{u\le y}h_R(u)=\sum_{d\le y}a_{R,n}(d)\sum_{v\le y/d}h_\infty(v).\tag{11.37}$$

For $d \le D_y$, we have $\log(y/d) \asymp \log y$ and

$$\frac{1}{\sqrt{\log(y/d)}} = \frac{1}{\sqrt{\log y}} + O\left(\frac{\log d}{(\log y)^{3/2}}\right). \tag{11.38}$$

The factor $\log d$ in (11.31) is needed precisely to sum the error in (11.38). Indeed,

$$\sum_{d\le D_y} \frac{|a_{R,n}(d)|}{d}\left|\frac{1}{\sqrt{\log(y/d)}} - \frac{1}{\sqrt{\log y}}\right| \ll \frac{1}{(\log y)^{3/2}} \sum_{d\ge 1} \frac{|a_{R,n}(d)|}{d}\log d \ll \frac{(\log R)^{O(1)}}{(\log y)^{3/2}}.$$

The error term in (11.28) gives the same bound, by (11.31) with $J = 0$.

For $d > D_y$, we use $h_\infty \le \tau_2$, and hence

$$\sum_{v\le x} h_\infty(v) \ll x\log(2x).$$

Together with (11.36), this shows that the contribution of $d > D_y$ to (11.37) is $O_K(y\log y\,(\log N)^{-K})$. Choosing K sufficiently large, and using

$$R = (\log N)^C, \qquad \log y \ge \frac{\log N}{\log\log N},$$

we conclude that

$$\sum_{u\le y} h_R(u) = \left(\frac{c_\infty}{\sqrt{\pi}}\sum_{d\ge 1}\frac{a_{R,n}(d)}{d} + o\left(\frac{C_R(n)}{B_R(n)\sqrt{\log R}}\right)\right)\frac{y}{\sqrt{\log y}}. \tag{11.39}$$

Here we also used $C_R(n)/B_R(n) \ge 1$, which follows from (11.21) and the non-negativity of h_R.

It remains to evaluate the coefficient in (11.39). By (11.31) with $J = 0$, the series $\sum_d \frac{a_{R,n}(d)}{d}$ converges absolutely. We may therefore set $s = 1$ in (11.33), obtaining

$$\sum_{d\ge 1}\frac{a_{R,n}(d)}{d} = \prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}} \frac{1+\sum_{j\ge 1} h_R(p^j)p^{-j}}{1+\sum_{j\ge 1} h_\infty(p^j)p^{-j}}.$$

Since $h_R(p^j) = h_R(p)$ and $h_\infty(p^j) = h_\infty(p)$ for every $j \ge 1$,

$$\sum_{j\ge 1}\frac{h_R(p^j)}{p^j} = h_R(p)\sum_{j\ge 1}\frac{1}{p^j} = \frac{h_R(p)}{p-1},$$

and similarly $\sum_{j\ge 1}\frac{h_\infty(p^j)}{p^j} = \frac{h_\infty(p)}{p-1}$. Consequently,

$$\sum_{d\ge 1}\frac{a_{R,n}(d)}{d} = \prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}} \frac{1+h_R(p)/(p-1)}{1+h_\infty(p)/(p-1)} = \frac{\displaystyle\prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}}\left(1+\frac{h_R(p)}{p-1}\right)}{\displaystyle\prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}}\left(1+\frac{h_\infty(p)}{p-1}\right)}.$$

By (11.21), the product in the numerator is

$$\prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}}\left(1+\frac{h_R(p)}{p-1}\right) = \frac{C_R(n)}{B_R(n)}.$$

Multiplying by c_∞, we conclude that

$$c_\infty \sum_{d\ge 1} \frac{a_{R,n}(d)}{d} = \frac{C_R(n)}{B_R(n)} \frac{c_\infty}{\prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}} \left(1+\frac{h_\infty(p)}{p-1}\right)}. \tag{11.40}$$

By (11.29),

$$\prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}} \left(1+\frac{h_\infty(p)}{p-1}\right) = \big(c_\infty + o(1)\big) \prod_{p\le R} \left(1-\frac{1}{p}\right)^{-1/2}.$$

Mertens' theorem therefore gives

$$\prod_{\substack{p\le R\\ p\equiv 1 \pmod 4}} \left(1+\frac{h_\infty(p)}{p-1}\right) = \big(e^{\gamma/2} c_\infty + o(1)\big)\sqrt{\log R}. \tag{11.41}$$

Combining (11.40) and (11.41), we obtain

$$c_\infty \sum_{d\ge 1} \frac{a_{R,n}(d)}{d} = \big(e^{-\gamma/2} + o(1)\big) \frac{C_R(n)}{B_R(n)\sqrt{\log R}}.$$

Substitution into (11.39) proves (11.26).

It remains to prove the upper bound (11.27). Shiu's theorem for non-negative multiplicative functions [45, Theorem 1], applied with modulus 1, gives

$$\sum_{u\le y} h_R(u) \ll \frac{y}{\log y} \prod_{p\le y} \left(1+\sum_{j\ge 1} \frac{h_R(p^j)}{p^j}\right).$$

It therefore suffices to show that

$$\prod_{p\le y} \left(1+\sum_{j\ge 1} \frac{h_R(p^j)}{p^j}\right) \ll \frac{C_R(n)}{B_R(n)} \left(\frac{\log y}{\log R}\right)^{1/2}. \tag{11.42}$$

For $y \ge R^2$, (11.20) and (11.21) give

$$\prod_{p\le y} \left(1+\sum_{j\ge 1} \frac{h_R(p^j)}{p^j}\right) = \frac{C_R(n)}{B_R(n)} \prod_{\substack{R<p\le y\\ p\equiv 1 \pmod 4}} \left(1+\frac{1}{p-2}\right) \ll \frac{C_R(n)}{B_R(n)} \left(\frac{\log y}{\log R}\right)^{1/2},$$

by Mertens' theorem in the progression 1 (mod 4). This proves (11.42), and hence (11.27).

Step 4: The weighted sum over ℓ. Next, we prove an estimate for a weighted sum involving h_R. For every fixed $A_1 > 0$, uniformly for $N^2(\log N)^{-A_1} \le m \le N^2$, one has

$$\sum_{\substack{\ell = 2up_2 \in L\\ \ell < m/2}} \frac{h_R(u)}{\ell(\log(m/\ell))^2} \le \left(\frac{2\sqrt{2}e^{-\gamma/2}}{\sqrt{\pi}} \mathcal{W}(\alpha) + o(1)\right) \frac{C_R(n)}{B_R(n)} \frac{1}{\sqrt{\log R}(\log m)^{3/2}}. \tag{11.43}$$

Fix $N^2(\log N)^{-A_1} \le m \le N^2$. We use partial summation twice. Put

$$r = \frac{\log u}{\log m}, \qquad v = \frac{\log p_2}{\log m}. \tag{11.44}$$

Since $\log m = 2\log N + O_{A_1}(\log\log N)$ and $\alpha = \frac{9}{8}$, the conditions defining L,

$$u < \frac{N^2}{2z^2}, \qquad z \le p_2 < \left(\frac{N^2}{2u}\right)^{1/2}, \qquad z = N^{1/\alpha},$$

become, up to $o(1)$-changes in the endpoints,

$$0 \le r \le 1 - \frac{1}{\alpha} = \frac{1}{9}, \qquad \frac{1}{2\alpha} = \frac{4}{9} \le v \le \frac{1-r}{2}.$$

Moreover, $\ell = 2up_2 \ll N^{10/9}$, whereas $m \ge N^2(\log N)^{-A_1}$. Hence $\ell < m/2$ is automatic for all sufficiently large N. Writing out the condition $\ell = 2up_2 \in L$, the left-hand side of (11.43) is

$$\sum_{\ell=2up_2\in L} \frac{h_R(u)}{\ell(\log(m/\ell))^2} = \frac{1}{2} \sum_{\substack{u<N^2/(2z^2)\\ p\mid u\Rightarrow p\equiv 1 \pmod 4}} \frac{h_R(u)}{u} \sum_{\substack{z\le p_2<(N^2/(2u))^{1/2}\\ p_2\equiv 3 \pmod 4}} \frac{1}{p_2(\log(m/(2up_2)))^2}. \tag{11.45}$$

Put

$$\mathcal{H}_R(x) = \sum_{\substack{u\le x\\ p\mid u\Rightarrow p\equiv 1 \pmod 4}} h_R(u), \qquad \pi_3(y) = |\{p \le y : p \equiv 3 \pmod 4\}|.$$

By partial summation, (11.45) equals

$$\frac{1}{2}\int_{1^-}^{N^2/(2z^2)} \frac{1}{x}\left(\int_{z^-}^{(N^2/(2x))^{1/2}} \frac{d\pi_3(y)}{y(\log(m/(2xy)))^2}\right) d\mathcal{H}_R(x).$$

Applying (11.26) to the outer integral and the prime number theorem in the progression 3 (mod 4) to the inner integral gives

$$\sum_{\substack{\ell=2up_2\in L\\ \ell<m/2}} \frac{h_R(u)}{\ell(\log(m/\ell))^2}$$
$$\le \left(\frac{e^{-\gamma/2}}{4\sqrt{\pi}} + o(1)\right) \frac{C_R(n)}{B_R(n)\sqrt{\log R}} \int_1^{N^2/(2z^2)} \frac{dx}{x\sqrt{\log x}} \int_z^{(N^2/(2x))^{1/2}} \frac{dy}{y\log y\,(\log(m/(2xy)))^2}.$$

With (11.44) in mind, we get

$$\sum_{\substack{\ell=2up_2\in L\\ \ell<m/2}} \frac{h_R(u)}{\ell(\log(m/\ell))^2} \le \left(\frac{e^{-\gamma/2}}{4\sqrt{\pi}} + o(1)\right) \frac{C_R(n)}{B_R(n)} \frac{I_\alpha}{\sqrt{\log R}(\log m)^{3/2}},$$

where

$$I_\alpha = \int_0^{1-1/\alpha} \frac{dr}{\sqrt{r}} \int_{1/(2\alpha)}^{(1-r)/2} \frac{dv}{v(1-r-v)^2}.$$

The factor $1/4$ comes from the prefactor $1/2$ in $\ell = 2up_2$, together with the density $1/2$ of primes $p_2 \equiv 3 \pmod 4$. For fixed r, put $t = 2\alpha(1-r)$ and $v = (1-r)y$. Since

$$\int_{1/t}^{1/2} \frac{dy}{y(1-y)^2} = \frac{t - 2 + (t-1)\log(t-1)}{t-1},$$

we get

$$I_\alpha = 2\alpha \int_2^{2\alpha} \frac{t - 2 + (t-1)\log(t-1)}{t^2(t-1)(1-t/(2\alpha))^{1/2}}\, dt = 8\sqrt{2}\,\mathcal{W}(\alpha).$$

This gives (11.43) on the range $\log u \ge \log N/\log\log N$.

Next consider $R^2 \le u < \exp(\log N/\log\log N)$. Repeating the preceding partial-summation argument, but using the upper bound (11.27) in place of the asymptotic (11.26), shows that this range contributes

$$\ll \frac{C_R(n)}{B_R(n)} \frac{1}{\sqrt{\log R}(\log m)^{3/2}} \int_0^{1/(2\log\log N)+o(1)} \frac{dr}{\sqrt{r}} \int_{1/(2\alpha)}^{(1-r)/2} \frac{dv}{v(1-r-v)^2}.$$

The inner integral is $O_\alpha(1)$, since $v \ge 1/(2\alpha)$ and $1-r-v \ge (1-r)/2 \gg_\alpha 1$ on this range. Hence the last display is

$$\ll \frac{C_R(n)}{B_R(n)} \frac{1}{\sqrt{\log R}(\log m)^{3/2}} \int_0^{1/(2\log\log N)+o(1)} \frac{dr}{\sqrt{r}} \ll (\log\log N)^{-1/2} \frac{C_R(n)}{B_R(n)} \frac{1}{\sqrt{\log R}(\log m)^{3/2}}.$$

This is $O((\log\log N)^{-1/2})$ times the right-hand side of (11.43). The remaining range $u < R^2$ is even smaller; indeed

$$\sum_{u<R^2} \frac{h_R(u)}{u} \ll (\log R)^{O(1)},$$

and the corresponding p_2-sum is $\ll (\log N)^{-2}$. Since $R = (\log N)^C$, this contribution is also

$$o\left(\frac{C_R(n)/B_R(n)}{\sqrt{\log R}(\log m)^{3/2}} \right).$$

This completes the proof of the claim.

Step 5: Replacing $H_{\ell,R}$ by H_ℓ. We now return to (11.25). First note that

$$\sum_{\ell=2up_2\in L} \frac{h_R(u)}{\ell} \le \frac{1}{2} \sum_{u<N^{2/9}} \frac{h_R(u)}{u} \sum_{z\le p_2<N} \frac{1}{p_2} \ll (\log N)^{O(1)}. \tag{11.46}$$

Indeed h_R is divisor-bounded uniformly in n, so $\sum_{u<N^{2/9}} h_R(u)/u \ll (\log N)^{O(1)}$, while $\sum_{z\le p_2<N} p_2^{-1} \ll \log\log N$. On the other hand, (10.13) gives, uniformly in ℓ,

$$|H_{\ell,R}(n-1) - H_\ell(n-1)| \ll NR^{-1/2}.$$

Therefore

$$\sum_{\ell\in L} \frac{h_R(u)}{\ell} |H_{\ell,R}(n-1) - H_\ell(n-1)| \ll NR^{-1/2}(\log N)^{O(1)}. \tag{11.47}$$

We now insert this error into (11.25). As discussed above, $z_\ell = (N^2/\ell)^{1/4} \gg N^{2/9}$. Thus $\log z_\ell \gg \log N$ uniformly in $\ell \in L$. Therefore the total contribution of this error term is

$$\ll B_R(n) \log R \sum_{\ell\in L} \frac{h_R(u)}{\ell \log z_\ell} |H_{\ell,R}(n-1) - H_\ell(n-1)| \ll B_R(n) NR^{-1/2} (\log N)^{O(1)} \frac{\log R}{\log N}.$$

By (11.24) and (11.17),

$$X_n \Theta_n(z) \gg C_R(n) N (\log N)^{-O_A(1)} \left(\frac{\log R}{\log N} \right)^{1/2}.$$

Since $B_R(n) \le C_R(n)$ and $R = (\log N)^C$, the contribution of the error from replacing $H_{\ell,R}$ by H_ℓ is $o(X_n\Theta_n(z))$ after choosing C sufficiently large.

Step 6: Integrating the weighted ℓ-sum. We now expand the integral defining $H_\ell(n-1)$. Put $m = m(y) := n - 1 - y^2$. Then

$$\begin{aligned} \sum_{\ell\in L} \frac{h_R(u) H_\ell(n-1)}{\ell \log(N^2/\ell)} &= \int_{\substack{2\le y\le N \\ m(y)>0}} \frac{dy}{\log y} \sum_{\substack{\ell=2up_2\in L \\ \ell<m(y)/2}} \frac{h_R(u)}{\ell \log(N^2/\ell) \log(m(y)/\ell)} \\ &\le \int_{\substack{2\le y\le N \\ m(y)>0}} \frac{dy}{\log y} \sum_{\substack{\ell=2up_2\in L \\ \ell<m(y)/2}} \frac{h_R(u)}{\ell (\log(m(y)/\ell))^2}, \end{aligned}$$

since $m(y) \le N^2$. Using (11.46) and

$$\big|\{y : 0 < n-1-y^2 < N^2(\log N)^{-A_1}\}\big| \ll \frac{N^2(\log N)^{-A_1}}{\sqrt{n}} \ll_{A_0} N(\log N)^{-A_1+A_0/2},$$

the contribution of $m(y) < N^2(\log N)^{-A_1}$ is

$$O_{A_0}\big(N(\log N)^{-A_1+A_0/2+O(1)}\big),$$

which is negligible after choosing A_1 sufficiently large. On the retained range $m(y) \ge N^2(\log N)^{-A_1}$, we have $\log m(y) = 2\log N + O_{A_1}(\log\log N)$. Thus (11.43) gives, uniformly on this range,

$$\sum_{\substack{\ell=2up_2\in L\\ 2\ell\le m(y)}} \frac{h_R(u)}{\ell(\log(m(y)/\ell))^2} \le \left(\frac{e^{-\gamma/2}}{\sqrt{\pi}}\mathcal{W}(\alpha)+o(1)\right)\frac{C_R(n)}{B_R(n)}\frac{1}{\sqrt{\log R}(\log N)^{3/2}}.$$

Here the factor $2\sqrt{2}$ in (11.43) cancels with the factor $2^{3/2}$ arising from $(\log m(y))^{3/2}$.

For these y, we have $2 \le n-y^2 = m(y)+1 \le N^2$, so the retained range is contained in the integration domain defining $H_1(n)$. Since $\log(n-y^2) \le 2\log N$, it follows that

$$\int_{\substack{2\le y\le N\\ m(y)\ge N^2(\log N)^{-A_1}}} \frac{dy}{\log y} \le 2\log N \int_{\substack{2\le y\le N\\ 2\le n-y^2\le N^2}} \frac{dy}{\log y\,\log(n-y^2)} = 2\log N\, H_1(n).$$

Integrating the preceding bound therefore gives the required estimate on the retained range.

To include the discarded range, recall that $C_R(n)/B_R(n) \ge 1$ by (11.21), while (11.17) gives $H_1(n) \gg_{A_0} N(\log N)^{-A_0/2-2}$. Since $R = (\log N)^C$, choosing A_1 sufficiently large in terms of A_0 makes the previously estimated error satisfy

$$N(\log N)^{-A_1+A_0/2+O(1)} = o\left(\frac{C_R(n)}{B_R(n)}\frac{H_1(n)}{\sqrt{\log R\log N}}\right).$$

Combining the two ranges, we obtain

$$\sum_{\ell\in L}\frac{h_R(u)H_\ell(n-1)}{\ell\log(N^2/\ell)} \le \left(\frac{2e^{-\gamma/2}}{\sqrt{\pi}}\mathcal{W}(\alpha)+o(1)\right)\frac{C_R(n)}{B_R(n)}\frac{H_1(n)}{\sqrt{\log R\log N}}. \tag{11.48}$$

Step 7: Concluding the proof. We now sum (11.25). The smoothing error (11.47) allows us to replace $H_{\ell,R}$ by H_ℓ at a cost $o(X_n\Theta_n(z))$. Since $\log z_\ell = \frac14\log(N^2/\ell)$, we get

$$\begin{aligned}\sum_{\ell\in L} X_n(\ell)\Theta_{n,\ell}(z_\ell) &\le \big(6+o(1)\big)B_R(n)\log R\sum_{\ell\in L}\frac{h_R(u)H_\ell(n-1)}{\ell\log z_\ell} + o(X_n\Theta_n(z))\\ &= \big(24+o(1)\big)B_R(n)\log R\sum_{\ell\in L}\frac{h_R(u)H_\ell(n-1)}{\ell\log(N^2/\ell)} + o(X_n\Theta_n(z)).\end{aligned}$$

Using (11.48), this gives

$$\sum_{\ell\in L} X_n(\ell)\Theta_{n,\ell}(z_\ell) \le \left(\frac{48e^{-\gamma/2}}{\sqrt{\pi}}\mathcal{W}(\alpha)+o(1)\right)C_R(n)H_1(n)\left(\frac{\log R}{\log N}\right)^{1/2}. \tag{11.49}$$

Lemma 10.11 and (11.17) give

$$H_{1,R}(n) = (1+o(1))H_1(n).$$

Since $\log z = \alpha^{-1}\log N$, (11.24) becomes

$$X_n\Theta_n(z) = \big(3\sqrt{\alpha}+o(1)\big)C_R(n)H_1(n)\left(\frac{\log R}{\log N}\right)^{1/2}.$$

Dividing (11.49) by this expression proves (11.19), since

$$\frac{48e^{-\gamma/2}/\sqrt{\pi}}{3\sqrt{\alpha}} = \frac{16e^{-\gamma/2}}{\sqrt{\pi\alpha}}.$$

□

Proof of Proposition 9.20. By Lemma 11.2, it is enough to estimate the fixed sum over $\ell \in L$ and W_ℓ. For each such ℓ, apply Lemma 11.13 to (11.11), with

$$D_\ell = \frac{(N^2/\ell)^{1/2}}{12(\log N)^B}, \qquad z_\ell = (N^2/\ell)^{1/4}.$$

Since $\ell \ll N^{10/9}$, we have $z_\ell \gg N^{2/9}$. Together with $D_0 = N^{o(1)}$ and $D_\ell = z_\ell^2/(12(\log N)^B)$, this verifies the level conditions in Lemma 11.13 uniformly in ℓ, with $A = 2$. The outer dimension is 1, by (11.12), and

$$\frac{\log(D_\ell/D_0)}{\log z_\ell} = 2 + o(1)$$

uniformly in $\ell \in L$. It follows that

$$S(M_n(\ell), W_\ell, z_\ell) \le X_n(\ell)\Theta_{n,\ell}(z_\ell)\big(F_1(2) + o(1)\big) + \sum_{d<D_\ell} c^+_{d,\ell} r_{12d,\ell}(n-1),$$

where $|c^+_{d,\ell}| \le 1$ and the coefficients are independent of n. Summing over $\ell \in L$ and applying Lemma 11.18 to the main terms gives

$$\sum_{\ell\in L_n} S(M_n(\ell), W_{\ell,n}, z_\ell) \le \big(c_\alpha F_1(2)\mathcal{W}(\alpha) + o(1)\big)X_n\Theta_n(z) + \mathcal{E}_2(n),$$

where

$$\mathcal{E}_2(n) = \sum_{\ell\in L}\sum_{d<D_\ell} c^+_{d,\ell} r_{12d,\ell}(n-1) + O(N^{1/4}).$$

The bounds defining L and (9.15) give $\ell \ll N^{10/9}$. Thus Lemma 9.10 applies with $L_0 = N^{6/5}$ and, for example, $\delta = 3/4$, and yields

$$\sum_{n<N^2} |\mathcal{E}_2(n)|^2 \ll_{A_0,K} N^4(\log N)^{-K}.$$

Here we used (11.4) with $\varepsilon = 1/4$. The added term contributes $O(N^{5/2})$ after squaring and summing over $n < N^2$, and is therefore absorbed by the preceding mean-square bound. This proves the proposition. □

We finish by proving the size of the common main term and the numerical inequality used in Section 9.

Proof of Lemma 9.22. For $p > 3$, every factor occurring in $C_R(n)$ is positive and is at least $1 - c/p$ for an absolute constant c. Mertens' theorem therefore gives

$$C_R(n) \gg (\log R)^{-c}. \tag{11.50}$$

Lemma 10.11 gives $H_{1,R}(n) = H_1(n) + O(NR^{-1/2})$. Taking C sufficiently large in terms of A_0, (11.17) therefore implies $H_{1,R}(n) \ge H_1(n)/2$.

Combining (11.24), (11.50), and (11.17), and recalling that $\log R = C\log\log N$ and $\log z = (8/9)\log N$, we obtain

$$X_n\Theta_n(z) \gg_{A_0,K} N(\log N)^{-C_{A_0}}$$

for some $C_{A_0} > 0$, uniformly in (11.1). This proves Lemma 9.22. □

Proof of Lemma 9.24. With the normalization of [24, (1.8)–(1.10)],

$$f_{1/2}(s) = \left(\frac{e^{\gamma}}{\pi s}\right)^{1/2} \int_1^s \frac{dt}{\sqrt{t(t-1)}}, \qquad 1 < s \le 3, \qquad F_1(2) = e^{\gamma}.$$

At $s = 9/8$, the substitution $t = \cosh^2 u$ gives

$$f_{1/2}(9/8) = \left(\frac{8e^{\gamma}}{9\pi}\right)^{1/2} \log 2.$$

It remains to bound $\mathcal{W}(9/8)$. The change of variables $r = 1 - 4t/9$ in (9.19) gives

$$8\sqrt{2}\,\mathcal{W}(9/8) = \int_0^{1/9} \frac{1}{\sqrt{r}} \int_{4/9}^{(1-r)/2} \frac{dv}{v(1-r-v)^2}\, dr.$$

On the inner interval, both v and $1-r-v$ are at least $4/9$. The inner integrand is therefore at most $729/64$, and its interval has length $(1/9 - r)/2$. Consequently,

$$8\sqrt{2}\,\mathcal{W}(9/8) \le \frac{729}{128} \int_0^{1/9} \frac{1/9 - r}{\sqrt{r}}\, dr = \frac{9}{32}.$$

Using the stated value of $c_{9/8}$ and $F_1(2) = e^{\gamma}$, we obtain

$$\Delta \ge \frac{e^{\gamma/2}}{\sqrt{\pi}} \left(\frac{2\sqrt{2}}{3} \log 2 - \frac{3}{8}\right) > 0.$$

The last inequality follows from $\log 2 \ge 1/2$ and $\sqrt{2} > 9/8$. This proves the numerical comparison. □

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